

Policy and Sustainability

Public Policy for Engineers

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Preface

These notes were created as support material for the course in *Technology, Sustainability, and Public Policy* at EPFL.

Disclaimer: these notes have been built from scratch. We have tried to eliminate typos and errors. Let us know if you spot a problem.

Thanks

We are grateful to Vanessa El Khoury, Emma Mantel, and Sandro Tissi for feedback, catching typos, etc. As always, all errors are ours.

Overview

This book is an introduction to the study of public policy. Its target audience are engineers and STEM students who may or may not have been exposed to topics in (formal and quantitative) social science. We focus on the topics of technology and sustainability (primarily *environmental* sustainability). There are two reasons for this. One is that they are normatively important. Another is that they lend themselves well to rigorous analysis. Indeed, the aim of this book is to equip readers with (passive) analytical and (active) problem-solving skills with a thematic focus on social problems that can affect our welfare over the long term. This stands in contrast to problems that are purely private (e.g., competition between two firms) or that are short term (e.g., a war).

This, in a nutshell, is what this book is trying to do: we want to analyze how social (environmental) problems emerge, what causes them, and identify effective solutions.

Pedagogical approach

This book builds heavily on (formal) models (S. E. Page 2018). We believe that models (abstract representations of real-life phenomena, often structured into a set of mathematical expressions) help readers in several ways:

1. Models train people to think in analytical ways and focus on the essential elements of a problem.
2. Models help people identify more effective solutions, given that they have an explicit statement of the problem.
3. Models make the assumptions of analysts more explicit.

In other words, thinking in models helps you understand problems and solve them. It also helps you understand where things might go wrong. Two examples:

Example. You want to solve the problem of over-use of a local resource (e.g., your public swimming pool). What will be the most effective policy? An ad campaign to ask people to limit the number of times they go to the pool? Maybe one that triggers guilt? Or a new pricing strategy? It is hard to tell without having a mental model of the causes of overuse.

Example. A local politician argues that a new highway project is justified because it passes a rigorous cost-benefit analysis. Is it ethically right to build it, then? Can you read the report and see if there are arbitrary choices? It is hard to be critical without understanding the models on which cost-benefit analyses are built.

For the most part, the math in this book is not particularly difficult, although knowing calculus and optimization will help. Sometimes (rarely), we will include differential equations, but in general, the reader will be able to understand the logic of the argument without knowing how to solve them.

Aside from this, we try to link theoretical models to empirical evidence as systematically as feasible. Having good models alone isn't enough for good policy. We must confront our models with reality. Whenever feasible, we will do so by drawing on recent (or canonical) quantitative analyses.

Part I

Building blocks

1 Post-industrial economy

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Summary

Public policy refers to the idea that public authorities intervene in society and markets to change the state of the world. We start by putting this in its historical context and the emergence of the post-industrial ‘growth’ economy.

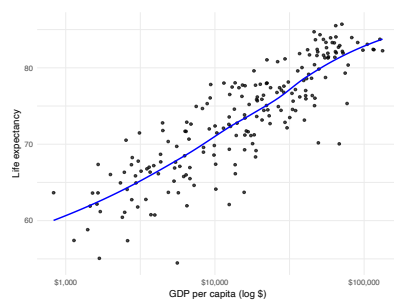
1.1 Growth and human welfare

One of the central objects of politics is the economy. Gross domestic product (GDP) receives particular attention from businesses, politicians, and citizens (see Note 1 for definitions).

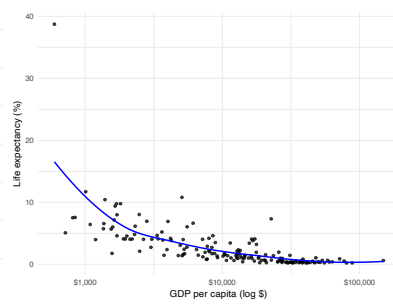
There are good and bad reasons to value GDP. Growth in GDP means more goods and services, and if happiness increases with these, then more GDP is desirable. GDP also correlates positively with things we generally value: life expectancy (Figure 1.1a), child mortality (Figure 1.1b), protein supply (Figure 1.1c), education (Figure 1.1d), interpersonal trust (Figure 1.1e), and self-reported life satisfaction (Figure 1.1f).

Several caveats apply:

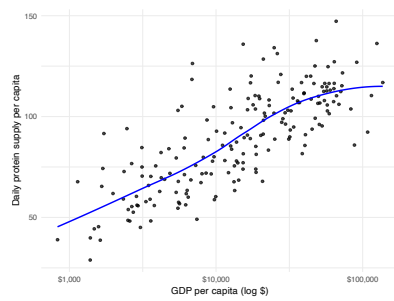
- These figures show *correlations*, not causal relationships. Reverse causation (e.g., education driving GDP) and spurious correlations are possible.
- These figures show *country-level averages*; it doesn’t tell us about *within-country* variation (e.g., two countries can have a GDP per capital of \$10,000, but the benefits of these resources might not be distributed in the same way).



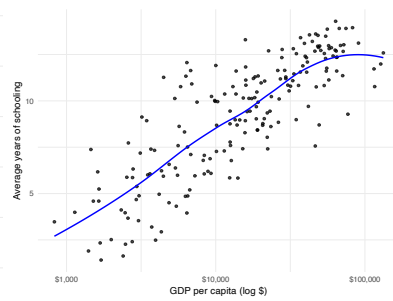
(a) GDP/capita and life expectancy.



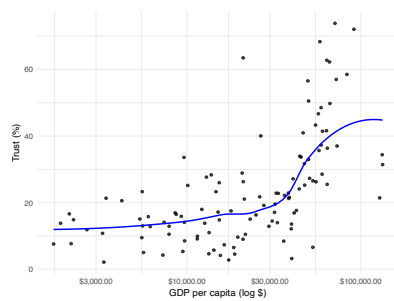
(b) GDP/capita and child mortality.



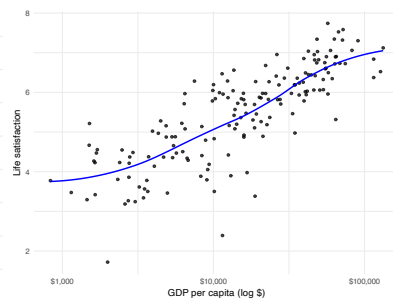
(c) GDP/capita and protein supply.



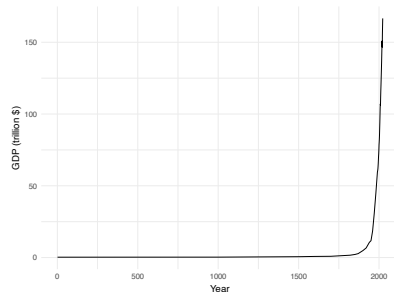
(d) GDP/capita and schooling.



(e) GDP/capita and trust.



(f) GDP/capita and life satisfaction.



(g) GDP over time.

Figure 1.1: GDP per capita and multiple social indicators. Country-level data. Source: World Bank (via *Our World in Data*).

- The literature on happiness and income shows conflicting findings, with some studies finding no long-run relationship (Easterlin et al. 2010) (though others do; (Stevenson and Wolfers 2013)).
- GDP does not capture all dimensions of human welfare.
- GDP does not measure well the emergence of new goods and of improvements of quality of products. For instance, someone living in a country with a GDP of \$10,000 fifty years ago does not magically get access to the medicine available to someone living in an equally rich country today (Trammell and Jones 2026).

Yet the correlations between GDP per capita and these outcomes are hard to ignore, at least for material welfare. Pritchett and Lewis (2022) note that there are no countries with high GDP per capita that are doing badly on material welfare indicators, and conversely, no poor country is scoring highly on material welfare either. High GDP per capita appears therefore necessary and sufficient for material wellbeing (keeping in mind the caveats above).

Interestingly, GDP growth is also a recent phenomenon: economic historians have reconstructed pre-20th century data showing just how new this era of growth truly is Figure 1.1g. This raises several key questions:

- Is output growth unambiguously good for human welfare?
- Can output growth be sustained permanently?
- What are the side effects of the unconstrained growth model inherited from the Industrial Revolution?
- How can technological development be steered to support wellbeing and sustainability?
- Can societies thrive beyond growth?

Note 1: GDP and income

A bit of terminology:

- **GDP** is the value of all final goods and services produced in a given geographic region (e.g., a country) over a given time period (e.g., a year). Equivalently, it is the sum of added value within a given region and time period. For instance, an artisan making a chair gets wood for free, transforms it into a chair, and sells it to a furniture shop for \$15, which then sells it to a consumer for \$25. Thus, two firms

add value: the chair-maker (+\$15) and the furniture shop (+\$10), giving a total added value (GDP) of \$25.

- **(Gross economic) output** is the value of all goods and services produced in a given geographic region over a given time period, including intermediate sales. Using the same example, output is \$15 plus \$25, for a total of \$40.
- **Gross national product (GNP)** is the value of goods and services produced by a given set of citizens, regardless of where they are located.
- **Goods and services** are quantified by their market value, initially measured in domestic currency, but often converted into another currency (especially US dollars), using either the exchange rate or purchasing power parity (PPP).
- **GDP per capita** is GDP divided by population.
- **Real GDP** adjusts **nominal GDP** for inflation. For example, if all goods double in price (100% inflation), nominal GDP doubles, but the quantity of goods does not. Since we are generally interested in the quantity (and quality) of goods and services, we often use *real* GDP.

1.2 The modern era

Where, then, does growth in economic output and technological innovations come from?

Human societies have not developed linearly (Comin, Easterly, and Gong 2010). Two critical junctures explain where we are today:

1. The Neolithic Revolution
2. The Industrial Revolution

1.2.1 The Neolithic revolution

The Neolithic Revolution (globally, 12,000–5,000 BC) marked the transition from hunter-gatherer to agricultural societies (Matranga 2024). Partly driven by climate change, societies became sedentary, sustained larger populations, and developed complex social organization (Bowles and Choi 2013, 2019):

- An increase in the size of society, as farming could produce food for a larger population over a smaller area.
- An increase in the division of labor: not everyone was engaged in food collection or childcare. New occupations emerged, including craftspeople, merchants, civil servants, and artists.
- New modes of ownership, with the emergence of private property.
- The creation of laws and state institutions. Larger populations and new occupations required general rules and people to implement and enforce them, such as tax collectors, law enforcement, and armies.

Societies continued to evolve—Roman water and sewage systems, for instance, were clear advances. Yet many changes remained marginal, and the overall scale of human societies did not shift dramatically. A person transported from 1 CE to 1000 CE would notice little difference in technology or welfare, and might even perceive decline in some areas.

1.2.2 The Industrial revolution

The next critical juncture was the Industrial Revolution, which is commonly said to have started around 1750 in the United Kingdom. It unleashed an unprecedented era of growth in the production and consumption of goods and services (Mokyr 1990, 2016, 2018; Allen 2009). Across much of Europe, wages then started to grow as well by the mid-1800s, leading to unprecedented levels of economic welfare.

The Industrial Revolution was not just about a change in economic production. Society and the state changed as well. Elements of *society* asked for new freedoms (e.g., voting rights or the right to move), for the end of aristocratic privileges, and so forth (Ebert et al. 2025).

Unified *states* with a centralized administration emerge around the same time (Polanyi 1957; Skowronek 1982; Mokyr 2011). For instance, Switzerland emerged as a federal state in 1848, Italy emerged in 1861, German was unified in 1871. These states all built up a new structure: a professional bureaucracy that provided all kinds of services (e.g., public schooling) and developed all kinds of resources to produce them (e.g., collecting statistics, designing new taxes, etc.). In Switzerland, for example, the modern state emerged alongside the

need for roads and rail networks to transport goods across the Alps. The state required engineers (leading to the creation of ETH Zürich and the École spéciale de Lausanne, EPFL's predecessor), as well as officials to collect statistics, draw maps, and promote innovation.

1.3 Technology, capital, and output

Here's what's key: The Industrial Revolution shifted (at the time mostly European and American) society from a situation of *steady state* (with stagnating output and welfare levels) to a situation of continuous (if unstable) growth. Economic historians have spent a lot of time debating what *caused* the Industrial Revolution. The main categories of explanations are (Mokyr 1990):

- **Capital** ("Solowian" growth): savings fueled investments in capital goods, such as factories, increasing production capacity.
- **Technological innovation** ("Schumpeterian" growth) (Schumpeter 1934): innovations made workers and capital more effective.
- **Institutions** ("Northian" growth) (North 1991): more democratic and capable institutions improved public goods and services, including infrastructure, education, and the legal system.
- **Trade** ("Smithian" growth): declining trade costs integrated markets and expanded production.
- **Scale** (division of labor): a larger workforce allowed greater specialization and efficiency.

Economists post-World War II have tried to develop mathematical models to understand what causes output growth. Since we will use a lot of models, let's start by a canonical model of output to get our feet wet. We begin with the *Solow-Swan* model that links various factors of production (labor, capital) to output (Aghion and Howitt 2009). This model is now dated, but it will give us a sense of how these models work.

1.3.1 Solow-Swan growth model

The Solow-Swan model (Robert M. Solow 1956; Swan 1956) was developed independently by Bob Solow and Trevor Swan after World War II to understand postwar recovery and development. Its goal is to explain how the level of economic activity Y – the total value of goods and services produced – changes over time. Note that since output generates income through wages (to workers) and returns to capital (to investors), Y also corresponds to total income.

The simplest version of the Solow-Swan model connects Y to capital K (tools, factories, offices, and other productive assets). To get traction, we formulate three assumptions:

Assumption 1. The production function of output Y is (solely) a function $F(\cdot)$ of capital K :

$$Y = F(K).$$

In the appendix, we include labor L as an additional factor of production; since the results remain qualitatively similar, we don't expand on it here.

Assumption 2. Output exhibits *diminishing marginal returns*.¹ This means that Y is increasing but concave in K .

$$\frac{dF}{dK} > 0, \quad \frac{d^2F}{dK^2} < 0$$

Assumption 3. The *Inada* conditions hold:

$$F(0) = 0 \quad \text{and} \quad \lim_{K \rightarrow \infty} F'(K) = 0 \quad \text{and} \quad \lim_{K \rightarrow 0} F'(K) = \infty$$

Since we will talk about growth, we are interested in change over time. Define $\dot{x}$ as the change of x over time (which we assume to be continuous):

$$\dot{Y} \equiv \frac{dY}{dt} \quad \dot{K} \equiv \frac{dK}{dt}$$

¹Diminishing returns mean that, as you increase the amount of input (here: capital), the amount of output (here: GDP or some similar measure) increases less and less. Intuitively: as you build more and more factories and machines, they produce less and less.

To understand how Y grows over time, we must understand how K changes, since that's the only source of growth. Capital grows (or not) according to the following law:

$$\dot{K} = sY - \delta K = sF(K) - \delta K,$$

where s is the saving rate (i.e., how much saved from total income Y) and δ is depreciation (how obsolete and less useful your capital becomes because of the passage of time). Savings sY can be used to build new factories, tools, etc. The implication: capital will grow ($\dot{K}$ will be positive) as long as savings are greater than depreciation. When the two are equal, we say that K will be in a **steady state** (or equilibrium).² See Figure 1.2.

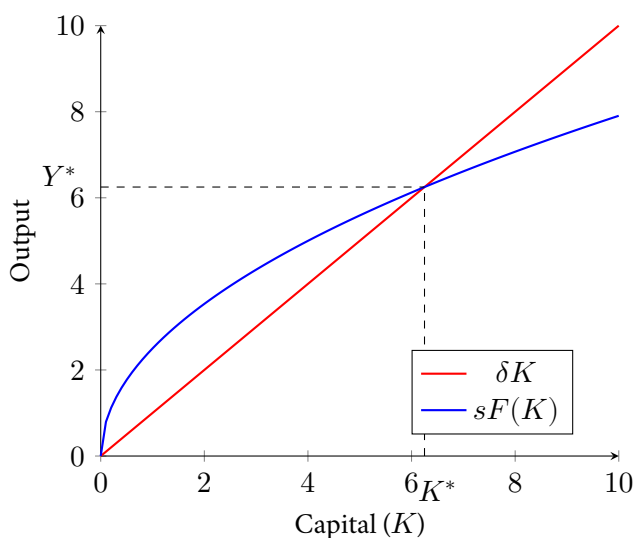


Figure 1.2: Solow-Swan dynamics. K^* is the equilibrium amount of capital, which corresponds to a steady-state output level of Y^* . At that point, new capital is created in equal amount to capital that is depreciated, meaning that no growth is happening.

To solve the model, assume an initial capital stock K_0 such that $0 < K_0 < K^*$, where K^* is the level of capital that satisfies $sF(K) = \delta K$.

²In general, we use *equilibrium* to denote a system that doesn't change (and therefore is more likely to be used for static problems) and *steady state* to refer to a dynamic system that settled at an equilibrium.

At K_0 , $sF(K) > \delta K$ due to diminishing returns and the Inada conditions: the concave function $sF(K)$ initially lies above δK . As a result, capital grows over time ($K_1 > K_0$) because savings exceed depreciation. Repeating this process over time, the economy eventually reaches a point where $sF(K) = \delta K$, where all new capital is offset by depreciation. At this steady state, K is constant, and output Y remains constant as well (by Assumption 1). We have reached an equilibrium, and in the steady state, growth is zero. See Note 2 for a discussion of the policy implications of these results.

i Note 2: Policy implications of Solow-Swan

In the Solow-Swan model, growth is driven by capital. This is why some call growth driven by capital accumulation “Solowian” growth. The only policy levers are:

1. Increasing the savings rate s , or
2. Reducing depreciation δ .

In Figure 1.2, this corresponds to shifting the blue line up or the red line down. However, neither lever generates sustained, long-term growth. For example, if the savings rate s rises from 4% to 6%, output increases from $Y(s = 0.04)$ to $Y(s = 0.06)$, but then growth eventually ceases at the new steady state. Sustaining growth would require continuously increasing s or decreasing δ , which is infeasible because savings cannot exceed 100% and depreciation cannot fall below zero.

1.3.2 Solow-Swan with technology

The model presented so far yields a weird result: eventually, growth gets ‘eaten’ by diminishing returns. Yet when Solow and Swan wrote their studies, many countries were experiencing large and seemingly persistent growth. Here’s an important insight: a model that is obviously wrong can still be interesting because it forces you to understand where your logic breaks down.

To make the model more realistic, the next step is to add two inputs: labor (workers) and technology. Let L denote labor and A a productivity parameter that augments it, so AL can be thought of as *effective*

labor. The production function becomes:

$$Y = F(K, AL),$$

By *technology*, we mean a parameter that makes capital and labor more productive, expanding total production. This excludes leisure-oriented innovations (e.g., a new musical instrument) that do not affect A . In general, we can think of two types of technologies:

1. **“Conventional” technology**: new physical items used to produce goods and services (e.g., machines, computers).
2. **Human capital**: ideas, knowledge, and skills.

Solow and Swan primarily had the former in mind. Subsequent growth models introduced human capital as a separate input (Mankiw, Romer, and Weil 1992; Barro 2001; G. S. Becker 2009; Squicciarini and Voigtländer 2015). For our purposes, we can treat A loosely as encompassing both types.

We now make one additional assumption regarding returns to scale, i.e., how output changes when all inputs increase proportionally.

Assumption 4. Assume **constant returns to scale**, namely:

$$F(\lambda K, \lambda AL) = \lambda F(K, AL) \quad \forall \lambda, K, L, A > 0$$

This simply states that if all inputs are increased by a factor λ , total output will increase by the same factor.³ Equivalently, this means that the production function $F(\cdot)$ must be homogeneous of degree 1.

Adding labor L alone does not change the key results obtained earlier (see appendix), because diminishing returns on capital continue to stabilize the economy at a steady state with no growth.

This, however, is not the case for technology: A can fight the weight of diminishing returns. How? To figure out, we need now to be more explicit about the output function $F(\cdot)$. Let us assume a Cobb-Douglas

³Diminishing *marginal* returns (Assumption 2) states that output increases at a decreasing rate when a single input rises, while constant returns to scale means that output increases proportionally when *all* inputs increase by the same factor. It’s the same difference as between the partial derivative with respect to inputs (*marginal returns*) and the expected change when all inputs are multiplied by a constant (*returns to scale*).

production function. Cobb-Douglas is widely used to capture such problems. We have:

$$Y = (AL)^{1-\alpha} K^\alpha.$$

Technology grows at rate $g \equiv \dot{A}/A$. For now, we simply assume such a process exists.

Next, we define new quantities:

$$\kappa \equiv \frac{K}{AL}, \quad \phi \equiv \frac{Y}{AL} = \kappa^\alpha,$$

where κ is capital per unit of effective labor and ϕ is output per unit of effective labor.⁴ We want to know how κ behaves, since that's what drives ϕ (which in turn is an important quantity, since it's close to GDP per capita).

What we want to know: can output per capita (Y/L) grow in the long run (and not just from time to time when savings increase)? Yes! To see this, we can rearrange the expression of output per effective labor:

$$\frac{Y}{AL} = \kappa^\alpha \Rightarrow \frac{Y}{L} = A\kappa^\alpha$$

Does this quantity grow over time? Yes, it can:⁵

$$\left[\frac{\dot{Y}}{L} \right] = \underbrace{\frac{\dot{A}}{A}}_{\text{Tech growth}} + \alpha \frac{\dot{\kappa}}{\kappa}$$

⁴To see how we obtained this, divide $F()$ by AL :

$$\frac{Y}{AL} = \frac{(AL)^{1-\alpha} K^\alpha}{AL} = (AL)^{1-\alpha} K^\alpha (AL)^{-1} = (AL)^{-\alpha} K^\alpha = \frac{K^\alpha}{(AL)^\alpha} = \left(\frac{K}{AL} \right)^\alpha.$$

⁵We can do the math. Take the log:

$$\log \left(\frac{Y}{L} \right) = \log(A) + \alpha \log(\kappa)$$

Then differentiate with respect to time:

$$\frac{d}{dt} \log \left(\frac{Y}{L} \right) = \frac{\dot{A}}{A} + \alpha \frac{\dot{\kappa}}{\kappa}$$

The left-hand side expression is growth of output per worker.

But $\kappa \rightarrow \kappa^*$ in the long term (to the steady state), so the second term will go to 0. The first term, however, could stay positive. This ultimately is the source of long-term growth in a model based on labor and capital.

And indeed, empirical studies struggled to find evidence that growth can be driven by capital (or labor) accumulation (Easterly and Levine 2002). See Note 3 and Note 4.

i Note 3: Growth from capital accumulation versus technology

Easterly and Levine (2002) build a Solow-Swan model to do some back-of-the-envelope calculations regarding the role of technology:

$$y = Ak^\alpha n^{1-\alpha},$$

Here, y is output per capita, n is labor per capita, which accounts for improvements in skills and human capital over time, and α is the share of income attributed to capital. The parameter A represents technology, although it is also commonly referred to as *total factor productivity* (TFP), a concept broader than just technology (Robert M. Solow 1957). Others also call it the “Solow residual” to highlight that A is a catch-all parameter capturing our ignorance.⁶

$$Y = AMK^\alpha H^{1-\alpha},$$

Here, AM represents total factor productivity (TFP), split between A (knowledge) and M (all other factors influencing TFP). K and H are physical and human capital, respectively.

Now, they engage in the following accounting exercise. First, we can rewrite the same production function in terms of growth rates:

$$\frac{\Delta y}{y} = \frac{\Delta A}{A} + \alpha \frac{\Delta k}{k} + (1 - \alpha) \frac{\Delta n}{n}$$

Suppose that per-capita output y grows by 2%, that $\alpha = 0.4$ (based on data), that per-capita capital grows by 3%, and that there is no human capital improvement ($\Delta n = 0$). We can then infer that TFP grows by 0.8% to account for the observed output growth. Note that $0.8/2 = 0.4$: 40% of per-capita output growth would come from technological improvements (or, more broadly, TFP growth). This kind of observation has triggered considerable research in the 1990s aimed at collecting bet-

ter data to distinguish the contributions of labor, capital, and productivity (Romer 1990; Mankiw, Romer, and Weil 1992; Barro and Sala-i-Martin 1992; Klenow and Rodriguez-Clare 1997; C. I. Jones 2016). Among the most recent studies, C. I. Jones (2016, 10) finds that TFP has been the dominant contributor to output growth in the US since 1948.

Is this just an academic debate? Not at all. It has two types of policy implications:

1. **Policy targeting.** Policies should focus on the appropriate domain: in particular, if you want to increase output, then your policies should focus on increasing A . This could include education, patents, etc.
2. **Policy design.** Technology exhibits public good features: it is non-rival (one person's use does not preclude others from using it) and sometimes non-excludable (new ideas and innovations spread). Capital, on the other hand, is generally privately owned (Klenow and Rodriguez-Clare 1997). Thus, technology-focused policies need to balance rewarding innovators with ensuring the optimal dissemination of innovations across the economy.

More recent models of economic growth have devoted considerable effort to endogenize A , that is, to explain what drives technology (or TFP). These models include endogenous growth models (Romer 1990, 1994; Charles I. Jones 2022). Other studies have investigated the role of institutions in shaping economic growth, including via their effect on innovation (Acemoglu, Johnson, and Robinson 2005; Juhász 2018). We will not explore these modern developments further until later when we discuss obstacles to (sustainable) innovation.

i Note 4: Policy implications of Solow-Swan with technology

This model predicts:

- Growth of *GDP per efficiency labor unit* ($\Delta Y / AL$) is zero in the long term.
- Growth of *GDP per labor unit* ($\Delta Y / L$) is positive if and only if $\dot{A}/A > 0$.

⁶Additionally, C. I. Jones (2016, 8) makes further distinctions.

- Changes in *savings* and *depreciation* lead to one-time changes in the level of Y , as in the Solow-Swan model.

Thus, technological progress (i.e., a positive $\dot{A}$) is the main driver of sustained long-term growth.

The key takeaway: **sustained economic growth since the Industrial Revolution is best explained by continuous technological progress.** Trade, division of labor, and capital accumulation generate one-time boosts in output *levels* but not ongoing *growth*; only technological advance sustains growth over the long run.

The questions, then, are:

- Can technological improvement continue forever? Concerns over the **sustainability** of economic growth cast doubt on this claim, at least without proper qualification.
- Where do technological innovations come from? What drives them? How can technologies with little or benign impact on human welfare be encouraged, and harmful ones discouraged?

Additional readings

There are several textbooks on economic growth, including Acemoglu (2009) and Aghion and Howitt (2008). On institutional economics and economic development, see Acemoglu and Robinson (2012). On the history of economic development, see Galor (2022) (for a popular audience treatment) or country-specific studies, such as Allen (2009) (on the British experience).

Appendix

Solow-Swan with labor

We can augment the Solow-Swan model to include labor L . We then obtain results that are quite similar to the baseline Solow-Swan model. Let us add labor to the production function:

$$Y = F(K, L)$$

Let's solve the model. Let's assume that L is the same as the population (no unemployment, children, etc) and that L grows at a rate of n ($\dot{L} \equiv n$). Let's also assume that each individual produces a single unit of labor per time period. Let's also focus on *per-capita* quantities. We can then write (keeping in mind Assumption 4):

$$y \equiv \frac{Y}{L} = \frac{F(K, L)}{L} = F\left(\frac{K}{L}, 1\right) = f(k)$$

We say that $f(k)$ is the *per-capita production function*.

To continue, we need to make a further assumption about $F(\cdot)$. We assume that it is Cobb-Douglas:

$$Y = K^\alpha L^{1-\alpha}, \quad 0 < \alpha < 1$$

This is a very common functional form that has all kinds of advantages. It is structured by a single parameter, α , which measures the share of Y that goes to K as income (often around 0.4 or so in real life).⁷

Note that if you divide this expression by L , you obtain:

$$y = k^\alpha$$

We ask a similar question to before: how does k change over time ($\dot{k}$)?

To work this out:

$$\begin{aligned} \dot{k} &\equiv \left[\frac{\dot{K}}{L} \right] \\ &= \frac{\dot{K}L - K\dot{L}}{L^2} && \text{by quotient rule} \\ &= \frac{\dot{K}}{L} - \frac{K\dot{L}}{L^2} \\ &= \frac{\dot{K}}{L} - \frac{Kn}{L} && \text{since } \dot{L}/L \text{ is pop growth} \\ &= \frac{\dot{K}}{L} - kn \end{aligned}$$

⁷In a competitive market, workers are paid their marginal product (i.e., the value of an additional unit of labor), or $\partial Y / \partial L$. If you sum this up over all workers, you obtain $L \cdot \partial Y / \partial L$. But this is equal to $Y(1 - \alpha)$. Thus, the marginal product of labor, summed up over all workers, equals $Y(1 - \alpha)$. The term $1 - \alpha$ is the share of total income that goes to workers, assuming that they are compensated at their marginal product. Likewise, capital is compensated at its marginal contribution, $\partial Y / \partial K$ and its share of income is α .

We can plug in our earlier definition of $\dot{K}$ in the first term:

$$\begin{aligned}\dot{k} &= \frac{sY - \delta K}{L} - kn \\ &= sy - \delta k - kn \\ &= sf(k) - k(\delta + n)\end{aligned}$$

Thus:

$$\dot{k} = \underbrace{sf(k)}_{\text{savings per cap}} - \overbrace{(\delta + n)k}^{\text{depreciation per cap}}$$

i Policy implications of Solow-Swan with labor

We are back to square one: adding labor does not open new policy prescriptions. There will be no growth, because excess savings will eventually lead to less productive capital, up to the point where the marginal added capital will be soaked up by the combined effect of depreciation and population growth.

2 Social and collective goals

This version: September 28, 2026

Summary

Public policies are interventions by public authorities in collective and private affairs to achieve a certain state of the world. But what should this ideal state of the world be? That's fundamentally a moral question.

2.1 Overview of ethical views

We organize ethical views for public policy along two main axes (Bueno de Mesquita 2016): **consequentialist** perspectives, which assess decisions based on their outcomes, and **deontological** perspectives, which focus on duties — whether an action is right in itself, regardless of its consequences. We largely skip other schools (e.g., virtue ethics); for a review see Williams (2012).

2.2 Deontological views

Deontological ethics tells people what their *duties* are – what they shall or shall not do – independently of outcomes. Rules can be prescriptive (what to do) or proscriptive (what not to do). The defining feature is that they are *unconditional*: an action is right or wrong in itself, not because of what it produces. If killing is deontologically forbidden, it cannot be justified even when it would prevent harm. Information about consequences does not change the moral verdict.

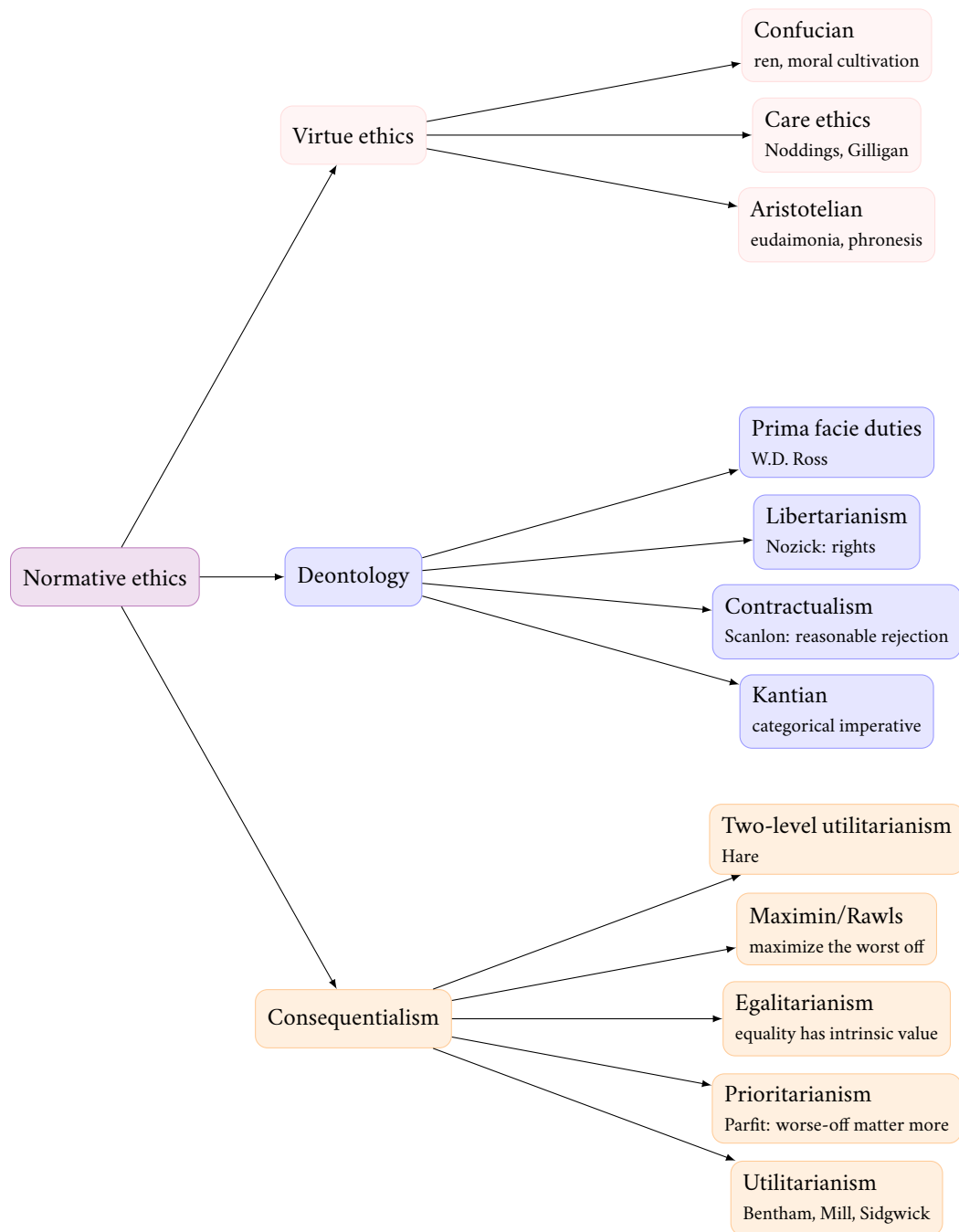


Figure 2.1: Broad overview of normative perspectives.

Deontological approaches can be based on a diverse set of foundations, including tradition (e.g., Hippocratic oath), religion (e.g., Ten Commandments), or reason (e.g., Kantian deontology, human rights). See Note 5.

Note 5: Oaths

Physicians are sometimes expected to take the Hippocratic oath, which includes several deontological requirements, including (1923 Loeb translation, as reported on Wikipedia):

Into whatsoever houses I enter, I will enter to help the sick, and I will abstain from all intentional wrongdoing and harm, especially from abusing the bodies of man or woman, bond or free. Source: [link](#).

Engineers have their oaths as well. EPFL students launched the *Serment d'Archimède* (Archimedian oath) in 1990. Whether it is deontological or consequentialist is not entirely clear:

In choosing and completing my projects, I will pay attention to their context and their consequences, particularly with respects to technical, economic, social, and ecological aspects, and will strive for the respect of the environment. Source: [link](#).

Deontological rules are often linked to religions. This makes sense, since religions tend to establish stark differences between what their deity wants and what it doesn't want. But there have been deontological systems that are rooted in rational Enlightenment as well. Kant (1900) for instance developed a system that is not directly connected to a religious system. At its core is the notion of **categorical imperative** (i.e., an unconditional necessity), which (simplifying a lot) states that we should act according to principles that can be universal (and thus independent from their specific impact).

There is little doubt that deontological principles constrain real-life policymaking. People who pushed to promote human rights, abolish the death penalty, fight against slavery often did so because they held it as evident that this is how humankind ought to behave. Warning 1 gives an example of a deontologically-inspired legal decision in the insurance sector.

 Warning 1: Deontologic public policies and gender-based price discrimination

Deontological-inspired views often shape public policies. One example comes from insurance. The idea behind an insurance is to pool risks. Take a population of 100 individuals who are at risk of some kind of costly accident. Without insurance, each individual has, say, a 1/1000 chance of getting affected by the accident within a calendar year. Unlucky individuals will experience large losses and others none. With insurance, each agent pays a premium to the insurance company, and the insurance company compensates whoever experienced the accident.

Now, the insurance company would like, ideally, to be able to identify which agents are bigger risks. It seems unlikely that all individuals have the same odds of having an accident. High-risk agents should pay higher premia (fees) compared to low-risk ones. Let's suppose that an insurance company found an observable vector of variables $\mathbf{X}$ that allows it to predict with high precision whether someone is a high- or a low-risk agent:

$$\mathbb{P}[\text{Accident}] = f(X_1, X_2, \dots, X_k)$$

Should the company be allowed to use this vector to bill people in a differentiated manner? It turns out, sometimes the law forbids it. This is the case in the car insurance market in the European Union since 2012 [\[link\]](#). The EU's Court of Justice ruled that gender-based premia discrimination was not legal. Concretely, an insurance company cannot send different bills to two people who are identical except for their gender. This meant that women (who tended to pay less than men until then, since they were lower risk customers) should have seen their bills increase in the aftermath of the ruling, whereas men's bills should have gone down.

Such rules have strong deontological flavors (it was deemed as wrong on principles). But note that these laws do not forbid all types price discrimination: EU law lets insurance companies use other criteria, such as age. Policymaking often combines deontologically-inspired rules with other, more consequentialist guidelines.

Did the EU Court of Justice's decision work (i.e., reduce inequal-

ity)? According to reports, no [\[The Guardian\]](#). The reasons brought up are interesting: insurance companies use other variables in $\mathbf{X}$ that are highly correlated with gender, thus mitigating the effect of removing gender from that vector. For instance, they include occupations. Knowing if the customer is, say, a construction worker or a nurse (two statistically gendered jobs) means that gender will be reflected in the premia *even* if the insurance company comply with the law. Furthermore, men have increasingly bought cars that are different from women's (larger, more expensive). Thus, the two groups – female and male customers – remain statistically distinct for insurances.

Yet at the same time, deontological views often lack subtlety to guide policymaking. Many problems of policymaking involve **ambiguity** and **tradeoffs**. There are situations where there are reasons in favor and against the same action. See examples in [Warning 2](#) and [Warning 3](#).

Warning 2: Climate change and poverty

Consider the problems of poverty and climate change. Suppose that you have two deontological principles guiding you:

1. We ought to mitigate climate change.
2. We ought to reduce poverty.

Now, there are reasons to believe that giving people access to electricity in developing countries reduces poverty (Dinkelman 2011; Khandker, Barnes, and Samad 2013). The cheapest and practical way to do so may be to use coal for thermal plants. Yet this clashes against the precept of fighting against climate change. It is not straightforward to see how to handle conflicting imperatives.

Warning 3: Air pollution and inequality

Let us now consider two other problems: air pollution and inequality. Like before, we have two guiding sets of duties:

1. We ought to reduce air pollution.

2. We ought to reduce inequality.

These are two principles that seem intuitively to match the preferences of members of the urban, left-leaning, middle class. Yet consider a policy such as congestion charges, in which access to city centers requires the payment of a fee. This policy has been used in cities such as London, New York, and Paris. The idea is that you discourage the use of (polluting) cars in urban areas by making it more expensive (Cook et al. 2025). Assuming that this is a flat fee applied to people coming from outside the city center, it is likely to be regressive (i.e., it will represent a higher financial burden for poorer individuals). (Note: this doesn't have to be true. It could be that folks using cars are richer than those who don't.)

Beyond its (occasional) impracticality, the problem with using deontological rules as a guide to public policy is that it can be quite arbitrary and (almost by definition) inefficient.¹ Consider the following (silly) example. Suppose that Country A elects a new president who claims to be Christian. They decide that there is a new rule that they wish to enforce: *Do not wear clothes of wool and linen woven together*. This is a quote from the Ancient Testament (Deuteronomy 22:9–11). Suppose that 40% of Country A happen to really like wearing clothes with wool and linen woven together. Is it really desirable to prevent them from being happy? Especially if the other 60% is not really affected by it?

This line of questioning brings up the notion of **consequentialism**. Maybe public policy should be designed according to its *consequences* and not solely because of it being a *duty*. There are several forms of consequentialism, including utilitarianism, egalitarianism, and so forth. We will see them next. But note that, in practice, policymaking will obey to a mixture of deontological and consequentialist principles. While it has to establish empirically, it seems that policymakers behave *as if* they sometimes have positions based on deontological principles (“it is our duty to do XYZ”) and sometimes based on consequentialist principles (“we will be better off if we do XYZ”).

¹We will come back to the notion of *inefficiency*, but for now, suffice to say that this refers to a situation in which there are unnecessary welfare losses.

2.3 Utilitarianism

Utilitarianism is the most influential consequentialist framework, rooted in Bentham and Mill (Bentham 1823; Mill 1871). In its simplest form, it identifies morally desirable actions as those producing ‘good’ consequences — Bentham’s “greatest happiness of the greatest number.” Happiness was later generalized to *utility*, broadly understood as welfare.

2.3.1 Utility functions

The ‘greatest happiness of the greatest number’ is not yet a precise statement. Researchers interested in formal models of decision-making have spent considerable efforts to clarify what this means and developed the concept of **utility function** (see also Note 11). There are two key ingredients in such a function.

1. There is a set $\mathcal{S}$ of alternatives, with an arbitrary alternative denoted as s_i . An alternative could be a single good (or technology): $\mathcal{S} = \{\text{oranges}\}$. More likely, it will be a bundle (or vector) of goods/technologies. For instance: $\mathcal{S} = \{\text{solar PV, wind turbine, diesel generator}\}$. $\mathcal{S}$ may or may not be countable.
2. Agents (firms, consumers, voters, governments, etc.) must have well-defined preferences over elements of $\mathcal{S}$. By “well-defined,” we typically mean that three assumptions must be met. First, the **completeness** assumption states that agents can rank all elements of $\mathcal{S}$. We write $A \succsim_i B$ to indicate that agent i prefers bundle A to B or is indifferent between the two. By completeness, we mean:

$$A \succsim_i B \text{ or } B \succsim_i A \quad \forall A, B \in \mathcal{S}.$$

Second, we also generally assume that agents have **transitive** preferences:

$$\text{If } A \succsim B \text{ and } B \succsim C \text{ then } A \succsim C.$$

Third, we sometimes require the **monotonicity** assumption to be met. It states that additional inputs change utility in a monotonic way (e.g., if we have a choice over bundles of fruits, then adding a banana cannot

reduce utility). However, we will encounter cases where we don't want to make this assumption.

A utility function, then, is a map from the set of alternatives to the real line:

$$U : \mathcal{S} \rightarrow \mathbb{R},$$

such that:

$$A \succsim B \Leftrightarrow u(A) \geq u(B) \quad \forall A, B \in \mathcal{S}.$$

In other words, a utility function puts a real number on elements of $\mathcal{S}$ that preserve the agent's preference ordering, with higher values denoting a higher ranked alternative. A utility function can be *ordinal* (the ordering is preserved). It can in theory also be *cardinal*, which means that you can interpret intensities.

Warning 11: Utility functions

Why do we care about utility functions? Isn't this just an academic exercise? There are at least two reasons to care. First, they help us make analytical sense of what it means to follow consequentialist ethics. Second, they can be useful to organize our thinking regarding agents' preferences. For instance, Lim et al. (2025) write down the utility function of workers in coal-producing regions and then quantify its parameters in order to design new job strategies.

There are a few things to pay close attention to when we handle utility functions.

1. The starting point of utility functions is an **ordinal** representation of preferences. This is analytically the least demanding approach, since it only requires us to be able to order alternatives and enforce transitivity.
2. When we work with ordinal preferences, we don't say anything about the strength of preferences (*intrapersonal comparison*), nor do we engage in comparisons between agents (*interpersonal comparison*). For instance, a utility of 3 isn't twice as strong as a utility of 1.5 (*intrapersonal comparison problem*).
3. The problem of **interpersonal comparison** of utility has been around for as long as social scientists have been

thinking about the practical implications of utilitarianism (Harsanyi 1955; Hammond 1989; Hausman 1995). The question is: can we compare two individuals' utility? In theory, economists are often reluctant to do so. But to generate policy recommendations, we often have to do so (e.g., in cost-benefit analysis).

4. Analysts often set problems of interpersonal comparison aside and work with **cardinal** utility functions. We are not just ranking alternatives, but we are also making claims about their relative desirability (the “intensive” margin). Note that we can use cardinal utility functions *without* making the assumption that interpersonal comparison is acceptable. Yet in practice people regularly make these two assumptions jointly. Whether this is legitimate is tricky, but it allows us to go further in making claims about the social welfare implications of different actions.
5. Utilitarianism does not assume that everyone is identical. In economic research, this is referred to as models with **heterogeneous agents**. But sometimes, it is in fact easier to deal with homogeneous agents, in which case you will see references to a **representative agent**.
6. Utility is defined over alternatives, that is, outcomes. But often, we will be interested in the effect of different public policies, business strategies, etc. We will then often write about the “utility of engaging in action a_i .” But note that this is somewhat imprecise: strictly speaking, utility comes from the outcome generated by the action, not from the action itself.

A utility optimizer is an agent that maximizes their utility:

$$\max_{s \in \mathcal{S}} u(s),$$

subject to constraints (if any),

where constraints could have all kinds of origins (e.g., a budget constraint).

In a later chapter, we will be particularly interested in the way in which a specific set of agents – *consumers* – make decisions in the market place. The utility function will be about ranking different bundles of goods to be consumed, subject to a budgetary constraint.

2.3.2 Utilitarian social welfare function

The next step to understand how utilitarianism maps onto public policy is to go from individual to collective utility. To do so, we must aggregate individual utilities into a **social welfare function**. The social welfare function is a mapping from individual preferences to an aggregate (i.e., *social*) preference over a set of alternatives (Burk 1938).

What, then, is utilitarianism's social welfare function? That is, how do we “glue” together individual utilities across competing alternatives?

One way that seems consistent with Bentham is to *add up* individual utilities. At the aggregate (collective) level, for a society with N agents, we have:

$$\max_{s \in \mathcal{S}} \sum_{i=1}^N u_i(s),$$

subject to constraints (if any).

Mathematically, this is one way to implement Bentham's idea of the “greatest happiness of the greatest number”: it is to select an alternative that maximizes the sum of utilities, where the summation is done over all agents included in the social group under consideration.² For instance, at the level of a city, it could be the sum of utilities of all inhabitants of this city. At the level of a state, it could be the utility of all adult citizens. We can also assume a continuum of agents, in which case we replace summation with integration.

 **Tip 7:** Utility maximization, Pareto efficiency, and Kaldor-Hicks

A challenge to utilitarianism is the following case. Suppose you have a world with 10 individuals and a single source of welfare (e.g., food). Now suppose that individuals 8, 9, and 10 *really* like this good. You could build utility functions such that it would be socially optimal to redistribute food from the majority (individuals 1 to 7) to these three gluttonous people. This doesn't seem right.

There are two ways to address this. One is to add a restriction

²Nothing guarantees that there is a single alternative that maximizes aggregate utility, hence “an” alternative rather than “the” alternative.

on justifiable utility functions. We could include an exogenous constraint on what is acceptable and what isn't. For instance, we could rule out 'perverse' utility functions, such as ones in which "I enjoy the pain you receive."³

Another is to use a different criterion for decision-making: the **Pareto criterion**. Under a Pareto requirement, we would allow an intervention (e.g., a public policy π') to be implemented if, and only if, no agent becomes worse off (in terms of utility) compared to the status quo π , and at least one agent becomes better off:

$$u_{-i}(\pi') \geq u_{-i}(\pi) \text{ for all } -i \quad \text{and} \quad u_i(\pi') > u_i(\pi) \text{ for at least one } i$$

The Pareto criterion protects individual rights. However, in practice, it is unusable: there are virtually no policies that don't negatively affect at least one agent. There are very few cases of "free lunches."

A modification of this criterion is **Kaldor-Hicks efficiency** (Kaldor 1939; Munger 2000; Bueno de Mesquita 2016). We say that a policy meets this criterion if no agent becomes worse off *if transfers were completed*. Thus, this means that a policy can be launched if you could, in principle, compensate losers enough to make them whole again. Note that Kaldor-Hicks does not require the transfer to actually happen. This underpins the logic of cost-benefit analysis.

There is an important element here that sometimes gets overlooked. In its strictest sense, utilitarianism aggregates (here, via summation) individual utilities. The sum of individual utilities is not the same as the utility of a sum of agents. Analysts sometimes can be ambiguous about this, even though it can affect the desirability of policies in a counter-intuitive manner. See Note 8.

³This connects to (but is distinct from) a distinction made by utilitarians between "act" and "rule utilitarianism" (Brandt 1959). *Act* utilitarianism evaluates specific actions based on their immediate utility effect, whereas *rule* utilitarianism considers the general implications of these acts. For instance, for act utilitarians, it might be okay to kill a patient to use their organs to save four other patients. However, a rule utilitarian would take into account the fact that this would lead to a decline in trust in hospitals which would have nefarious utility effects down the line. See Lyons (1965) for a critique.

i Note 8: Utility of aggregates versus aggregates of utilities

If we think about the welfare of a group (say, a country), should we consider this group as a unitary actor or as a sum of individual agents? This matters, as the following example shows.

Imagine a society of ten identical individuals, all of whom only care about wealth (c). Suppose that this society has a stock of wealth of 100 units. Let us further suppose that it is universally the case that utility follows a square root function in c .

If we consider this society as a single, unitary agent, then its utility is:

$$u(c) = \sqrt{c} = \sqrt{100} = 10.$$

If total wealth increases, then it is always strictly true that utility increases (since square root is a monotonic transformation).

Now, let us consider the case where utility is not measured at the aggregate level, but at the individual level, and then summed up. Let us compare two scenarios: one in which aggregate wealth is 100 units that are equally distributed, and another in which there are 120 units that are unequally distributed. Note that under the society-as-a-single-actor approach, the second wealth alternative is strictly superior to the first ($\sqrt{120} > \sqrt{100}$). Is this still the case?

Let's start with 100 units of wealth, equally distributed among all ten individuals. We have:

$$\sum_{i=1}^{10} u_i(c) = 10 \cdot \sqrt{10} \approx 32.$$

Note that we cannot compare this with the earlier utility function that considers this society as a single actor.

Next, let's increase wealth to 120 units, and let's say that two individuals have 60 units each, and the rest have nothing. We get:

$$\sum_{i=1}^{10} u_i(c) = 2 \cdot \sqrt{60} \approx 15.$$

Thus, even though total wealth *increased* (from 100 to 120), aggregated utility *decreased* (from 32 to 15).

What does this tell us? First, even under utilitarianism, distribution of inputs matters when agents' utility functions are non-linear. Our counter-intuitive result obtains because individuals

have concave utility functions. This means that giving to the rich has less and less of an effect on aggregate utility. This captures the idea of **decreasing marginal utility**. A lower level of wealth that is better distributed will increase aggregate utility.

Second, we cannot infer that a policy is utility-maximizing if it increases an aggregate metric. Think of a free trade policy that is known to increase GDP. If you believe that countries have a single utility function (and one that is increasing in GDP), then this policy is always ethically preferable. However, if you believe that what counts is the sum of individual utility functions, then it does not automatically follow that this trade policy is desirable. To make a call, we must know (a) each agent's utility function, and (b) how the policy's profits are distributed.

This is why the politics of distribution and redistribution (of wealth, of pollution, etc.) are so important and contentious.

A few important points:

- Utilitarianism is *consequentialist*, not inherently *materialistic*. Utility can be over immaterial objects, and the utility function can take any shape — including one that is declining in consumption of goods.
- Bentham's utilitarianism is typically associated to the sum of utilities. However, utilitarianism could aggregate utilities in other ways. Other form of utilitarianism could, for instance, use a multiplicative function:⁴

$$\max \prod_{i=1}^N u_i$$

A good exercise is to think through the substantive meaning and implication of summation versus multiplication.

- Who counts in the maximization is assumed exogenous — a contested question. Should a government include non-citizens? Future generations? At what discount rate?
- The mathematical form is a choice: we could maximize the *product* of utilities rather than the sum, or assign different weights to different agents (as affirmative action policies effectively do).

⁴Sometimes referred to Nash's SWF.

2.3.3 Dynamic utilitarianism

Public policy decisions often involve costs and benefits over years or decades — solar power plants, for instance, require large upfront investment but generate returns over their lifetime. These **dynamic** problems require us to account for the **time value of money**: a dollar today is worth more than a dollar in the future for two main reasons:

- A dollar today can be invested to earn a return (the *interest rate*).
- People typically prefer to receive benefits sooner and defer costs (we are alive now and maybe not tomorrow).⁵

How can we take into account the time value of money to compare different actions? **Net present value (NPV)** is a method for taking time into account. It converts future costs and benefits into present-day equivalents. Concretely, we apply a **discount rate** r to future values (costs and benefits).⁶ The present value (at time $t = 0$) of a cost or benefit x_t received at time t is:

$$\text{PV}(x_t) = \frac{x_t}{(1 + r)^t}$$

The discount rate captures our *impatience* and how we value the future. High discount rates mean that we give less weight to the future. For instance, \$100 a year from now is worth \$90.9 at a 10% discount rate and only \$83 at a 20% discount rate.⁷

The choice of a discount rate is critical if you make decisions based on NPV. A higher discount rate places lower weight on future benefits, often disadvantaging long-term investments (like, say, renewable energy infrastructure). So how they are computed matters a lot. In private sector projects, a natural discount rate is the market interest rate (e.g., the rate on borrowing capital). In contrast, public sector projects often use a **social discount rate**, which may be lower than market rates. This is done to “correct” the market rate, since the latter does not (we assume)

⁵One of us, while in college, asked the late Ellen Wiegandt if people sometimes valued the future *more* than the present. She responded that she could only think of the *Père Goriot*.

⁶Textbooks sometimes use δ instead of r .

⁷That is, $90.9 + 0.1 \cdot 90.9 = 100$ and $83 + 0.2 \cdot 83 = 100$.

take into account future generations and thus undervalues the future. See Note 10.

The **net present value** of a stream of benefits B_t and costs C_t from time $t = 0$ to $t = T$ is:

$$\text{NPV} = \sum_{t=0}^T \frac{B_t - C_t}{(1+r)^t}$$

If the NPV is positive, the investment is considered worthwhile from an economic perspective. Of course, it doesn't mean that it is the *best* among available options. In fact, it does not even mean that it is a *desirable* investment. For instance, it could have hurt disadvantaged groups in an unacceptable manner. We return to this when we discuss public policy criteria. See Note 9 for an example.

i Note 9: NPV example: solar power plant

Suppose a government is evaluating whether to subsidize the construction of a solar power plant. The costs and benefits are estimated as follows:

- **Time horizon:** from $t = 0$ to $t = 25$ (assuming that the solar plant is dismantled in 25 years).
- **Initial construction cost:** \$50 million (at $t = 0$)
- **Annual maintenance cost:** \$2 million per year (from $t = 1$ to $t = 25$)
- **Annual benefits** (avoided carbon emissions, energy savings): \$5 million per year (from $t = 1$ to $t = 25$)
- **Discount rate:** 5% annually

We can compute the NPV as:

$$\text{NPV} = -50 + \sum_{t=1}^{25} \frac{5 - 2}{(1.05)^t}$$

This simplifies to:

$$\text{NPV} = -50 + 3 \cdot \sum_{t=1}^{25} \frac{1}{(1.05)^t}$$

The sum is a standard geometric series. Using the formula for a finite geometric sum:

$$\sum_{t=1}^T \frac{1}{(1+r)^t} = \frac{1 - (1+r)^{-T}}{r}$$

We plug in the values:

$$\begin{aligned} \text{NPV} &= -50 + 3 \cdot \frac{1 - (1.05)^{-25}}{0.05} \\ &\approx -7.7 \end{aligned}$$

Since the NPV is negative, the project would not be worth it economically. But this does not automatically mean that it is not worth doing *ethically*. It might benefit some communities that ‘deserve’ it morally speaking. Hence the importance of thinking hard about ethical viewpoints.

We can also express NPV in continuous time as:

$$\text{NPV} = \int_0^T e^{-rt} [B(t) - C(t)] dt$$

This is useful when costs and benefits are not discrete or occur continuously over time.

And finally, we can generalize this last expression to obtain its utility-maximization version:

$$\max_{s \in \mathcal{S}} \int_0^T \underbrace{u(s_t)}_{\text{utility}} \overbrace{e^{-rt}}^{\text{decay}} dt,$$

where $u(s_t)$ is the utility from strategy s at time t .

This expression might look a bit arcane, but it essentially is an exponential decay function e^{-rt} that weights the utility $u(s_t)$ obtained at that time. See Figure 2.2. The idea behind impatience and dynamic programming go back to Koopmans (1960), Bellman (1972), and others.⁸

⁸We can see here that NPV is a special case of utilitarianism in which utility is defined as economic benefits minus costs.

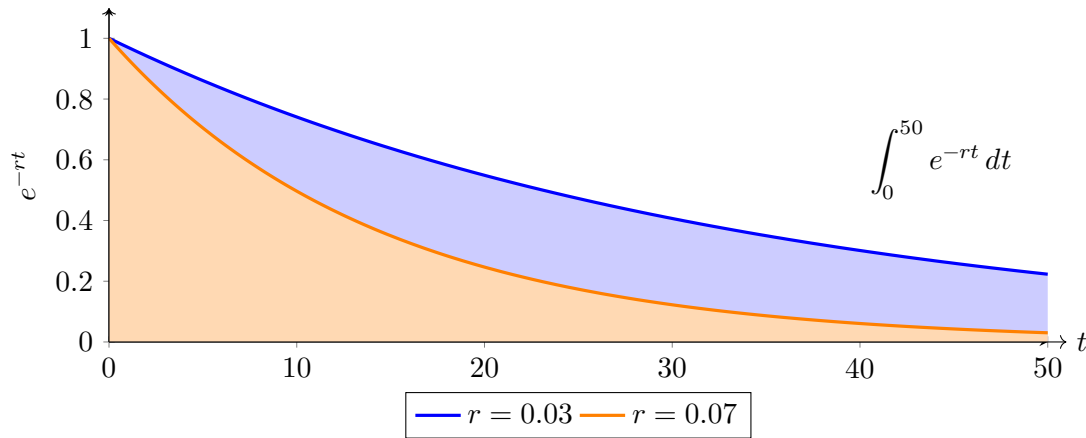


Figure 2.2: Dynamic utility function, with discount rate r assumed to be 3% and 7%, assuming that $u(s_t) = 1$ at every period of time t until $T = 50$. Total utility is then the (integrated) surface under the curve over this time window.

If we then integrate over all agents (and assuming that it is a continuous set as well of measure N), we obtain the **dynamic utilitarian** optimization problem:

$$\max_{s \in \mathcal{S}} \int_0^N \left[\int_0^T u_i(s_t) e^{-rt} dt \right] di.$$

The discrete time and population version is:

$$\max_{s \in \mathcal{S}} \sum_{i=1}^N \sum_{t=0}^T \left[\frac{u_i(s_t)}{(1+r)^t} \right].$$

💡 Tip 10: Discounting climate change

The **social discount rate** is often an extremely important parameter to decide what course of action is the best. This has generated heated debates, notably in climate economics. In 2007, Sir Nicholas Stern was tasked by UK Prime Minister Gordon Brown to prepare a report (“Stern Report”) on what to do about climate change. Stern (2007) argued for aggressive action that went beyond what others, such as William Nordhaus, typically recommended (Nordhaus 2007; Dietz 2008).

Partly, these differences in policy prescriptions are the result of different choices of social discount rates. For instance, Nordhaus and Boyer (2000, 16) uses a social discount rate that slides from 3% in 1995 to 1.8% in 2200.

Stern (2007) uses a slightly more complicated social discount factor. He splits the social discount rate into (adapting to our notation):

$$r = \delta + g\eta,$$

where r is the social discount rate, δ is the utility discount rate (how much time reduces value of utility), g is the economic growth rate, and η is the elasticity of the marginal utility of consumption (roughly: how fast marginal utility diminishes as consumption rises). The second term on the right-hand side captures the possibility that we will be richer in the future (large g) Stern picked $\delta = 0.1\%$, $\eta = 1\%$, and g depended on various scenarios, but on average is $g = 1.3\%$. Thus, for Stern, the social discount rate is $r = 1.4\%$ (and thus quite lower than Nordhaus').

Does this matter? According to Dietz et al. (2007) and Tol (2011), yes. Consider the present (i.e., discounted) cost of an additional ton of CO₂ on our societies (better known as the **social cost of carbon**). In a world in which we continue as now ('business as usual' scenario), the social cost of carbon in the Stern model is reduced by 70% if the discount factor goes from 0.1% to 1.5%. Likewise, Tol (2011) finds that the present marginal cost of an additional ton of CO₂ (tC) goes from \$24/tC to \$317/tC as the social discount rate goes from 3% to 0%. You can imagine that this may have a large impact on whether we should act.

F. C. Moore et al. (2024) reviews 147 studies and, after corrections, estimates the social cost of carbon to be \$283 per tC. They highlight the importance of the selection of discount rates to generate competing estimates of the social cost of carbon.

And we see that these debates spill over into public policy. After President Trump's election, the US Environmental Protection Agency decided to use 3% and 7% discount rates to calculate the NPV of greenhouse gas emission regulations. Under President Biden, the typical social discount rate was 2% (US Government 2023, 76). You can guess how this affects the value of climate regulation.

2.3.4 Uncertainty and risk

Most real decisions are taken under uncertainty. Rather than assuming known outcomes, we treat actions as lotteries over outcomes.⁹ The decision problem becomes one of maximizing **expected utility** (Neumann and Morgenstern 1947; Savage 1951): for action a with discrete payoffs $a_1, a_2, \dots, a_f$,

$$\text{EU}(a) := \mathbb{E}[u(a)] = \sum_{i=1}^f \mathbb{P}(a_i)u(a_i).$$

For instance, a could be the decision to implement a law to implement a congestion charge in Lausanne. Then, $a_1, a_2, \dots$ is the number of drivers per day inside the city, each of which has a probability $\mathbb{P}$ attached to it.

We can of course extend this to continuous utility functions:

$$\text{EU}(a) = \int_{-\infty}^{\infty} u(a)f(a)da,$$

where $f(a)$ is the probability density function of a . Here, the policy could be the decision to implement a carbon tax in the UK, and a could be its GDP.

2.4 Egalitarianism

Egalitarianism is another consequentialist ethical perspective, in the sense that it cares about outcomes rather than, say, decisions. In contrast to utilitarianism, which adds up utilities, egalitarianism prioritizes certain *distributions* of welfare (in a way, one must add, that isn't always explicitly stated).

Egalitarianism has historically been a powerful engine of change. It is embedded in many public policies, whether they are about wealth redistribution, social safety nets (e.g., unemployment benefits), public education policies, or else. You find it even in constitutions: the Swiss constitution, for instance, includes the following in its preamble:

⁹*Risk* refers to situations with known probability distributions; *uncertainty* to situations where we don't know them (Knightian uncertainty, Knight (1971)). In practice we rarely know true distributions, so expected utility is best used as a heuristic.

*the strength of a people is measured by the well-being of its weakest members [...]*¹⁰

The operationalization of egalitarian views, in practice, depends on its more precise definition. There exists a range of subdivision of egalitarianism (Bueno de Mesquita 2016).

Equality of outcomes. We can think of two version of equality of outcomes. One version is equality of (material) *resources* (R. Dworkin 1981). For instance, we all ought to have access to clean water, electricity, and so forth. Another version is equality of *welfare*. We may see deviations from equality of possession: I may have the same ‘level’ of utility as you, even though I don’t have electricity at home, if I don’t care for electricity.

Of course, equality of *welfare* raises the usual problems when making interpersonal comparisons: is there a meaningful scale? Can we measure it? Will people truthfully report it?

At the same time, equality of *resources* beyond basic needs seems highly inefficient. Not everyone wants a smartphone, so why should humankind feel obliged to produce one for everybody? Taken to its extreme, this approach may imply that we either all want the same stuff, or we are willing to indulge large efficiency losses by giving some people what they don’t want, possibly at the cost of not providing stuff that they have wanted instead.

A solution to this problem is to create a distribution of resources that is free of *envy* (R. Dworkin 1981; Brams and Taylor 1996). This would characterize an allocation of resources in which nobody would prefer someone else’s bundle of resources. How feasible this is in practice is unclear.

Equality of opportunity (Nussbaum and Sen 1993; Sen 1999). This perspective acknowledges the diversity of aspirations and preferences. However, it points out that the barriers to leading a fulfilling life are often rooted in inequality of opportunities that we have (or not). The typical example comes from people’s family wealth and social capital: a wealthy family can afford all kinds of opportunities to their children, either through money or through connections (Robert D. Putnam 2001).

¹⁰See [Federal Constitution of the Swiss Confederation](#).

Equality of opportunity, then, can be conceived as a form of *ex ante* perspective: I don't care where we end up (since we are all different), but we should all have similar chances in life. As a result, researchers interested in promoting equality of opportunity often study interventions in education and other parts of early life.

Risk minimization. Another take on egalitarianism, and one that also takes an *ex ante* viewpoint, is to say that we ought to minimize risk (R. M. Dworkin 2003).¹¹ Bad outcomes can happen to people because of bad luck. Risk minimization policies will, as a result, often mimic those improving opportunities.

Below, we will see how some thinkers such as J. A. Rawls (1971) try to combine elements of utilitarianism with egalitarian constraints. This happens notably via the the **maximin principle** in which inequality is tolerable as long as as it *maximizes* the *minimum* utility of a set of agents. We return to this below.

 Warning 11: Egalitarianism and efficiency

Imagine you had a society of 100 people and you find a treasure with \$1 million. What should you do with this sum? It seems plausible that an egalitarian would support a policy like this: *give everyone in the group \$10,000 in cash*. That's the idea of an unconditional cash transfer.

Is this efficient? Is it fair? Not necessarily. Some members of the group might be very wealthy to begin with. It seems reasonable to think that the poorer members would benefit more. This raises the *problem of intra-group allocation*. Furthermore, the group might be likely to become wealthy in the future, which would suggest that it should give the money to another group. That's the *inter-group allocation* problem.

This is not a hypothetical situation. The 154 students (*group*) who graduated from Indiana University's law school in 2026 were informed that they would each receive a \$10,000 cheque ([link](#)). This came from a \$1.6m gift (*treasure*) from anonymous donors.

On the one hand, this seems like a nice gesture. Law school stu-

¹¹This assertion of ours is not strictly true. You can also construct an argument based on *ex post* prioritizing the utility of people who are worst off (Arneson 2004). You may even end up doing a utilitarian-style additive social utility exercise, in which you weight people by how poorly they are doing.

dents in the US generally graduate with debt that can become a heavy burden. On the other hand, are lawyers really the most in need? And setting this point aside, did all students have an equal need for this money? About 61% of students at Indiana's law school take loans ([link](#)). This means that 39% can afford law school by paying in cash. It doesn't mean that they are wealthy, but it suggests that there is uneven needs among students. Beyond this particular case, the point is the following: operationalizing egalitarianism is not always straightforward.

2.5 Libertarianism

Egalitarian views have received considerable pushback. This pushback operates at several levels:

- Does it have a reasonable basis? For instance, if it is really about a state of the world free of envy, does this mean that the justification for egalitarianism is envy (Anderson 1999)?
- Is it justified? Consider the example of an incredibly talented medical doctor who worked hard on a new vaccine. Should they receive the same resources as someone who didn't?
- Does it respect individual freedom enough? This is near to a *deontological* view with Kantian inspiration that emphasizes the importance of individual rights.
- Is it inefficient? As we stated above, it is not easy to implement egalitarian distributions that are efficient, in the sense that there will be considerable welfare loss for some. Worse, it could be that the total utility declines (i.e., it is not just redistributing slices of the pie from agent A to agent B, but the size of the pie declines). We call these *deadweight losses*.
- Are the *transaction costs* worth it? Policies are costly: they must be designed, implemented, monitored, and enforced. This requires a large bureaucratic body that is expensive and, to libertarians, not self-evidently increasing welfare. In some states like Belgium or France, government spending as a share of GDP is over 50%. Some of this is consumption, but some of it is just to keep operating.
- The state is immensely powerful and therefore creates a threat when it is captured by special interests (e.g., creating rent-

seeking activities) (Olson 1965; Rowley, Tollison, and Tullock 1988; Murphy, Shleifer, and Vishny 1993).

To be clear, there are lots of debates regarding the theoretical and empirical validity of each of these claims. The point remains that egalitarianism involves some kind of public interventions to be implemented, and several ethical schools – including libertarianism – are skeptical about the desirability and effectiveness of such actions.

While these approaches encompass a diversity of views, we can generally predict that they will share a preference for (1) sharp limits to state action, (2) a prioritization of individual freedom, and (3) a state apparatus that focuses on protecting them (instead of, say, engaging in redistribution).

This doesn't mean that libertarians are equivalent to anarchists. Oftentimes, they will insist on a well-defined role of the state regarding the guarantee of private property rights (to safeguard individuals) and by extension the need for well-defined laws, courts, and law enforcement. Nozick (1974), for instance, calls for a *minimal state* that conducts these tasks.

2.6 Collective preferences and sustainability

Ethical views help us identify **policy problems**: situations where the state of the world *as it is* diverges from *what it ought to be*. “Ought” implies moral assessment, which requires an ethical framework — there is no justification of public policy without a normative statement.¹² For environmental issues, this means clarifying what, exactly, is wrong with the status quo.

A problem caused by environmental degradation can then be identified on several grounds.

- *An environmental issue reduces aggregate welfare.* This is for instance the rationale behind models that estimate the effect of climate change on economic output: if you assume that people's utility is strictly increasing in GDP, and if you assume that you can use a representative agent (thus you maximize

¹²Public policy exists regardless of ethics; what we are saying is that its *justification* requires one.

average GDP), then a decline in per-capita GDP compared to a counterfactual of no environmental issue signals a policy problem. This explains, for instance, why researchers are interested in the consequences of climate change on GDP (Bilal and Känzig 2024).

- *An environmental issue increases inequality.* Some types of environmental issues have strong distributional consequences (Sovacool and Dworkin 2015). For instance, locally, the siting of polluting industries may disproportionately affect minorities (Mohai and Saha 2015). People refer to this as *environmental racism* (Bullard 2018). Globally, greenhouse gas emissions (and the benefits associated to them) are highly unevenly distributed (Oswald, Owen, and Steinberger 2020; Chancel 2022). So are the impacts of global warming (Carleton et al. 2022).
- *Certain environmental behaviors are deontologically unacceptable.* Ethicists discuss problems such as the animal rights (Regan 2008). From a deontological standpoint, we may assign rights to animals akin to those of humans and thus acknowledge them as autonomous agents.

Note that similar ethical imperatives may curtail public policy. For instance, an environmental policy that disproportionately affects certain groups may increase inequality and therefore be normatively unacceptable. Put another way: normative issues can arise both because of the status quo (“the state of the world is ethically bad”) *and* because of the way policymakers intervene to change it (“this policy to change the state of the world is ethically bad”).

2.7 Decision criteria for public policy

Drawing on the ethical frameworks above, we can identify practical criteria for choosing among policies.

2.7.1 Defining public policies

Let’s start with the setup. We will assume that there is a *policymaker* in the truest sense of the term: a person who makes decisions about which

public policies to implement. We have a set of policies, with arbitrary element π :

$$\mathcal{S} = \{\pi_1, \pi_2, \dots, \pi_k\},$$

for k different policies. The set of policies available could also be continuous (e.g., the level of a carbon tax).

For now, we make no assumptions regarding how the set $\mathcal{S}$ is created. We just assume that the policymaker shows up to her office, and a set of k memos are waiting for her. Later, when we discuss the politics of policies, we will think about how the choice set is established, including the role of agenda-setters.

A **public policy** π is then defined as an organized intervention by the policymaker to (try to) bend the real world toward a desired outcome:

- It is *organized*, in the sense that it has an internal logic that is rational (though not necessarily omniscient).
- It is about *changing* or *maintaining* something about the world.
- It has a well-defined *goal*.

A public policy can therefore be a single intervention. It can also be a bundle of interventions. We then sometimes talk of a **policy program**.

A policymaker might implement a policy for selfish goals. If, however, it is a normatively acceptable goal, and if we use a consequentialist framework, then we will be interested in the effect that this policy has on agents (e.g., voters). We thus rely on utility theory. Each policy delivers utility to agent i , such that:

$$U_i = u_i(\pi_k),$$

for some utility function u that may be common to everyone (homogeneous agents) or specific to i (heterogeneous agents).

Note that $u_i(\pi_k)$ sometimes means what it says, and sometimes it is shorthand to say that the policy π_k has an effect on utility via its impact on a set of variables w_l that enter i 's utility function. For an example of a direct effect of policy on utility, consider the problem of taxpayers

and the policy to be the income tax rate τ . We might have something like this:

$$u_i = \overbrace{(1 - \tau)\text{Wage}}^{\text{net wages}} + \underbrace{\alpha\tau\text{Wage}}_{\text{public services}}, \quad \tau \in [0, 1], \alpha \in (0, 1)$$

But sometimes, the effect of policies will be indirect and operate via their effect on other variables that are inputs to utility functions:

$$u_i(\pi_k) = u_i(w_1(\pi_k), w_2(\pi_k), \dots, w_l(\pi_k)).$$

This is captured in Figure 2.3 and an example is given in Figure 2.4. For instance, your utility is (presumably) not affected directly by the existence of a carbon tax, but because this carbon tax affects stuff that you care about (e.g., consumption, traveling, etc.).

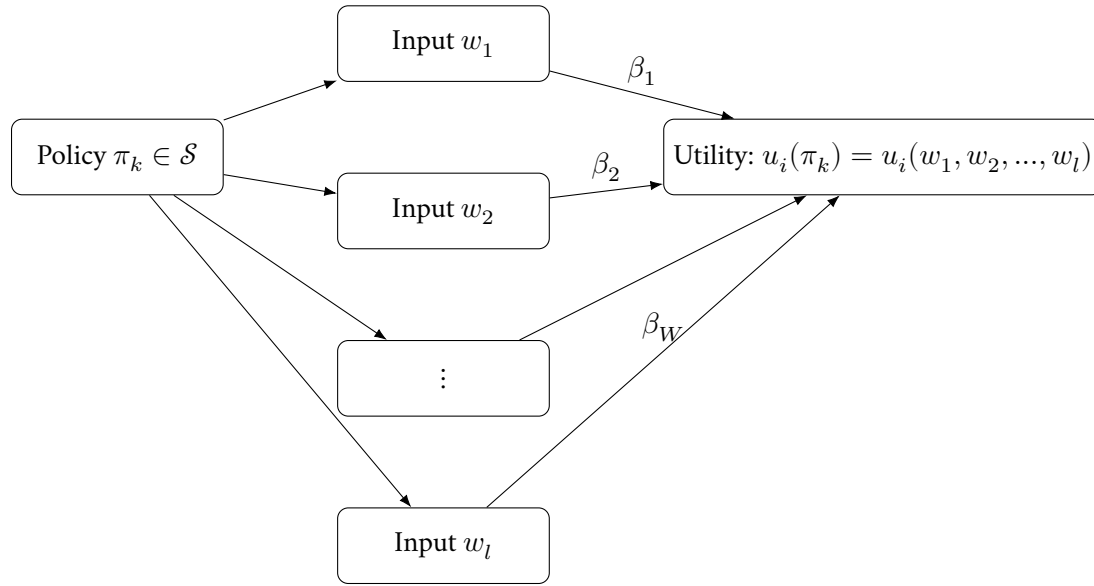


Figure 2.3: From policy to utility. Parameters $\beta_1, \beta_2, \dots, \beta_W$ are utility weights for each input variable. The functional form of $u(\cdot)$ may or may not be linear in inputs and weights.

In a dynamic context, each policy π_k is a **policy program** (or sequence) from time 0 to T , such that:

$$\pi_k = [\pi_{k,0}, \pi_{k,1}, \dots, \pi_{k,T}]$$

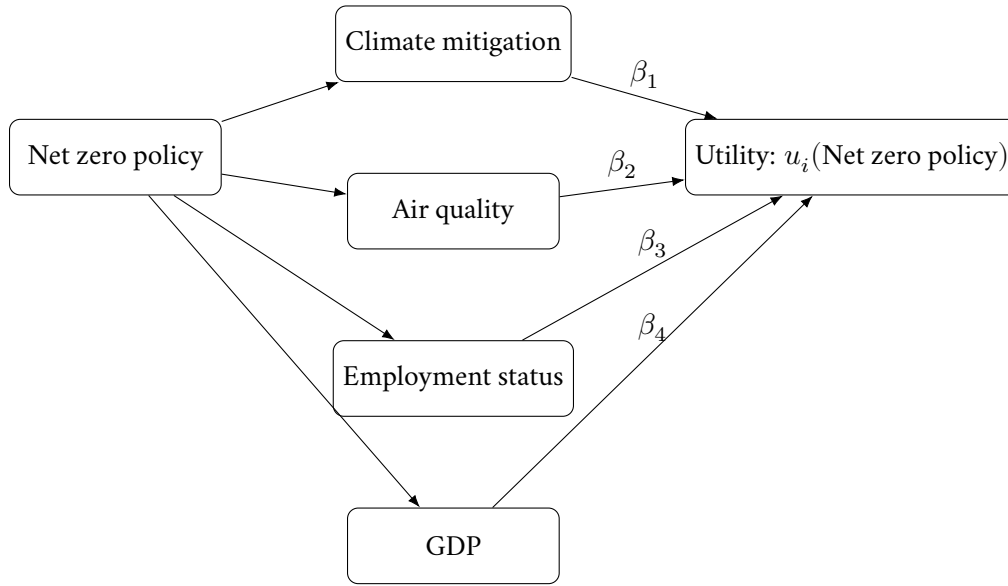


Figure 2.4: Example of the link from net zero policy to utility.

We would then document how utility is delivered at each time period:

$$u_i(\pi_{k,0}), u_i(\pi_{k,1}), \dots, u_i(\pi_{k,T})$$

And if there is uncertainty, we could further claim that each realized input value (and therefore utility payoff) depends on a stochastic draw from some distribution $\mathcal{D}$ (e.g., the normal distribution).

$$w_{l,t}(\pi_{k,t}) \sim \mathcal{D}.$$

Let us then see how each ethical approach affects policymaking over $\mathcal{S}$.

2.7.2 Policy choices and deontology

First, a deontological approach is one that *filters* the set of available public policies. For instance, consider a set of national security policies:

$$\mathcal{S} = \{\text{espionage, torture, military intervention}\},$$

and imagine that *torture* is held as (deontologically) inadmissible. This is the equivalent of applying a weight of 0 to this particular strategy, filtering the final set of available policies to $\mathcal{S}_D \subset \mathcal{S}$:

$$\mathcal{S}_D = \{\text{espionage, military intervention}\}.$$

When deontology is proscriptive, as it is in this example, then we are left with no further guidance regarding which policy to use among those in the filtered set.

To sum up, deontological criteria operate like filters. These filters are constructed based on the intrinsic appropriateness of policy strategies, regardless of what outcome they generate. Thus, the deontological leader should not use filters that are conditional on the impact of policies. In our example, torture is eliminated even if it allows to prevent a major security threat. Put another way, if I had good intelligence and told the policymaker: “you should use torture because I know it will avoid a terroristic attack,” this should have no bearing on her decision. It is truly *unconditional* on outcomes.

As discussed earlier, there exist a wide range of deontological filters, including (but not limited to):

- Cultural and religious (e.g., policies over abortion, marriage)
- Human Rights (e.g., policies over security)
- Philosophical (e.g., policies over freedom)
- Environmental rights (e.g., animal rights)

2.7.3 Policy choices under utilitarianism

In line with utilitarianism, a policymaker may wish to maximize aggregated expected dynamic utility (ignoring constraints):

$$\max_{\pi \in \mathcal{S}} \mathbb{E} \left[\int_0^N \left(\int_0^T u_i(\pi_t) e^{-rt} dt \right) di \right],$$

where the integration takes place over time (from 0 to T , with T possibly infinity) and over all N agents of utility $u_i(\pi_t)$ delivered at time t and discounted at a rate r . The expectation is included if the realization of π_t (or rather its effect on input variables w_l) is stochastic.

Notice the components of this function:

- The utility function captures what agents want ($u_i(\pi_t)$).
- The integration over time and the discount rate r captures inter-generational relations ($\int_0^T$ and r).
- The integration over the population captures the way we aggregate individual preferences and who is a member of the “polity” ($\int_0^N$).
- The expectation captures uncertainty over the effect of policies ($\mathbb{E}$).

In practice, using this formula raises all kinds of challenges, not limited to:

- How do you measure utility? How do we obtain information regarding preferences ordering of people?
- Should all rankings of outcomes be admissible? What if some people like outcomes our intuition tells us are bad?
- Can utilities of two individuals be compared?
- Who should be included in the calculation? Adults and children? Citizens and non-citizens? Future generations? How do we decide who counts in the population?
- How do we set the discount rate?

In practice, the closest thing to utility maximization is **cost-benefit analysis**, which we discuss below, but whose guiding idea is to replace $u_i(\pi)$ with economic profits (benefits net of costs, possibly discounted, and possibly taking into account uncertainty).

We review policy criteria in Note 12

i Note 12: Policy criteria

Utilitarian-type criteria for what is the ‘right’ policy can be a bit confusing. Several concepts are related by need to be distinguished.

(Pareto) efficiency. An allocation of resources (i.e., an ‘outcome’) is *Pareto efficient* if there is no other allocation that would yield similar welfare (utility) to all agents while leaving at least one agent strictly better off. For a society with N agents endowed with $\mathbf{x}$, a policy π achieves a Pareto efficient outcome if:

$$u_i(\mathbf{x}_i + \pi_i) \geq u_i(\mathbf{x}_i) \quad \forall i \in N$$

and

$$u_i(\mathbf{x}_i + \pi_i) > u_i(\mathbf{x}_i) \text{ for at least one agent.}$$

Inefficiency, by contrast, means that there is a way to reallocate resources that would make someone better off without hurting anyone. This captures the idea of ‘free lunch’: there is no way in which resources can be redistributed that doesn’t bother agents while making at least one agent happier. A policy that is efficient has therefore no slack.

(Pareto) optimal. Often used interchangeably with *Pareto efficient*. Note that people who are not using a utilitarian framework might mean something else. For instance, it may be used colloquially for a policy that is the “best among a set of good policies.” It can also be used to say “a policy that achieves several goals at once.”

Pareto-superior (-inferior). An outcome A is Pareto-superior to an outcome B if it increases the utility of at least one agent without hurting anyone. Note that these relations aren’t always transitive. Consider the following three outcomes for two agents:

$$A \equiv (3, 3) \quad B \equiv (1, 4) \quad C \equiv (2, 2)$$

Then:

- A is neither Pareto-superior nor Pareto-inferior to B
- A is Pareto-superior to C
- But B is not Pareto-superior to C

Kaldor-Hicks efficiency. A policy is efficient if, after (hypothetical) compensations, those who benefit could compensate those who lose by a sufficient quantity of utility to make them indifferent. In other words, for a society with N agents, each of whom have a pre-policy endowment $\mathbf{x}_i$ and who would gain/lose π because of the policy, there exists a vector of transfers (some being positive and some negative):

$$\mathbf{t}^T = [t_1, t_2, \dots, t_N],$$

such that:

$$u_i(\mathbf{x}_i + \pi_i + t_i) \geq u_i(\mathbf{x}_i) \quad \forall i \in N$$

This is a criterion that is used in cost-benefit analysis, where utility is replaced by financial flows.

Effectiveness. An *effective* policy is one that achieves its goals. As such, there is no built-in ethical dimension to saying that a policy is “effective.” We don’t have clear measures of effectiveness. In climate policy, Hovi, Sprinz, and Underdal (2003) have defined a policy’s effectiveness to be the ratio:

$$\text{Effectiveness}(\pi) = \frac{\text{Realized outcome} - \text{Starting point}}{\text{Ideal outcome} - \text{Starting point}}$$

But there is no widespread accepted formal definition of effectiveness.

Fairness. A *fair* policy can refer to (a) the distribution of its consequences (*ex post*), or (b) the way it was constructed (*ex ante*). What is fair *ex post* varies along the lines of what we discussed earlier: is it fair if each agent contributes the same amount (e.g., of utility, of taxes, etc.)? Or to their ability (e.g., as in progressive taxation)? For *ex ante* fairness, we generally think of *procedural fairness*: were all stakeholders included in the political debate?

2.7.4 Policy choices under egalitarianism and Rawlsian compromise

As we saw above, we could also wish to reduce inequality. In practice, if we think about equality of utility, this could mean to select the policy alternative that minimizes variance in utility:

$$\min_{\pi \in \mathcal{S}} \mathbb{E}[u_i(\pi) - \mu_u(\pi)]^2,$$

where $u_i(\pi)$ is agent i ’s utility under policy π , and $\mu_u(\pi)$ is the average utility under alternative π .¹³

¹³You could generalize this to a dynamic setting, where you want to minimize variance over time. For instance, a policy that invests in education may have an effect in 10-20 years and thus immediate minimization of variance does not make sense.

Now, you can guess why this type of egalitarianism might not always be a very desirable criteria. For instance, assuming that economic profit is the only input and that utility increases linearly in it, then inequality is minimized when nobody gets anything. Is being poor but equal really better than being somewhat richer but more unequal?

This question is the one J. A. Rawls (1971) tried to answer. Rawls **difference principle** suggested that collective welfare should obey two principles:

1. Rational individuals ought to prefer equality over inequality. To justify this, Rawls suggests to imagine yourself having to decide what the best distribution of income (or wealth) should be, assuming that you don't know where in this distribution you will end up (a decision made behind a "veil of ignorance"). Surely, Rawls says, you would prefer the flatter distribution, since it reduces your downside risks.
2. Inequality is tolerable as long as it increases the welfare of the least well-off. In cash terms, a policy that increases inequality is therefore ethically justifiable as long as it improves the utility of the agent that is at the bottom of the income distribution.

Policies such as free trade are sometimes justified on Rawlsian ground: it increases overall welfare, and if the losers are properly compensated, then it is justifiable to reduce trade barriers. (But note that you can make the same argument on more classical utilitarian grounds, assuming concave utility functions and redistribution.)

Rawlsian criteria are sometimes characterized – and Rawls himself wrote about it – as a **maximin** approach: you want to maximize the minimum level of utility (J. Rawls 1974). Formally, you want to pick the allocation of resources $\mathbf{x}$ out of the set of all possible distribution of resources $\mathcal{X}$, such that it maximizes the minimum level of utility:

$$\max_{\mathbf{x} \in \mathcal{X}} \left[\min_{i \in N} \{u_i(\mathbf{x}_i)\} \right].$$

i Backlash against egalitarian policies

Our suspicion is that a majority of people reading these notes is, in principle, favorable to egalitarianism. But egalitarian policies can generate strong pushback. One example comes from

California. In 2014, San Francisco's School Board (which supervises public schools of the city) voted to stop teaching algebra in 8th grade (end of middle school). The backstory is complicated ([San Francisco Killed 8th-Grade Algebra. Now It's Set to Come Back.](#), *New York Times*, March 26, 2026), but essentially, by the time high school arrived, many students from disadvantaged backgrounds were lagging behind in algebra. As a result, these students avoided taking math in high school, which in turn reduced the probability that they would go into STEM subjects at university. That's not great if you want to boost STEM enrollment.

How can you help these students? San Francisco's School Board came up with the following idea: if you don't teach algebra in 8th grade to anyone, then surely all students would be at about the same level at the beginning of high school. All students could then catch up the missed year together.

This can be interpreted as an egalitarian policy. However, it was quickly criticized. The *New York Times* writes that "*San Francisco 'tried to achieve equity not by raising the floor, but by lowering the ceiling,' said Thomas S. Dee, a Stanford University economist who studied the policy with colleagues.*" Given widespread public hostility and a nonbinding vote in favor of 8th grade algebra in 2024, the policy was finally overturned by the School Board in 2026.

Does this mean that one must abandon the goal of helping underrepresented groups? San Francisco and other places are now trying other policies. For instance, one could try to autoenroll high-achieving students. Or you can create accelerated sequences teaching several math classes in parallel.

Recommended reading: Durlauf (2026) formalizes the problem of meritocracy in the classroom and asks how students should be assigned to classrooms (maximize the minimum level of skills? maximize the maximum level of skills? maximize the average level of skills?).

i Note 13: Other consequentialist views: uncertainty and survival

Utilitarianism and egalitarianism don't exhaust consequentialist views. A third perspective, influential in environmental and financial contexts, focuses on avoiding catastrophes. The concern is that human societies are subject to lotteries with fat-tailed outcomes — distributions with non-trivial probabilities of extreme, irreversible harm, including collapse or extinction (Taleb 2007). This has spurred research on systemic financial crises (Laherrere and Sornette 1998), nonlinear climate responses (Pindyck 2011; Weitzman 2014), and societal collapse (D. H. Meadows et al. 1972; Diamond 2005). The policy implication is to avoid strategies that increase fragility and to build robustness against catastrophic tail risks (Taleb 2012).

Additional readings

We didn't exhaust the set of all ethical perspectives. The ones we discussed, as we presented them, originated in Europe and Northern Americas. We didn't talk much about frameworks developed in other parts of the world and thus the perspectives we discuss are not globally representative (Wargo 1990; Gyekye 1995; Dhand 2002). Our aim was not so much to provide a universal set of perspectives; it was simply to show that views can vary quite a bit even within one region.

A general introduction on moral philosophy and ethics is Swift (2014) and its application to public policy is discussed in an accessible manner in Bueno de Mesquita (2016) (chapter 1). Another introduction is Williams (2012). Debates over utilitarianism, which started in the 1970s and continued on and off since then, have generated work that have had an impact on (mostly Western) policymaking in a way that is rare these days for philosophically-minded work. This includes work by folks such as J. A. Rawls (1971), Nozick (1974), MacIntyre (2013), Sen and Williams (1982), or Sen (1999).

Readers interested in the theory of utility and utility functions should refer to microeconomic textbooks, such as Mas-Colell, Whinston, and Green (1995), Jehle and Reny (2001), Bowles (2004), Kreps (2020), or Rubinstein (2022).

A widely used textbook on cost-benefit analysis is Boardman et al. (2018). Its application in the United States is structured by Circular A-4 ([link](#)).

Workshop: cost-benefit analysis

Cost-benefit analysis (CBA) is the closest to a real-life application of consequentialism (especially utilitarianism) in public policy. At its core, it compares the (net present) value of a given policy (e.g., an investment in a new infrastructure, such as a power plant or a train connection) compared to alternatives or to the status quo.

The following steps outline how to conduct a CBA from start to finish.

Step 1: Define the policy, baseline, and alternatives

Start by specifying the policy intervention π you want to evaluate. Clearly describe its features, duration, and scope. Since you will ultimately compare it to some benchmark, make sure to do the same for it, whether it is the status quo (“business as usual”) or a competing policy. This framing (a) determines what counts as a cost or a benefit and (b) defines the opportunity set.

Step 2: Identify relevant costs and benefits

List all effects that the policy causes on welfare. In a consequentialist framework, this would be the set of input variables $\{w_l\}$. In practice, these will often be measured in monetary terms. It typically includes

- Direct costs (e.g., construction costs)
- Direct benefits (e.g., time saved by commuters)
- Indirect effects or externalities (e.g., changes in pollution levels or labor market effects).

Thus, this may include material and non-material variables.

Step 3: Quantify costs and benefits over time

Translate each identified effect into a quantity (indexed by time $t = 0, 1, \dots, T$). For instance, if the policy reduces carbon emissions, estimate the tons avoided per year. If it affects travel time, estimate hours saved per person per year. Let B_t and C_t denote the total benefits and

costs, respectively, occurring at time t . Note that we want to think of benefits and costs in terms of real resources. For instance, we do not include information about who pays for the project. Thus, if the policy-maker must borrow money at an interest rate, then this will not show up here (since it's a transfer). We only want to include stuff that requires real resources.

Furthermore, determine the time horizon T for the analysis based on the policy's expected lifecycle.

Step 4: Monetize all items

Assign monetary values to each quantity. You could use market prices when available (e.g., wages, construction costs). For non-market goods (e.g., willingness to pay for a beautiful landscape), you will need to rely on estimates from revealed or stated preference methods. For instance, reduced mortality is sometimes quantified using the value of a statistical life (VSL). The value of a natural park could be proxied via contingent valuation by local inhabitants.

Of course, this can be challenging, both in spirit (how can you quantify the value of a year of life) and in practice (how can you reliably get good estimates of these quantities). But note that people do this all the time, even if they do not realize it. If you smoke, you put a value on your health. If you accept the construction of solar panels in Alpine regions, you put a value on that site.

We return to the quantification of non-market variables (e.g., willingness to pay for natural goods or public infrastructures) later in this book.

Step 5: Discount future values

To compare costs and benefits across time, convert them to present value using a social discount rate r . Recall from our discussion of dynamic utility and net present value that:

$$\text{NPV} = \sum_{t=0}^T \frac{B_t - C_t}{(1+r)^t}$$

This formula defines the **net present value**. In general, a positive NPV implies the policy improves overall welfare compared to the business-as-usual scenario (at least in its simplest formulation in which benefits and costs are computed as departures from the status quo).

If the effects occur in continuous time and are smooth, you can use the continuous-time analogue:

$$\text{NPV} = \int_0^T [B(t) - C(t)]e^{-rt} dt$$

Choose r to reflect social time preferences. Government guidelines often recommend using a constant rate (e.g., 3% to 7%), but there's no universal consensus. Most of the time, analysts often test alternative rates, including declining discount schedules. As we saw from climate change, the choice matters a lot and can be very contentious.

Step 6: Aggregate and compare alternatives

Compute the NPV for each policy option, including the baseline (which will often have NPV of 0 by construction). Rank alternatives by NPV and rule out dominated options. This step provides the efficiency basis for comparison, assuming all relevant impacts have been captured and properly valued.

Step 7: Assess distributional effects

Aggregate NPV ignores who gains and who loses. Disaggregate benefits and costs by group—such as income level, geographic region, age cohort, or ethnicity. For each group g , calculate:

$$\text{NPV}_g = \sum_{t=0}^T \frac{B_{gt} - C_{gt}}{(1+r)^t}$$

You may then present these disaggregated NPVs alongside the overall total. This helps identify regressivity or unanticipated inequalities.

To go further, you could apply distributional weights to give more importance to benefits or costs going to disadvantaged groups. Of course, this is also very challenging ethically: which group deserves more weight? But notice again that we do so implicitly when we treat everyone as equal: we are putting equal weights on individuals, even though it is not self-evident that this is just. For instance, adding costs to someone very wealthy might have a lower effect on utility than adding benefits to someone very poor, which would suggest the need for non-equal weights.

Step 8: Conduct sensitivity and uncertainty analysis

You should test how results change when key parameters vary. Start with one-dimensional sensitivity analysis: change one input (e.g., the discount rate, cost estimate, etc.) and observe how the NPV shifts. Then consider multi-way sensitivity analysis to explore combined changes.

To assess uncertainty more formally, you can use probabilistic methods. Assign probability distributions to uncertain parameters (e.g., a lognormal distribution over costs) and run a **Monte Carlo simulation**. Create a large number of simulated values for uncertain parameters (e.g., 10,000 simulations). For each simulation iteration, draw parameter values at random from said distribution and compute the NPV. This produces a distribution of NPVs, allowing you to report the mean and median NPV, the probability that $NPV > 0$, and confidence intervals (e.g., 95% CI of NPV centered on its mean).

You can also do versions in which you modify simultaneously several parameters. In such case, you will need to make assumptions regarding their correlation with each other.

Step 9: Document qualitative factors

This is an extra step. Not all important effects can be quantified in monetary terms. Furthermore, political feasibility, institutional constraints, and cultural impacts often matter as well. Document these clearly and explain their potential implications. While these factors don't enter the NPV formula, they may rightly influence the final policy decision.

Step 10: Communicate your results and limitations

Communicate your results, and highlight transparently:

- The assumptions you made (e.g., over the discount rate).
- The robustness (or lack thereof) of your estimates.
- What you were not able to include (Step 9).
- The core results, depending on ethical models (e.g., including the distributional impact of policies).

3 Causes of environmental issues

This version: September 28, 2026

Summary

Why are environmental problem happening? And why might they qualify as a *policy* problem?

3.1 A model of the economy

There are three complementary ways to model an economy. The first treats it as a **complex system** connecting consumers/households, producers/firms, the state, other economies, and input owners (labor, capital, natural resources). A simple representation is shown in Figure 3.1. The **market** is the institution — or set of rules — through which agents trade goods and services, embedded within a broader physical, political, and social environment.

Now, markets are not the only way in which goods are produced and exchanged. Economic systems may leave the production of goods and services (or the management of natural resources) to other agents—such as the state and related entities (local governments, state-owned enterprises, etc.)—as illustrated in Figure 3.2. This is just one of many possible ways resources can be managed.

Given how important market-based actors are, both in their share of production and in their contributions to environmental issues, the next part of this book focuses on interventions in markets (see also Note 14). The next two chapters therefore examine the demand and supply sides of markets. Later, we will discuss interventions that transform the

Physical, political, social environment

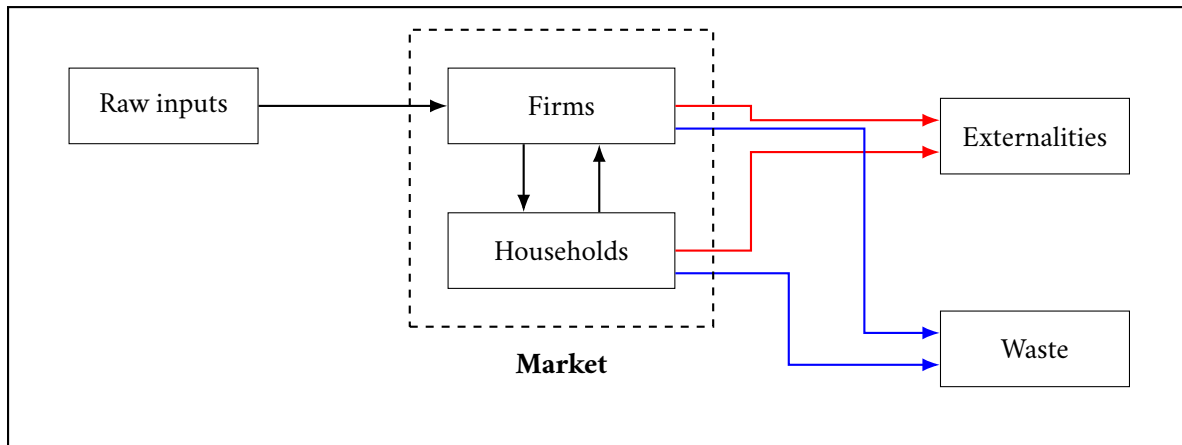


Figure 3.1: Simplified model of an economic system with markets as the main mechanism of allocation of resources. The flow from producers to households includes goods and services. The flow from households to producers includes cash payments.

structure of markets themselves. These are *institutional* reforms, in the sense that they modify the rules of access to and use of resources, as well as the flows within an economic system (flows of goods, flows of money, flows of inputs, etc.).

The second way to think about an economy is to model it as a **macroeconomic** process, whose simplest version is a production function that takes a set of inputs and transforms them into outputs. Following earlier discussions, we could write:

$$Y = F(\text{labor, capital, natural resources}).$$

The third approach is rooted in **microeconomic** models. The basic version of such models focuses more narrowly on markets as an abstract space where “demand” and “supply” meet (at least in theory). Given that, in this part of the book, we want to understand how agents behave (and how they could be incentivized to change their behaviors), we will focus on this approach for now.

Physical, political, social environment

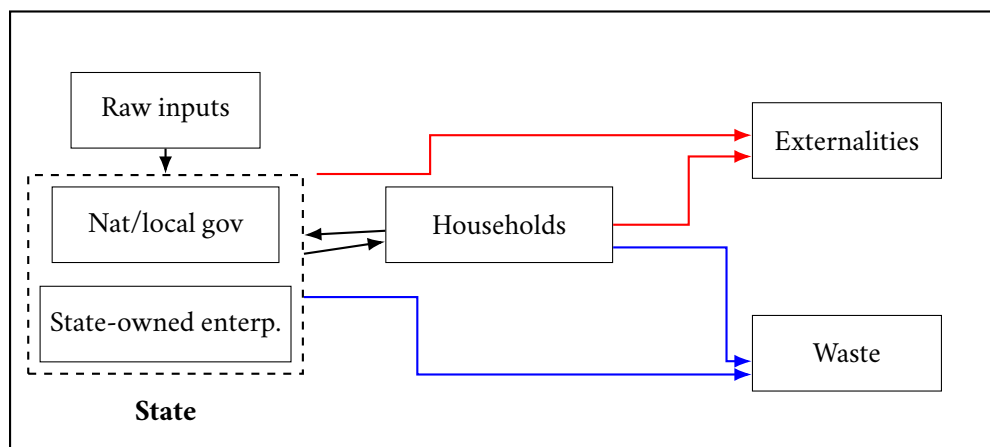


Figure 3.2: Simplified model of an economic system in which state or state-related entities are the primary producer of goods and services. The flow from the state to households includes goods (public goods, club goods, etc.). The flow from households to the state include taxes, payments for specific services (e.g., utility bills), etc.

Regardless of the model, we can distinguish two kinds of environmental issues:

1. Overuse of natural resource inputs.
2. Overproduction of negative externalities.

Why do these happen? A fundamental result of economic theory is that perfectly functioning markets produce a **general equilibrium** that is Pareto efficient (Kenneth J. Arrow and Debreu 1954). If true, there would be no obvious case for intervention. Before designing policy responses, we must therefore identify the conditions that cause this result to fail.

Note 14: Markets

Crudely, a *market* is a (physical or virtual) area in which agents exchange stuff (in general: goods and services), for money or for other stuff (though barter is rare and complicated). It can take place in the “market square”; quays and city gates were also places where business happened (W. F. Leemans 1968, 190). Thus, wherever we find foreign trade, we have a market. And this means that markets have been around for a very long time (Feinman and Garraty 2010; Le Goff 1992). Historical evidence shows that foreign trade occurred in Mesopotamia by 3000 BC already (Wilhelmus François Leemans 1960, 5).

However, when people these days talk of a (*modern*) *market economy*, they think of something different. This is a system that is much more complex and goes beyond the occasional trade of goods across long distances. The market economy is about an economic system. To understand what this means, we must consider the range of modes of economic interactions. This includes (Malinowski 1922; Polanyi 1957; Temin 2012):

- Reciprocity: goods are exchanged by individuals or groups without obligations.
- Redistribution: goods are produced, taken by a central authority (e.g., the head of the family or the head of the manor), and redistributed within the group (e.g., family members).
- Exchange: goods are exchanged for other goods or for cash.

The *modern* market economy, following Polanyi (1957) and others, corresponds to a change in the scale and depth of exchanges. It represents a change of scale, because exchange takes place in an abstract space where buyers and sellers do not necessarily interact. It is not just a matter of geographic distance: trade over long distances has existed for a long time, and traders were part of networks (Greif 1993). It is also a matter of depth: in the modern economy, labor and capital—the inputs that generate economic output—are exchanged as well. For instance, instead of working for your family or tribe (a “redistribution” system), you work for a wage. Thus, not only are goods traded, but so are your labor and your capital—and the same is true for other inputs such as land.

You can see links between a modern market economy and the Industrial Revolution. Technological evolutions facilitated long-distance and abstract exchanges (e.g., the telegraph sped up the dissemination of information). And the demand for workers and the need for cash pushed workers and savers to sell their labor (via wages) and capital (via loans and investments).

3.2 Competitive markets and the First Welfare Theorem

The **First Welfare Theorem** states that if every market is *competitive* and *complete*, any competitive equilibrium is Pareto efficient. The intuition: households maximize utility, firms maximize profits, markets clear — so no good is wasted and no reallocation can improve anyone’s situation without hurting someone else. This result requires several assumptions:

- *Complete markets*: every good can be traded, that is, each good has a price and transaction costs are negligible. This must be the case under any contingencies (there are no states of the world where this isn’t the case).
- *Perfect competition*: no agent has market power, so prices reflect true marginal trade-offs (this rules out monopolies, for instance, where prices can be set higher than marginal production costs).

- *Local nonsatiation*: each consumer always prefers more of something to less, so budgets are fully spent.
- *No externalities*: production and consumption don't have side effects on people's welfare (utility), either positive or negative.

These assumptions are never perfectly met in reality. This being said, we may sometimes be in the presence of markets that are “*close enough*”: the assumptions may not be met, but they are sufficiently plausible for us to think that we are near the Pareto optimal outcome. This may be especially true if we compare to a counterfactual in which goods are not provided by markets, but by some kind of social planner (as in canonical communist systems).

But sometimes—and often whenever the environment is involved—these assumptions are so far from being met that we face a Pareto *inefficient* situation. Examples include:

- A natural resource is owned by a monopolist who either overcharges (if privately owned) or undercharges (if publicly owned and pressured to keep prices low).
- A natural resource has no legal owner, triggering a race toward extraction.
- A technology generates side effects (externalities) that create a gap between its private cost (to the agent using the technology) and social costs (to the population at large).
- Consumers or producers lack information about the impact of their actions. For instance, they may not know how their behavior generates pollution.

We then say that we have a **market failure**. And as we will see, market failures are often connected in one way or another to poorly-defined **property rights**.

3.3 Hotelling's Rule

A second reason why we may hope that self-interested actors use the environment rationally was derived by Harold Hotelling (1931). His idea was the following: can we construct an optimal dynamic path of extraction of a finite, non-renewable resource? What does such a path

mean and look like? And how does the market then coordinate the extraction of resources?

The steps that we will follow are:

1. The Hotelling rule will tell us what a natural resource's price path p_t over the future will look like.
2. The market will take that price path and produce an aggregate quantity of resource that is extracted at each time period.

The setup is the following. Many firms each own a fixed share of the total stock of an exhaustible resource, $S_0 > 0$. Consider one of them, firm i , which owns $S_{i0} > 0$ (so that $\sum_i S_{i0} = S_0$). At time t , the firm extracts $q_{it} \geq 0$. It does so at a constant marginal cost c ; the extracted good is immediately sold at the market price p_t . This is a large market, so that p_t doesn't depend on whether a single firm sells or not ("price-taker" assumption). There is an interest rate r which tells you how much any other asset gets as a return.

The firm's stock of the resource evolves as:

$$\dot{S}_{it} = -q_{it}, \quad S_{it} \geq 0.$$

The firm wants to extract in a way that maximizes the net present value of the profits it will make over time. That is:

$$\max_{\{q_{it}\}} \int_0^T \underbrace{e^{-rt}}_{\text{discounting}} \underbrace{(p_t - c)q_{it}}_{\text{profit}} dt \quad \text{s.t.} \quad \underbrace{\int_0^T q_{it} dt}_{\text{natural limits}} \leq S_{i0}.$$

Hotelling's intuition is that any unit of resource left in the ground (i.e., not extracted and sold) is an asset that has a potential market value. To see this, think in discrete time for a moment. If you extract a unit today, you obtain $p_t - c$. This return can be invested into other assets to give $(1 + r)(p_t - c)$ next period. On the other hand, you could keep it in the ground and sell it for $p_{t+1} - c$ next time period. Here's the key: in equilibrium, the owner must be indifferent between extracting or keeping it in the ground:

$$p_{t+1} - c = (1 + r)(p_t - c).$$

If the right-hand side was larger than the left-hand side, the entire set of producers would extract now, pushing p_t down (remember that individual firms are price-takers, but the same isn't true for the entire market). If it were smaller, everyone would wait, pushing p_t up.

Now let's return to continuous time. At what prices will a firm be willing to extract? Well, it's a constrained optimization problem. We can use the Lagrangian method:

$$\mathcal{L} = \int_0^T e^{-rt} (p_t - c) q_{it} dt - \lambda \left(\int_0^T q_{it} dt - S_{i0} \right).$$

Here, λ is the present value of one more unit of stock in the ground. Since all firms face the same prices and the same cost, λ is the same for every firm. Maximizing with respect to q_{it} at each date (we focus on an interior solution, $q_{it} > 0$):

$$e^{-rt} (p_t - c) = \lambda.$$

Note that this is a condition on prices, not on the firm's quantity: the firm is willing to extract at date t only if the present value of the net price equals λ . When it does, the firm is indifferent about when to extract. We then rearrange:

$$p_t - c = \lambda e^{rt},$$

and differentiate with respect to t :

$$\frac{d}{dt} (p_t - c) = r \lambda e^{rt}.$$

We can replace λe^{rt} with $p_t - c$:

$$\frac{d}{dt} (p_t - c) = r (p_t - c),$$

which yields:¹

$$\frac{\frac{d}{dt} (p_t - c)}{p_t - c} = \frac{\dot{p}_t}{p_t - c} = r$$

¹The first two expressions are equal because c is assumed to be a constant.

This is the **Hotelling rule**: the net price (also referred to as *scarcity rent*) $p_t - c$ increases at the same pace as the rate of interest. Its present value λ is the same at every date. This gives us a price path over the future. What's striking is that it must have a constant present value, otherwise firms would increase or decrease extraction now.

From there, we can then transform this price path into a quantity path at the market level. We assume a well-behaved demand for this good; at each period, the market demands a total quantity $Q_t = D(p_t)$ at price p_t . Market extraction is the sum of what all firms extract, $Q_t = \sum_i q_{it}$. By assumption, there's some price $\bar{p}$ beyond which demand is zero. As a result, there's also a last period T by which time all resources will have been extracted, $\int_0^T Q_t dt = S_0$ (if anything was left in the ground, then it would be wasted). Price will gradually increase over time (as the resource gets scarcer) until it reaches $\bar{p}$ at T . Note that only the market quantity Q_t is pinned down: since each firm is indifferent about timing, how extraction is split across firms is not determined.

So that's what rational firms would do: collectively, they gradually extract the resource along a path that maximizes returns. Does it correspond to what a social planner or benevolent dictator would do? It turns out: yes! A social planner maximizes (discounted) surplus:

$$\max \int_0^T e^{-rt} \left[\int_0^{Q_t} P(x) dx - c Q_t \right] dt \quad \text{s.t.} \quad \int_0^T Q_t dt \leq S_0,$$

where $P(\cdot)$ is inverse demand (price as a function of quantity; see later chapter in the course notes). We can use the same derivation as above to find: $P(Q_t) - c = \lambda e^{rt}$. But in the market equilibrium, $p_t = P(Q_t)$, so this is exactly the condition that the competitive market satisfies! In that way, a competitive market of rational firms that own an exhaustible resource produces the socially optimal extraction path.

Now, to achieve this result, we had to make several assumptions, including:

- Known, fixed stock of the resource. But we discover new reserves all the time!
- Constant extraction cost. But costs can change as labor and capital costs change!
- No technological change. But extraction can become suddenly much cheaper (e.g., shale gas)!

- Perfect foresight. Who knows that demand will be like tomorrow?
- Competitive markets. What if we have monopolies?
- Secure property rights. But what if a resource is accessible to multiple producers?

What if these don't hold? The assumption of property rights, in particular, is often demanding.

3.4 Property rights

One of the fundamental assumptions behind the First Welfare Theorem is that property rights are well-defined. This means two things:

- Inputs, such as natural resources, have well-defined ownership.
- There are no *externalities*.

By *externalities*, we mean the effects that an agent's actions may have on other agents. For instance, if I smoke, people around me suffer from passive smoking. Externalities can be *negative* (as in the smoking example) or *positive*. If I have a smart idea (e.g., a new technology) and share it freely in the open, then my action generates positive externalities.

Note that property rights are not the same as *private* property rights. Property rights mean that one or more agents have a legal claim over some resource (a pen, a computer, an oil field, etc.). We may have:

- *Private property*: ownership is in the hands of a single entity (a person, a firm, etc.).
- *Collective property*: ownership is shared among a defined set of agents. This system is often used to manage common-pool resources in many regions.
- *State or public property*: ownership is controlled by the state (e.g., state-owned enterprises).

In the absence of well-defined property rights, the results that we obtained earlier fail. As a result, we cannot guarantee that markets achieve a Pareto optimal outcome. Recall our discussion on ethics: this means we may face a *policy problem* that can, but does not have to, be solved by policy interventions.

The problem of property rights is often connected to the nature of goods and resources. In environmental economics, goods are often classified based on two key characteristics: *rivalry* and *excludability*. Rivalry refers to whether one person's use of a good reduces its availability to others, while excludability indicates whether people can be prevented from accessing the good.

Based on these characteristics, four main types of goods exist (see Table 3.1):

- *Private goods*. These are both rival and excludable, like timber or fish sold in a market.
- *Club goods*. These are excludable but non-rival (up to a point), such as a private park or a toll road.
- *Public goods*. These are neither rival nor excludable, such as clean air or climate stability, making them susceptible to free-rider problems.
- *Common-pool resources*. These are rival but non-excludable, like fisheries and groundwater basins. The phenomenon in which common-pool resources are overused and depleted is known as the *Tragedy of the Commons* (Hardin 1968).

Table 3.1: Classification of goods (based on rivalry and excludability).

	Excludable	Non-excludable
Rival	Private goods	Common pool
Non-rival	Club goods	Public goods/bads

The relationships between humans and natural resources are generally defined by a set of rights called property rights, which maps onto the *excludability* dimension. Resources that have no assigned property rights are typically non-excludable. Policies that aim to address ecological impacts on ecosystems generally alter property rights regimes. For instance, the protection of ecosystems may involve the implementation of national parks, which may lead to a transformation of ownership from local communities to the State.

Property rights regimes determine who holds rights and responsibilities over a resource, under what conditions, and how those rights are enforced. A useful framework for understanding property rights is provided by Schlager and Ostrom (1992). Property rights regimes are

systems of rules, institutions, and social arrangements that define how rights to access, use, manage, exclude, and alienate resources are distributed among groups.

- *Access rights* are the rights to enter a defined resource area. These determine who is allowed to physically be in the resource space. For instance, having the right to enter fishing grounds or to walk through grazing lands.
- *Use rights* (also called withdrawal rights) are the rights to obtain and use specific resource units from the system. These determine what activities you can perform and what benefits you can take from a specific resource and space, e.g., the right to cut a certain amount of firewood, extract water for irrigation, or catch a certain number of fish.
- *Management rights* refer to the “right to regulate internal use patterns and transform the resource by making improvements.” For example, in Switzerland, a *Bergschaft* (mountain pasture association) decides on grazing rotations, reseeded degraded areas, and maintaining irrigation channels.
- *Exclusion rights* refer to the right to determine who will have access rights and how those rights may be transferred. On Lake Geneva, a fishing association grants fishing rights only to members who hold a local permit, excluding non-members.
- *Alienation rights* refer to “the right to sell or lease either or both of the above collective-choice rights.” Local authorities often lease part of their lands, such as communal forests, to certified logging companies for selective timber harvesting.

Bundles of rights are hierarchical: to hold management rights, one must first hold access and use rights. Furthermore, property rights regimes are not only technical arrangements for managing resources — they also reflect and reproduce power relations within societies. Decisions about who can access or exclude others from valuable ecological goods often shape patterns of wealth, poverty, and inequality. For example, during colonial times, the establishment of national parks frequently involved displacing Indigenous communities from their ancestral lands to create protected areas managed by the state. While these parks aimed to conserve biodiversity, they also transferred control over natural resources from local communities to colonial or state authorities, undermining traditional livelihoods and access rights.

In addition to the types of goods and their implications, it is important

to distinguish between *de jure* and *de facto* property rights. *De jure rights* are formally recognized and enforced by governments through legal and administrative systems. Holders of *de jure rights* can rely on official institutions to uphold their claims if challenged. In contrast, *de facto rights* arise from customary practices and cooperation among resource users themselves, without formal government recognition. For example, peasant communities' informal arrangements against the crown *livrée* system in Medieval France maintained resource use rules internally. These rights are enforced socially within the user community and can be less secure legally, though they may eventually gain formal recognition. Often, within a single resource system, both *de jure* and *de facto* rights coexist, sometimes complementing or conflicting with each other.

How does this relate to environmental issues and market inefficiencies? The key point is: **many environmental issues involve situations in which a good is non-excludable (or, similarly, a behavior's consequences are non-excludable):**

- An exhaustible natural resource may have open-access characteristics. Without property rights, anyone can use the resource. If it is valuable, users may extract it without considering the impact on others — a Tragedy of the Commons (Hardin 1968). Competition over resources may accelerate depletion.
- A behavior that generates negative externalities (e.g., pollution from a factory) creates a public bad: affected people cannot be excluded. The agent generating the externality does not internalize the costs imposed on others, and no property rights are attached to the negative externality.

We next present two related approaches to model the problem of excessive extraction of natural resources and excessive production of negative externalities: a Prisoner's Dilemma and a modified Cournot model.

3.5 Resource extraction and externalities: a Prisoner's Dilemma-based approach

Another approach to model the problem of property rights comes from the *Prisoner's Dilemma* ("PD game" or just "PD") (Axelrod 1984; Taylor

1987; E. Ostrom 1990). The PD is a model from game theory, a branch of mathematics and the social sciences, used to analyze *strategic* interactions, that is, interactions among two or more decision-makers (see Note 15).

i Note 15: Game theory, prisoner dilemma, and global environmental issues

Game theory, a branch of mathematics and the social sciences, analyzes strategic interactions among rational decision-makers whose outcomes depend on both their own choices and those of others.

This framework is especially useful for understanding complex environmental issues. Consider international climate negotiations. In such settings, countries are often stuck in non-cooperative games, such as the Prisoner's Dilemma, where individual incentives to free-ride on others' efforts can undermine collective action. Sustainability games extend this analysis by modeling interactions among diverse actors—governments, firms, and consumers—whose decisions affect the environment and resources for present and future generations (Alessio and Manfredi 2022). These games feature public goods with key characteristics such as irreversible environmental damage, widespread impacts on all participants, global consequences, and inequalities that hinder cooperation. These dynamics often resemble the Prisoner's Dilemma.

In the context of climate diplomacy, these interactions are frequently simplified as 2×2 games. The classic Prisoner's Dilemma captures the core challenge of cooperation: two players independently decide whether to cooperate or defect without knowledge of the other's choice. While mutual cooperation yields the best collective outcome, rational self-interest drives both to defect, resulting in a collectively suboptimal outcome. This tension between individual incentives and collective benefits helps explain why cooperation in international environmental agreements is difficult. Although the global benefits of environmental protection are shared, the costs of mitigation fall unevenly on different countries, making binding agreements hard to achieve.

As the table below shows, the payoff matrix represents the classic Prisoner's Dilemma, assuming that preference orderings are

$T > R > P > S$. In the context of climate negotiations, countries face a similar dilemma: while mutual cooperation (e.g., committing to reduce emissions) yields the best joint outcome (Reward, R), individual incentives to defect (free-ride on others' efforts) lead to worse outcomes for all (Punishment, P). The temptation (T) to defect while others cooperate often undermines global climate action, as does the fear of being taken advantage of (S), highlighting the challenge of achieving binding agreements.

		Player 2	
		Cooperate (C)	Defect (D)
Player 1	Cooperate (C)	R, R	S, T
	Defect (D)	T, S	P, P

Payoffs shown as (Player 1, Player 2). R: reward, T: temptation to defect, S: sucker's payoff, P: punishment.

Moreover, international negotiations function as “two-level games,” in which leaders balance external diplomatic pressures with domestic political considerations (Robert D. Putnam 1988). This balancing act complicates both consensus-building and the implementation of agreements. Game theory sheds light on these layered dynamics, helping to explain the often slow progress in global environmental diplomacy.

The simplest version of the PD game can be illustrated with this story: two individuals committed a burglary involving a stolen car. They get caught, but the District Attorney lacks sufficient evidence to charge them for the burglary. However, she can charge them for the stolen car. The District Attorney meets both thieves separately and offers them a deal: if you admit to the crime, I will set you free and put your friend in jail for 5 years. If you both admit, you will both receive a 3-year sentence, as your confession is considered less valuable. If neither admits, the jury will probably send you to prison for 2 years. Depending on how the two thieves behave, this gives rise to a 2×2 matrix reported in Table 3.2.

We have defined the problem, but we don't yet know what will happen. One way to approach this is by considering each player's best responses (strategies), given what the other player might do. Put yourself in the shoes of Thief 1:

Table 3.2: Prisoner's dilemma. Game with two players (Thief 1, 2) and two strategies, Admit (the crime) or Deny. This yields four outcomes. Numbers in the cells indicate the number of years of prison (larger number are thus worse).

		Thief 2	
		Admit	Deny
Thief 1	Admit	3, 3	0, 5
	Deny	5, 0	2, 2

- If Thief 2 admits, then I should admit as well (3 years of prison is better than 5).
- If Thief 2 denies, then I should admit (0 years of prison is better than 2).

Thus, regardless of what Thief 2 does, I should admit. The same rationale holds for Thief 2. Therefore, both players have a dominant strategy to admit, with a payoff of {3,3} (3 years of prison each). A “dominant strategy” is a strategy that offers equal or superior payoffs regardless of what the other player(s) do. In other words, its payoff is equal to or greater than any other available strategy, regardless of the other player's actions.

Not all games have dominant strategies that can be used to formulate predictions. A more general concept to ‘solve’ games (i.e., identify sets of strategies that meet reasonable criteria) is the **Nash equilibrium** (and its variants) (Nash 1950). A Nash equilibrium is a set of strategies in which no player can improve their payoff by unilaterally changing their strategy, given the strategies chosen by the other player(s). In other words, once a Nash equilibrium is reached, each player's choice is the best response to the other's, and no one has an incentive to deviate on their own. This seems like a reasonable way to approximate how agents will behave: if you could do better by changing your strategy on your own, why not do it?²

²Two technical notes. (1) A Nash equilibrium doesn't always exist if we limit ourselves to *pure* strategies. However, a Nash equilibrium always exists if we allow players to use *mixed strategies*, in which player randomize over feasible strategies according to a defined probability distribution. (2) Since Nash's canonical definition of an equilibrium, much work has gone into refining it to avoid non-sensical equilibria and to accommodate other game structures (e.g., games in which agents select strategies in turns). We will discuss some refinements later.

Let's get back to the PD game. Since no player has any reason to deviate from {Admit, Admit} alone, this constitutes a Nash equilibrium. But note that this outcome is strictly worse for both players than {Deny, Deny}, assuming they dislike prison (which seems a fair assumption). Recall our definition of *Pareto efficiency*: an outcome is Pareto-superior to another if we can switch to it without hurting anyone and make at least one agent better off. Clearly, {Deny, Deny} is Pareto-superior to {Admit, Admit}: you can move there without hurting anyone and make at least one agent (here, both) better off.

The PD illustrates a fundamental tension: rational self-interest leads individuals to an outcome that is unambiguously worse for everyone — worse without requiring any interpersonal comparisons, since both players prefer {Deny, Deny} regardless. This framework applies directly to environmental problems:

- *Excessive extraction of natural resources*: {Slow extraction, Slow extraction} would be socially optimal by limiting overuse, but firms are incentivized to extract faster ({Fast extraction, Fast extraction} is the Nash equilibrium).
- *Provision of public goods*: {Contribute, Contribute} would be socially optimal, but {Not contribute, Not contribute} is a Nash equilibrium.
- *International environmental agreements*: It is socially optimal to reduce environmental degradation (e.g., mitigate greenhouse gas emissions), but every country has an incentive not to do so ("I pocket the profits associated with emissions, and you do the costly work of stabilizing the climate").

Note something important here: when we say that the Nash equilibrium is "suboptimal" (or Pareto-inferior), we speak from the perspective of the *players* involved, not necessarily from the perspective of society. It may be that the strategies and resulting outcomes for society at large—whose preferences are not directly included in this version of the PD—are even worse. Note 16.



Tip 16: Welfare-reducing cooperation successes

Something to keep in mind: when a cooperation problem is successfully solved by players, it doesn't necessarily mean that this is desirable for society at large. It is only a success for the play-

ers directly involved, not necessarily for broader society, which might still be left worse off.

One example of a successful solution to cooperation games that is welfare-reducing is employer collusion over recruitment. In the US, tech companies for a while had a “no-poaching” agreement, under which they did not actively recruit talent from other firms. One day, Adobe started recruiting employees from Apple. Steve Jobs, then CEO, promptly reminded Bruce Chizen (Adobe) that this was not an equilibrium (Figure 3.3). The emails were uncovered in a class-action lawsuit, where employees argued that several US-based employers had colluded to avoid competing for talent. The firms settled for \$415 million in 2015 ([Reuters \(link\)](#)).

From: Steve Jobs
Sent: Thursday, May 26, 2005 9:36 AM
To: Bruce Chizen
Subject: Recruiting

Bruce,

Adobe is recruiting from Apple. They have hired one person already and are calling lots more. I have a standing policy with our recruiters that we don't recruit from Adobe. It seems you have a different policy. One of us must change our policy. Please let me know who.

Steve

On May 26, 2005, at 4:15 PM, Bruce Chizen wrote:

I thought we agreed not to recruit any senior level employees (at Adobe this is Sr. Director/VP and represents about 2% of the population). I am pretty sure your recruiters have approached more junior ones.

I would propose we keep it this way. Open to discuss. It would be good to agree.

From: Steve Jobs
Sent: Thursday, May 26, 2005 6:27 PM
To: Bruce Chizen
Cc: Steve Jobs
Subject: Re: Recruiting

OK, I'll tell our recruiters that they are free to approach any Adobe employee who is not a Sr. Director or VP. Am I understanding your position correctly?

Steve

Figure 3.3: Email by Steve Jobs (Apple) to Bruce Chizen (Adobe).
Source: [link](#).

This type of behavior can happen more generally. For instance, if firms cooperate to fix prices, consumers lose. This type of *market-fixing* or *collusion*, often orchestrated by cartels, can generate substantial welfare losses. One example occurs in the power sector, notably in Italy (Proz and Huber 2025). Roughly, the Italian electricity market, like most power markets, is divided into (a) a day-ahead market and (b) a balancing market.

The day-ahead market provides the electricity that the market

predicts will be needed. This is generally done via an auction, where each utility announces the cost at which it would sell its electricity (in EUR or USD/MWh). The market operator then ranks bids from cheapest to most expensive and accepts bids until expected demand is met. Everyone is compensated at the level of the last, marginal bid.

The balancing market, in contrast, adds power on short notice if needed (for instance, if the day happens to be extra hot and everyone turns on their air conditioning unexpectedly). To incentivize utilities to provide balancing power, the market operator generally pays a premium.

Thus, one way for utilities to increase profits consists of attempting to sell power at the balancing rate rather than in the day-ahead market. This is precisely what happened in Italy. Proz and Huber (2025, 6) describe it as follows:

the units [utilities] intentionally lost MGP [day-ahead, auction] offers and took turns meeting the resulting MSD [balancing] demand, thus delivering a similar output at a higher price than the MGP [day-ahead] price. While the average MSD [balancing] price exceeded the MGP [day-ahead] price by two and a half times, GME [market operator] did not initially suspect a cartel, as MSD [balancing] units are generally more costly than MGP units.

This type of behavior is not limited to the power sector. Global oil supplies are partly controlled by agreements made by members of the Organization of the Petroleum Exporting Countries (OPEC). Member states meet periodically to collude on the amount of oil extracted, aiming to achieve specific price levels. Whether these agreements work, however, is unclear (Brémond, Hache, and Mignon 2012; Colgan 2014).

Typically, this kind of collusive behavior raises several problems (Keohane 1984; Barrett 2003):

1. **Cooperation.** Firms or states often have an incentive to renege on their collusive agreements. For instance, an OPEC member might secretly produce a little more oil to take advantage of the higher prices created by the cartel.

But if every producer does the same, the system moves toward a non-cooperative outcome.

2. **Monitoring.** If the incriminated behavior is hard to observe, monitoring by policymakers is difficult. Moreover, illegal behaviors are often difficult to detect. Proz and Huber (2025) notes that collusive actions in the power sector are particularly hard to identify. At the international level, agreements on sensitive topics, such as nuclear proliferation (where ‘cheating’ corresponds to producing nuclear weapons), are extremely difficult to monitor properly—for example, in the case of Iran.
3. **Punishment.** Suppose there is an agreement and monitoring is feasible. If you want the agreement to be effective (e.g., price-fixing in the construction sector), you must find a way to punish actors who deviate. Domestically, this might be feasible if you have a robust legal and law-enforcement system. But how do you enforce compliance when the ‘misbehaving’ player is another sovereign state?

Tip 17: Strategic famines

Eggertsson (1998) analyzes how Iceland’s internal institutional rules led, sometimes, to catastrophes.

Let’s unpack. First, some background. Iceland lost its independence in 1262, first controlled (from a distance) by Norway and then by Denmark, until it finally regained its independence after World War II. For a long time, the country was isolated because of its geography and because its colonial rulers wished to avoid undue inference by others. As a result, Iceland was long a stagnating island, with a population of about 50,000.

The question Eggertsson (1998) asks is: how does such a small, homogenous population, handle large systemic shocks, such as extreme weather (via the impact it has on farming)? What struck Eggertsson (1998, 25) is that Iceland occasionally suffered from dramatic famines, one of which killed 24% of the population (1784-1785). Why did Icelandic farmers not save sufficient amounts of hay to survive? Unlike continental Europe, answers such as theft and destruction by other countries and armies don’t

apply here.

To answer this question, Eggertsson (1998) considers the incentives of farmers and the rules imposed by local authorities. Simplifying things, a farmer has two strategies: s_a build up hay reserves (low risk) or s_b keep low reserves (high risk). The puzzle is: why not do s_a ? The reason might be the following: the authorities had a “good neighbor” rule, which meant that, in case of urgent need, excess hay had to be offered (against a (presumably cheap) payment) to other people living nearby.

Thus, under s_a , two things can happen:

- there is no negative climate event, and thus expenses to store hays are lost;
- there is a climatic crisis, but then your hay could be taken away and redistributed.

Thus, your dominant strategy is *not* to store hays and rely on others (in essence *free-riding*). Of course, if everyone does that, then you simultaneously should (a) see occasional famines (since the strategies are high-risk) and (b) see very little hay being distributed (since it's not rational to save, in equilibrium).

Historical evidence is scarce, so we don't know for sure if that's what happened (Eggertsson 1998, 22). But it would go some way to explain why even a small, homogenous population, can undermine itself in a way that may appear irrational at first.

What could Iceland have done? One solution would be for authorities to centralize hay storage (e.g., via a tax) instead of using a decentralized approach. But that can be expensive. And it creates political issues, such as how the hay ought to be distributed in case of a crisis (who gets it first? the friends of the rulers?).

3.6 Resource extraction and externalities: a Cournot-based approach

How does competition affect the use of resources? Recall that a key idea behind property rights is that the absence of well-defined rights may lead to overuse of a resource, since it becomes a freely available, open-access resource. Can we formalize this intuition more rigorously?

We can apply a model introduced by Antoine Cournot (1801–1877), who pioneered strategic models of firm behavior. He did *not* have environmental issues in mind, but we can tweak his canonical model, as presented by Gibbons (1992), to address our question: what changes when multiple agents have access to a resource instead of just one?

Consider two firms, A and B , producing a single good (say, chairs) using a single input (say, wood). We initially assume that both firms have unrestricted access to the input, which can be interpreted as a case with no clearly defined property rights over the resource; you can also think of this as a case where nature is (unfairly) priced at zero.

The (inverse) demand function faced by both firms is linear:³

$$P = a - Q, \quad \text{with } a > 0 \text{ and } Q = Q_A + Q_B,$$

where Q_A and Q_B are the quantities produced by firms A and B , respectively, and P is the price of the good. This means that the price declines linearly with total output, with slope 1 for each additional unit of the good (i.e., each chair). Or put in a more intuitive way (by flipping P and Q): the quantity demanded declines by one unit for every increase of the price by one unit.

On the supply (producer) side, we assume that the marginal cost of production (the cost of producing one more unit) is constant, c . Thus, producing one chair might cost CHF 30, the next chair the same, etc. This doesn't have to hold; oftentimes, marginal costs eventually increase: you first produce chairs with the cheapest inputs, and then you gradually have to hire more expensive workers, build more expensive factories, etc.

If we want to account for negative externalities, we can introduce an environmental cost function that increases with the quantity produced (e.g., deforestation caused by chair production):

$$E = f(Q), \quad \text{with } f'(Q) > 0.$$

³We will discuss demand functions later. For now, just think of it as the response of consumers (in terms of how many units of the good they want) to changes in price. A *demand* function is $Q(P)$ (quantity as a function of price), whereas an *inverse demand* function is $P(Q)$ (price as a function of quantity). The two are obviously connected.

The variable E can be thought of as the social cost caused by the production and consumption of this good. For instance, if making chairs reduces biodiversity, this would represent the impact of doing so (e.g., by reducing natural services).

Each firm sets the quantity of chairs it produces, considering the production choice of the other firm. Thus:

$$\max_{Q_i \geq 0} \Pi_i(Q_i, Q_j^*),$$

Where Q_j^* denotes the equilibrium quantity of the good produced by firm j (we use asterisks to indicate equilibrium strategies).

Accordingly, the profit functions can be expressed as follows:

$$\Pi_i = \underbrace{Q_i P(Q)}_{\text{Revenue}} - cQ_i.$$

Using the linear demand function, we can plug in the price equation:

$$\begin{aligned} \Rightarrow \Pi_A &= Q_A[a - (Q_A + Q_B)] - cQ_A \\ \Rightarrow \Pi_B &= Q_B[a - (Q_A + Q_B)] - cQ_B. \end{aligned}$$

We can solve this optimization problem via first- and second order conditions:

$$\frac{\partial \Pi_i}{\partial Q_i} = a - 2Q_i - Q_j - c = 0 \Rightarrow Q_i = \frac{a - c - Q_j^*}{2}.$$

This gives us two equations for Q_A and Q_B , which we can solve to find equilibrium quantities (or Cournot-Nash equilibrium):

$$Q_A^* = Q_B^* = \frac{a - c}{3},$$

and total production:

$$Q^* = \frac{2}{3}(a - c)$$

Consider the case of a single firm. This would mimic the situation in which the input resource is entirely owned (or, more precisely, there is no competitive pressure over it). This is the problem of the *monopolist*

M , where we solve the same profit-maximization problem, but set the other firm's quantity to zero:

$$\max_{Q_M \geq 0} \Pi_M(Q_M, 0),$$

which yields:

$$Q_M^* = \frac{1}{2}(a - c) \Rightarrow Q_M^* < Q^* \Rightarrow E_M^* < E^*.$$

Thus, the input resource will be used less intensively when it is owned by a single agent. Note that here, firms would prefer to be monopolists (to increase profit), and even those affected by the resource's overuse would prefer a single owner. When the resource becomes contested, the firm's welfare declines and environmental degradation increases. (Important warning here: this does not mean that monopolists are good for the environment or desirable more generally. Recall that this is a model to capture a specific dynamic and not a full-blown welfare analysis. For instance, you could also expect a monopolist to operate less efficiently and generate more waste.)

Alternatively, we could model the problem by noting that the unit cost of production, c , is lower than optimal when the input good has no assigned property rights. At the limit, if this input is free, costs could be close to zero (ignoring the cost of labor and capital). This would also increase production and generate similar results. This is why we often say that natural resources are *under-priced* due to weakly defined property rights.

3.7 Conclusion

Markets are powerful tools for achieving efficiency — but only when the First Welfare Theorem's conditions hold. When property rights are ill-defined, these conditions fail, producing **market failures** and a gap between the world as it is and as it ought to be.

Policy interventions can target markets directly — either on the demand side or the supply side (the next two chapters) — or restructure the underlying institutions and “rules of the game” (explored later in the book).

Additional readings

Classics on the study of cooperation and public goods include Axelrod (1984) and Taylor (1987). E. Ostrom (1990) and Barrett (2003) use game theoretic models to explain the lack of (domestic and global) environmental regulations. On a critical discussion of Hardin, see Matto Mildenberger's note in the *Scientific American* ([link](#)).

Appendix: Proof of the First Welfare Theorem

Let's start with the setup of this economy. An economy has:

- **L goods**, indexed by $\ell = 1, \dots, L$. Prices are a vector $p \in \mathbb{R}^L$ (one price for each good).
- **I consumers**, indexed by i . Consumer i has a consumption set $X_i \subseteq \mathbb{R}^L$ (typically $\mathbb{R}_+^L$: we don't consume negative numbers), a preference relation $\succsim_i$ on X_i (the consumer can weakly order bundles), and an endowment $\omega_i \in \mathbb{R}^L$. Write $\bar{\omega} = \sum_i \omega_i$ for aggregate endowment.
- **J firms**, indexed j . Firm j has a production set $Y_j \subseteq \mathbb{R}^L$, written in *netputs*: a vector $y_j \in Y_j$ has negative entries for inputs and positive entries for outputs, so $p \cdot y_j$ is exactly profit.

We will need a few additional concepts. An **allocation** is a pair $(x, y) = ((x_i)_i, (y_j)_j)$ with $x_i \in X_i$ and $y_j \in Y_j$. This allocation is **feasible** if:

$$\sum_i x_i = \bar{\omega} + \sum_j y_j$$

Then, a feasible allocation (x^*, y^*) is **Pareto efficient** if there is no other feasible allocation (x', y') with

$$x'_i \succsim_i x_i^* \text{ for every } i, \quad \text{and} \quad x'_k \succ_k x_k^* \text{ for at least one } k.$$

A price vector p , wealth levels (w_i) with $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$, and an allocation (x^*, y^*) produce a **Walrasian equilibrium (with transfers)** if:

1. **Profit maximization by firms.** For each j : $p \cdot y_j^* \geq p \cdot y_j$ for all $y_j \in Y_j$.
2. **Utility maximization by consumers** For each i : x_i^* is $\succsim_i$ -maximal in the budget set $B_i = \{x_i \in X_i : p \cdot x_i \leq w_i\}$.
3. **Market clearing.** $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$.

Setting $w_i = p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$ (with ownership shares θ_{ij}) sets the ordinary no-transfer equilibrium.

Finally: **local nonsatiation (LNS)**. For every $x_i \in X_i$ and every $\varepsilon > 0$ there exists $x'_i \in X_i$ with $\|x'_i - x_i\| < \varepsilon$ and $x'_i \succ_i x_i$. In plain English: regardless of x_i , you can always do better by ‘moving’ a bit. Note that it sounds a bit like monotonicity (e.g., utility function that is non-decreasing in some x). It’s true that monotonicity implies LNS, but LNS is weaker than monotonicity. The latter allows bads (goods with negative utility) and satiation in individual goods. But note that LNS can in theory be a pretty strong assumption.

We can then get to the First Welfare Theorem.

Theorem 1. Suppose every consumer’s preferences are locally nonsatiated. If (p, w, x^*, y^*) is a Walrasian equilibrium with transfers, then the allocation (x^*, y^*) is Pareto efficient.

At equilibrium prices, anything a consumer likes better than what she chose must have been *unaffordable* (otherwise she would have bought it!). So a Pareto improvement would have to cost strictly more, in aggregate, than the economy’s income. But the economy’s income is exactly the value of what it can produce, and firms are already producing the most valuable bundle available to them. So the improvement cannot be paid for. Prices act as a common accounting unit that makes “everyone gains” and “the budget balances” incompatible.

We can formalize this. We will need two lemmas. Both turn statements about *preference* into statements about *money*.

Lemma 1 (strictly better $\Rightarrow$ strictly more expensive). If $x_i \succ_i x_i^*$ then $p \cdot x_i > w_i$.

Proof. Suppose not, i.e. $p \cdot x_i \leq w_i$. Then $x_i \in B_i$, so x_i was available to consumer i . Since $x_i \succ_i x_i^*$, the bundle x_i^* was not $\succsim_i$ -maximal on B_i , contradicting utility maximization. ■

Notice this lemma needs no assumptions at all beyond “the consumer optimized.” It is pure revealed preference.

Lemma 2 (weakly better $\Rightarrow$ weakly more expensive). If $x_i \succsim_i x_i^*$ then $p \cdot x_i \geq w_i$.

Proof. Suppose not: $x_i \succsim_i x_i^*$ but $p \cdot x_i < w_i$. Since the inequality is strict, there is room to move: choose $\varepsilon > 0$ small enough that every point within distance ε of x_i still costs less than w_i (possible because $z \mapsto p \cdot z$ is continuous). By local nonsatiation, pick $x'_i \in X_i$ with $\|x'_i - x_i\| < \varepsilon$ and $x'_i \succ_i x_i$. Then $x'_i \succ_i x_i \succsim_i x_i^*$, so $x'_i \succ_i x_i^*$ by transitivity, and $p \cdot x'_i < w_i$ means $x'_i \in B_i$. This contradicts Lemma 1. ■

Lemma 1 handles the person who strictly gains from the proposed improvement. But a Pareto improvement only makes *one* person strictly better off; everyone else is merely *not worse off*. Lemma 2 is what lets us charge those people full price anyway. And Lemma 2 is exactly where local nonsatiation enters — it is the assumption that rules out a consumer who is indifferent over a fat region and could therefore have been “given” the same satisfaction more cheaply. Delete LNS and the theorem is false (see §2.4).

We can now complete the proof of the First Welfare Theorem.

Proof of Theorem 1. Suppose, for contradiction, that (x', y') is feasible and Pareto-dominates (x^*, y^*) : $x'_i \succsim_i x_i^*$ for all i , with $x'_k \succ_k x_k^*$ for some k .

Step 1: price the consumption side. Apply Lemma 2 to every $i \neq k$ and Lemma 1 to k :

$$p \cdot x'_i \geq w_i \quad (i \neq k), \quad p \cdot x'_k > w_k.$$

Summing over all i — one strict inequality is enough to make the sum strict —

$$p \cdot \sum_i x'_i > \sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*. \quad (*)$$

Step 2: use feasibility of the improvement. Since (x', y') is feasible, $\sum_i x'_i = \bar{\omega} + \sum_j y'_j$, so the left-hand side of $(*)$ equals $p \cdot \bar{\omega} + \sum_j p \cdot y'_j$. Substituting and cancelling $p \cdot \bar{\omega}$:

$$\sum_j p \cdot y'_j > \sum_j p \cdot y_j^*.$$

Step 3: contradict profit maximization. If a sum of J terms strictly exceeds another sum of J terms, at least one term must: there is some firm j with $p \cdot y'_j > p \cdot y_j^*$. But $y'_j \in Y_j$, so firm j was not maximizing profit at y_j^* . Contradiction. ■

Part II

Markets

4 Demand-side problems

This version: September 28, 2026

Summary

New (sustainable) technologies can help solve problems caused by incumbent (non-sustainable) technologies. Markets can plausibly help this transition, but their impact will depend on the relative demand for old and new technologies.

4.1 Theoretical models of demand

Demand drives economic transformation and innovation. Understanding it is essential for sustainability policy, since production ultimately responds to what consumers want. We focus on demand for green goods — both private (e.g., rooftop solar) and public (e.g., clean air). Two main modeling frameworks are useful:

1. **Discrete choice models** — for individual-level decisions (e.g., whether to buy a solar panel). These are the microfoundation: they model how each person chooses among a finite set of alternatives.
2. **Continuous demand models** — for aggregate market-level analysis (e.g., how a Pigouvian tax shifts quantity demanded). These emerge from aggregating individual choices across a population.

We start at the individual level with discrete choices, then show how aggregating them yields the demand curves used for policy analysis. We also review estimation techniques for quantifying demand — informative for both policy and business strategy.

4.2 Discrete choice models

4.2.1 Random utility models

Discrete choice models analyze situations where individuals choose among a finite set of mutually exclusive alternatives — predicting *which* option is selected rather than *how many* units are bought. When a consumer chooses between an ICE car, an EV, or the bus, the decision depends on attributes like price, convenience, and environmental impact. Modeling these choices informs both business strategy and policy interventions.

The theoretical foundation of most discrete choice models rests on the **random utility model (RUM)** (McFadden 1972; Train and Weeks 2005; Henscher, Rose, and Greene 2015). This framework assumes that each individual i derives utility U_{ij} from choosing alternative j from their choice set $\mathcal{C}_i$. A rational individual chooses the alternative that provides her with the highest utility:

Individual i chooses alternative j if $U_{ij} > U_{il}$ for all $l \neq j \in \mathcal{C}_i$

The key insight of RUM is that utility has both a *systematic* and a *stochastic* component:

$$U_{ij} = V_{ij} + \varepsilon_{ij},$$

where:

- V_{ij} is the *deterministic* (observable) component of utility.
- ε_{ij} is the *random* (unobservable) component representing unknown factors that vary from people to people.

i Note 19: The value of a technology

One way to think of the random utility model (RUM) is to go back to a utility function (which we saw earlier) and split it between a deterministic and a stochastic component. The stochastic part contains idiosyncracies about you and about the good. So far, so good. But where does the deterministic value come from? One way to think of it is as an objective item, such as profits. This could be appropriate if the item is used for productive use. An-

other way to think of it is as something subjective, like taste ('I like this electric bike a lot').

Once we model individual decision to buy (or not) a given technology, it's quite clear what kind of strategy one may want to use to influence potential consumers: you can try to push V up (for supporters) or down (opponents) to discourage consumers. Information campaigns, for instance, can be used to undermine the (subjective) value of a given new technology. And such campaigns don't have to be based on truth to be effective. One study found that about one-quarter of survey respondents in the United States, United Kingdom, and Australia had "contrarian" views about wind farms (Winter et al. 2024). Think of similar campaigns (whether truthful or not) about electric vehicles (limited range, risk of fires), nuclear (waste, risk of meltdown; although admittedly this isn't about individual consumers), wind turbines (risk to birds), solar PV (amounts of waste), etc.

As we will see below, if a sufficient number of consumers have low V for a technology, demand will soften thereby slowing down its diffusion. That's why observers are very quick to worry about disinformation campaigns.

The deterministic component is typically specified as a linear function of observed attributes:

$$V_{ij} = \alpha + \beta_1 x_{ij} + \beta_2 w_{ij} + \dots$$

where β_k is the coefficient measuring the marginal utility of that variable. Note that there are two types of variables:

- Variables that vary by decision-makers, i . For instance: you want to model choices based on the decision-maker's age, gender, origin, etc. This will typically lead to *multinomial logit* model.
- Variables that vary by item, j . For instance: you want to model choices based on a car's color, price, size, etc. This will typically lead to a *conditional logit* model.

Let us now build and solve the model. The probability that individual

i chooses alternative j is:

$$\begin{aligned}\mathbb{P}_{ij} &= \mathbb{P}(U_{ij} > U_{il} \text{ for all } j \neq l) \\ &= \mathbb{P}(V_{ij} + \varepsilon_{ij} > V_{il} + \varepsilon_{il} \text{ for all } j \neq l) \\ &= \mathbb{P}(\varepsilon_{il} - \varepsilon_{ij} < V_{ij} - V_{il} \text{ for all } j \neq l)\end{aligned}$$

The specific form of this probability depends on the distribution of the two error terms ε (and by extension their difference). A common assumption is that the error terms are independently and identically distributed with a *Type I extreme value* (or Gumbel) distribution.¹

Assuming a Type I error, we obtain:

$$\begin{aligned}\mathbb{P}_{ij} &= \frac{\exp(V_{ij})}{\sum_{j \in \mathcal{C}_i} \exp(V_{ij})} \\ &= \frac{\exp(\sum_k \beta_k x_{ijk})}{\sum_{j \in \mathcal{C}_i} \exp(\sum_k \beta_k x_{ijk})}\end{aligned}$$

If there is a binary choice (two items a and b), then this simplifies to:

$$\mathbb{P}_{ia} = \frac{\exp(V_{ia})}{\exp(V_{ia}) + \exp(V_{ib})},$$

and we obtain the logit model.

4.2.2 Estimation of random utility models

In a RUM setup, the probability of i selecting an option j , $\mathbb{P}_{ij}$, is a non-linear model that can be estimated with maximum likelihood. The parameters β_k are typically estimated using **maximum likelihood estimation (MLE)**. The log-likelihood function is:

$$\ln L(\beta) = \sum_{i=1}^N \sum_{j=1}^J y_{ij} \ln \mathbb{P}_{ij}(\beta)$$

where $y_{ij} = 1$ if individual i chose alternative j , and 0 otherwise.

¹Another approach to binary choices is to assume that the errors are distributed normal (which leads to probit models).

Sometimes, though, analysts will keep things simpler and estimate a **linear probability model**, in which:

$$\mathbb{P}_{ij} = \beta'_k \mathbf{X}_i + \varepsilon_i,$$

where $\mathbf{X}$ is the vector of choice predictors. See Note 19 and Note 20 for examples.

i Note 19: Example: household electricity technology

Consider a household's decision to adopt solar panels or improve energy efficiency. The choice set contains three alternatives:

- $j = 1$: install solar panels
- $j = 2$: continue with grid electricity only
- $j = 3$: invest in energy efficiency improvements

The utility functions might be specified as:

$$V_{i1} = \beta_1 \cdot \text{Income}_i + \beta_2 \cdot \text{ElectricityPrice}_i + \beta_3 \cdot \text{EnvironmentalConcern}_i + \beta_4 \cdot \text{SolarCost}_i$$

$$V_{i2} = 0 \quad (\text{reference alternative})$$

$$V_{i3} = \beta_5 \cdot \text{Income}_i + \beta_6 \cdot \text{HomeAge}_i + \beta_7 \cdot \text{EfficiencyCost}_i$$

Here, we expect:

- $\beta_1 > 0$ (higher income increases solar adoption),
- $\beta_2 > 0$ (higher electricity prices make solar more attractive),
- $\beta_3 > 0$ (environmental concern promotes solar), and
- $\beta_4 < 0$ (higher solar costs reduce adoption).

Once estimated, this model can predict adoption probabilities for different household types, simulate the impact of solar subsidies (reducing SolarCost_i), or evaluate how changing electricity rates affects market penetration. For instance, if a household has income of \$80,000, electricity price of \$0.15/kWh, high environmental concern (score = 8/10), and faces solar installation cost of \$20,000, we can calculate:

$$\mathbb{P}[\text{Solar}]_i = \frac{\exp(80\hat{\beta}_1 + 0.15\hat{\beta}_2 + 8\hat{\beta}_3 + 20\hat{\beta}_4)}{\exp(\hat{V}_{i1}) + \exp(0) + \exp(\hat{V}_{i3})}$$

i Note 20: Demand for air conditioning

Global warming is a public health hazard in several ways. One is by increasing temperatures to excessive levels, threatening people's health and life (Carleton et al. 2022). Air conditioning (A/C) is in many ways an unsustainable technology, as it may leak greenhouse gases and consumes a lot of electricity (which may come in some countries from fossil fuel). Yet it is also a technology that can improve well-being and, in extreme cases, even life. So what determines the diffusion of A/C?

Davis and Gertler (2015) search for an answer in Mexico. Using household-level data, they look at the predictors of A/C adoption. The first five models are linear probability models, such that:

$$\mathbb{P}[A/C = 1]_i = \alpha + \beta \text{Income}_i + \gamma \text{CDDs}_i + \dots + \varepsilon_i.$$

They estimate these parameters with least squares (they also use a nonlinear version estimated with maximum likelihood). The result is copied in Figure 4.1.

Using column 1, we see that for each increase in household income by 10,000 pesos (about \$690 at the 2015 exchange rate), the probability of having an A/C increased by 12 percentage points. Likewise, for every additional day with temperatures above 65 degrees, that probability increased by 7 percentage points. Columns 7-12 examine heterogeneity of the effect of household income based on whether the municipality in which households lived was above average in terms of heat. The results show that the effect of income was much larger in warm municipalities: an increase in household income of 10,000 pesos increases the probability of having A/C by 3 percentage points in cold municipalities, but by 27 percentage points (3+24) in warm villages.

Interestingly, the paper contained forecasts that the authors revisited ten years later (Davis and Gertler 2025). Using twelve years of additional data, they show that they underestimated the rate at which A/C would diffuse. The main reasons appear to be that electricity became cheaper than expected and A/C's became more efficient.

Table S1. Predicting air conditioning adoption

Variable	Income and climate					
	1	2	3	4	5	6
Annual household income (in US 2010 dollars, 10,000s)	0.12** (0.01)	0.17** (0.02)	0.15** (0.02)	0.15** (0.02)	0.15** (0.02)	0.06** (0.01)
Annual CDDs (Based on 65 °F, 1,000s)	0.07** (0.01)	0.07** (0.01)	0.05** (0.01)	0.04** (0.01)	0.04** (0.01)	0.02** (0.01)
R ²	0.18	0.17	0.32	0.37	0.37	0.46
Income, climate, and the interaction						
Annual household income (in US 2010 dollars, 10,000s)	0.03* (0.01)	0.03** (0.01)	0.03* (0.01)	0.03* (0.01)	0.03* (0.01)	0.04** (0.01)
Annual CDDs (Based on 65 °F, 1,000s)	0.02 (0.02)	0.01 (0.02)	0.04* (0.02)	0.04* (0.02)	0.04 (0.02)	0.02** (0.01)
1(warm municipality)	0.03 (0.06)	0.02 (0.06)	-0.12** (0.04)	-0.14** (0.04)	-0.14** (0.04)	-0.02 (0.02)
Annual household income* 1(warm municipality)	0.24** (0.02)	0.36** (0.03)	0.32** (0.03)	0.31** (0.03)	0.31** (0.03)	0.05** (0.01)
R ²	0.25	0.24	0.36	0.41	0.41	0.46
Proxy for income using expenditure	No	Yes	Yes	Yes	Yes	Yes
Region fixed effects	No	No	Yes	Yes	Yes	Yes
State fixed effects	No	No	No	Yes	Yes	Yes
Quadratic in precipitation	No	No	No	No	Yes	Yes
Probit model	No	No	No	No	No	Yes

Table S1 reports coefficient estimates and SEs from 12 separate regressions. Columns 1–5 report estimates from linear probability models and column 6 reports estimates from probit models. In all regressions the dependent variable is an indicator variable equal to 1 for households who have air conditioning at home. The indicator variable 1(warm municipality) is equal to 1 for municipalities with more than the mean number of annual CDDs. The sample for all regressions includes all 27,395 households in the 2010 ENIGH survey and all regressions are estimated using ENIGH sampling weights. SEs are clustered at the municipality level. Single and double asterisks denote statistical significance at the 5% and 1% levels, respectively.

Figure 4.1: Estimating the determinants of demand for air conditioning in Mexico. Results reported by Davis and Gertler (2015), Table S1. CDD are cooling degree days, the number of days with temperatures above a certain threshold (here: 65 degrees Fahrenheit).

Note 21: Case study: Electric cars

What are the barriers to EVs on the demand side? We can use a model by Springel (2021). This model has two main components in consumer utility:

- **Price of EV**
- **Availability of charging stations**

Should governments subsidize EVs or charging stations? It depends on...

- Consumer's valuation of easy charging (if high: subsidize stations)
- Elasticity of EV demand + cheap alternatives (if cheap: subsidize EVs)
- If station producers have elastic supply (if high: subsidize stations)

4.2.3 Data

Two main data sources exist: **revealed preferences** (inferences from actual behavior) and **stated preferences** (inferences from intentions or survey responses). Revealed preferences are generally preferred, since stated preferences are prone to social desirability bias, inattention, incomplete information, and strategic misreporting. That said, stated preferences are valuable when studying demand for goods that don't yet exist or scenarios not observed in real markets.

4.2.4 Willingness to pay

To quantify the demand for a product, a useful tool is to measure individual (potential) consumers' **willingness to pay** (WTP) (Breidert, Hahsler, and Reutterer 2006; Schmidt and Bijmolt 2020). WTP tells us how much consumers are willing to pay for a given good. For the reasons stated above, relying on stated preferences can be tricky and unreliable. Fortunately, there exist ways to do it using techniques such as the Becker–DeGroot–Marschak (BDM) method (G. M. Becker, DeGroot, and Marschak 1964).

Becker–DeGroot–Marschak (BDM) method for WTP

The G. M. Becker, DeGroot, and Marschak (1964) (BDM) mechanism is a strategy way to measure someone's WTP for a product. The fundamental problem for the analyst who wants to measure WTP is that people might not be truthful when asked straight up how much they would pay for, say, solar panels. BDM is a strategy that should, in principle, ensure truthful responses, thereby facilitating the construction of demand curves.

In BDM, each individual (potential customer) is shown an item and asked to write down the maximum amount of money they would be willing to pay for it. Let's call this value w . After everyone has stated their WTP, a random price p is drawn from a known distribution (for example, uniformly between CHF 0 and CHF 10).

The transaction rule is:

- If $w \geq p$: the participant buys the item and pays p .

- If $w < p$, no transaction occurs.

The selling price p is determined randomly and not influenced by the participant's stated WTP (which would be where misleading answers would matter). Thus, the optimal strategy is to report the true maximum WTP. If a person overstate their w , they risk buying at a price higher than what the item is worth to them. If they understate it, they risk missing a beneficial purchase. Therefore, telling the truth is a dominant strategy.

To implement this in practice, we would do something like this:

1. Explain the rule clearly and run one or two examples so participants see that truthful reporting is best. (It's not completely trivial for someone who has never done it.)
2. Have each participant privately record their w .
3. Randomly draw a price p (e.g., with a software, dice, etc.).
4. Apply the rule and, if real goods are involved, complete the transaction.

If you don't want to be on the hook for potentially up to n sales of the good (one for each participant in the study), then you can possibly transform the last step into a lottery, where the actual transaction depends on some separate random draw.

i Note 22: Problem with stated preferences

When asking people about WTP (or anything, really), we may face an important problem: what people *say* might differ from what they *do*. For instance, I might say that I would pay EUR 30,000 for an EV, but maybe I wouldn't actually do it. Economists therefore often prefer data that is produced by *revealed preferences* (based on actions) over *stated preferences*.

There are several reasons for this discrepancy:

- **Precision.** When asked about your WTP for some technology, you might not give it your full attention. Or it's a technology that you don't really know about. For instance, at the time of writing, most people have only a hazy un-

derstanding of what a heat pump is. How could you give a reliable WTP for something you don't really know? Your answer will therefore reflect your true WTP plus (mean-zero) noise.

- **Social desirability bias.** Suppose you're asked about your WTP on a controversial topic. You might feel pressured into answering in a way that meets society's expectation. For instance, if asked what your WTP is to save seals from clubbing, you might give an overly high number. We talk of *social desirability bias* (Grimm 2010). In contrast to the problem of (im)precision, here your reported WTP will be systematically different from your true WTP.
- **Preference falsification.** This is rarer in the cases that interest us but is worth knowing about. Imagine you live in an authoritarian country and you are asked about your support for some government initiative. You are likely to report incorrect preferences in a way that is aligned with the government. Thus, people sometimes *falsify* their (stated) preferences (Kuran 1995).

Another approach one can use is a **conjoint experiment**, a type of **discrete choice experiment** (P. E. Green and Srinivasan 1990; Orme 2005; Pelz, Aklin, and Urpelainen 2022). The idea behind a conjoint experiment is the following: you ask potential consumers to pick between two (or more) items. You then vary features of these items, including their cost. You can then see, statistically, how response to costs people are. Here, we won't get individual WTP; instead, we obtain group-level WTP.

An example will help. Consider that you want to see whether consumers have a positive WTP for low-emissions vehicles. You would set up the experiment as follows. You would generate two cars with differing colors (red, blue, green), sizes (3 doors, 5 doors), prices (CHF30k, CHF50k, CHF80k), and emissions level (low, high). Put in terms of a utility function, this corresponds to:

$$U_i = \alpha + \sum_c \beta_c \text{Color} + \sum_s \gamma_s \text{Size} + \sum_p \pi_p \text{Price} + \sum_l \lambda_l \text{Emissions} + \varepsilon_i$$

In the experiment, each feature is randomized (e.g., via a survey soft-

ware or on a portable tablet), offering a choice to consumers. For instance, they might be asked to decide between a red, 5-door, low emissions car at CHF50k versus a red, 3-door car, with high emissions at CHF30k. Each consumer then selects the car they prefer. Typically, these pairwise choices are repeated several times for each consumer. At the end, the analyst has a dataset with a series of choices: color, size, price, emissions, and whether this combination was selected (=1) or not (=0). From this dataset, the analyst can use econometric estimation tools (such as ordinary least squares or maximum likelihood) to estimate the vectors β_c , γ_s , λ_l , and π_p . Finally, WTP will simply be a ratio of the effect of a feature over the effect of the price. For instance, WTP for low emissions will be:

$$\text{WTP low emission car} = -\frac{\lambda_l}{\pi_p}.$$

If this quantity is negative, it implies that the consumer would need to be paid to take on a low emissions car. If it's positive, it would mean that they are willing to pay more to get it.

Things become more complicated if we add a third choice (e.g., a 3rd car), or if we include the choice of “neither,” but the general logic remains. Textbooks on discrete choice models are useful starting points for more complicated empirical measurement strategies (Train and Weeks 2005). The advantages of conjoint analysis include:

- It's a realistic setting.
- Social desirability bias is reduced, since nobody can connect your answer to a specific variable.
- It embeds tradeoffs (e.g., “you can have lower pollution levels, but you must pay for it”).

Willingness to pay for electric vehicles

The automotive industry is very interested in learning whether people are willing to pay for electric vehicles (EVs) (Hidrué et al. 2011; Ma, Xu, and Fan 2019). More precisely, they would like to know whether there is a penalty (all things equal, do people prefer internal-combustion engines) and whether there are specific things that people care about (e.g., the range of a battery).

This is the background for the study by Ma, Xu, and Fan (2019). They note that demand-side subsidies for EVs are expensive to

governments. Thus, the question is whether there are other things that are ideally cheaper that could be provided to potential EV users. To assess this, they fielded a discrete choice experiment in which consumers were asked about three potential policies (two new policies and the status quo). A fragment of that experiment is shown below. They examine whether policies that allow shorter charging times, that increase the number of urban charging stations, and that change the driving range would receive support, depending on the additional cost it would impose on consumers. They then estimate several choice models to quantify people's WTP for these policies.

				
Please choose a choice below	Charging time	Urban charging station	Driving range	Additional price (¥,thousand)
Choice 1	Shortening 1/3	Every 3km	/	45
Choice 2	Shortening 1/3	Every 5km	Increasing 100km	65
Choice3 (Status quo)	/	/	/	0

(a)

				
Please choose a choice below	Parking fee in public parking lots	Charging fee in public charging station	Highway toll	Additional price (¥,thousand)
Choice 1	Free of charge	50% off discount	Free of charge	65
Choice 2	Free of charge	/	/	26
Choice3 (Status quo)	/	/	/	0

Figure 4.2: Discrete choice experiment of Ma, Xu, and Fan (2019) (first part).

4.2.5 Demand for public goods

So far, we spoke primarily about private sustainable (or green) goods and technologies. Yet many problems that we care about concern *public goods* and *open access resources*. What does the demand for such goods look like? It is trickier to analyze it, because these are often goods that aren't traded on a conventional market. For instance, you cannot 'buy' a public park, nor can you 'buy' clean air. Later, we will discuss ways to transform these goods into *excludable* ones, but at this stage, we would

like to be able to say something about people's valuation of (or WTP for) such goods.

One approach to handle these types of good is to use **contingent valuation** (CV) techniques. CV is a stated preference method used to estimate the monetary value people assign to non-market goods – such as those we just discussed (public parks, clean air, etc.). Here, respondents are presented with a detailed hypothetical scenario (the *contingent* situation). This would entail either:

- A good to be *produced*, in which case we will assess people's WTP, or
- A good to be *preserved*, in which case we are interested in people's willingness to accept compensation (WTA).

For instance, we could use CV to assess how large compensations need to be for you to accept a polluting factory next to your house. The implementation would typically be done in a survey or an interview in which the scenario is detailed, as is the *payment vehicle* (a tax? a bill surcharge? a one-time fee? something else?). Finally, respondents would be asked their (stated) WTP. This can be done in various ways:

- **Open-ended:** "What is the maximum you would be willing to pay?"
- **Dichotomous choice:** respondents vote "yes" or "no" to paying a specified amount. "Would you be willing to pay CHF30,000 for this EV?"

The dichotomous choice can be expanded to an *iterative bidding* setup: after a "no," you could use a lower amount and see whether people now say "yes" (Boyle and Bishop 1988).

From there, you can do two things:

1. Build a demand curve.
2. Predict what affects someone's WTP (e.g., using a respondent's sociodemographic features).

The challenge with CV is the same as we discussed earlier, which is that it can be a challenge to obtain truthful responses.

Another approach is to use a more conventional WTP measurement strategy such as G. M. Becker, DeGroot, and Marschak (1964). This requires transforming the problem into one in which a private good

is involved. The Box, for instance, shows the problem of indoor air pollution (a public bad) and what people's WTP for air purifying technologies are. This technology is contributing a public good, but it can be exchanged on a market like a private good.

Willingness to pay for indoor air quality

R. D. Metcalfe and Roth (2025) conduct a study in London to measure the impact of knowledge about air quality on people's behavior (like ventilating rooms). In addition – and this is where it's relevant for us – they implement a WTP component to see how much people are willing to pay for an air quality monitor and an air purifier.

To do this, they implement a version of the G. M. Becker, DeGroot, and Marschak (1964) procedure to elicit WTP. Concretely, they do it as follows:

1. They tell participants that they are invited to participate in a lottery to win a GBP 100 Amazon voucher.
2. They are asked how much of this voucher they would be willing to give up to receive an air monitor and an air purifier (e.g., GBP 25). Information about the two items was shared to make sure the choice was informed.
3. Lastly, they told that if they won the voucher, the following would happen. A computer would select a number at random from a predefined list (e.g., 8). If their WTP (GBP 25) was higher than this number (here: 8, so yes, it is), they would receive the monitor + air purifier *and* the difference between their WTP and that number (here: $25 - 8 = \text{GBP } 17$). As a result, they have an incentive to give the highest WTP that they feel comfortable with. This makes them similar to mechanisms that elicit truth-telling as a dominant strategy (Roughgarden 2016).

In practice, they found an average WTP for the monitor and purifier of GBP 39-50 (\$52 to \$67).

4.3 Continuous demand models

4.3.1 From individual choices to aggregate demand

The discrete choice framework operates at the level of a single person facing a finite menu of options. But policy analysis often requires knowing how the *market* responds to a price change — that is, a demand curve. The connection is direct: if there are N consumers each facing the same choice, the expected quantity demanded for option j at price p is

$$D_j(p) = N \cdot \mathbb{P}_{ij}(p) = N \cdot \frac{\exp(V_j(p))}{\sum_k \exp(V_k(p))},$$

which is a demand curve — downward-sloping in p_j (since a higher price reduces V_j and thus $\mathbb{P}_{ij}$) and grounded directly in individual utility maximization.

This aggregation applies to *indivisible* goods where each consumer buys at most one unit (an EV, a solar panel). For *divisible* goods — where consumers choose a continuous quantity (how many kWh, how much fuel) — we use a parallel but distinct framework: utility maximization over quantities, which yields Marshallian demand functions. That is the subject of the next section.

4.3.2 Continuous demand functions

The simplest way to think of *demand* is as a function mapping inputs — especially prices — to quantities desired, derived from consumers' utility functions. Consider a single consumer choosing quantities of two goods, q_1 and q_2 . Let:

- p_1, p_2 be the prices of the two goods,
- $u(q_1, q_2)$ be the utility function (undefined for now, but we will use specific ones later),
- I be the consumer's income, which will operate as a constraint on their spending.

While this may look over simplistic (we make decisions over more than two goods), this problem can already be enlightening. Note that you

can think of good 2 as a composite of all other goods that consumers could ask for.

The consumer solves an optimization problem, namely how much to consume of the two goods while staying within its budget constraint:

$$\max_{q_1, q_2} u(q_1, q_2) \quad \text{subject to} \quad p_1 q_1 + p_2 q_2 \leq I.$$

Assuming the utility function $u(\cdot, \cdot)$ is strictly increasing in both goods, the consumer will spend her entire budget, and thus we can replace the inequality with an equality:

$$p_1 q_1 + p_2 q_2 = I.$$

This is not a trivial assumption. Oftentimes, we do not behave that way. In real life, utility is likely to *decline* at some point. If I buy more solar panels than I have space on my rooftop, the extra solar panels will be burden. But at the same time, it is also true that if your income went up, you would spend at least a fraction of it on consumption (instead of saving all of it), which suggests that we often have some slack.

To find the optimal bundle (q_1^*, q_2^*) , we solve the utility maximization problem using the Lagrange multiplier:

$$\mathcal{L} = u(q_1, q_2) - \lambda(I - p_1 q_1 - p_2 q_2),$$

which yields a system of equations that can be solved analytically.

The **Marshallian demand functions** are the solutions $q_{1,i}^*(p_1, p_2, I)$, $q_{2,i}^*(p_1, p_2, I)$ to the optimization problem by the individual consumer (see Note 23 on terminology). They tell us how much of each good the consumer will buy at given prices and income levels.

We can now aggregate individual demand into an aggregate demand function. In a market, there are many consumers, each with their own utility function and income. The **aggregate demand** for a good is the sum of individual demands:

$$Q_1(p_1, p_2, I_1, \dots, I_N) = \sum_{i=1}^N q_{1,i}(p_1, p_2, I_i)$$

Each consumer i has their own income I_i . Possibly, the underlying utility function could also be consumer-specific. For Cobb-Douglas, for instance, this could mean having a specific taste-parameter (e.g., α_i).

i Note 23: Demand and inverse demand

Note that graphical representations of demand in textbooks generally define the price (y-axis) as a function of quantity (x-axis). See figure 4.3. We thus want to distinguish between:

$$\text{Demand function: } Q = f(P)$$

$$\text{Inverse demand function: } P = g(Q) = f^{-1}(Q)$$

For instance, in the linear case, we have:

$$\text{Demand function: } Q = a - bP$$

$$\text{Inverse demand function: } P = \frac{a - Q}{b} = \frac{a}{b} - \frac{1}{b}Q$$

Notice that the slope of Q in the inverse demand function is the inverse (hence the name) of the slope of P in the demand function.

This can occasionally make things confusing. On the one hand, we generally think of the quantity as a function of the price (*demand function*). On the other hand, we sometimes use shortcuts that refer to the graphical representation of demand (*inverse demand function*). For instance, when we colloquially talk about the “steepness” or “flatness” of the demand curve, we often refer to the slope of Q in the inverse demand function. But note that whether you plot price on the x- or y-axis, the slopes of supply and demand have the same sign.

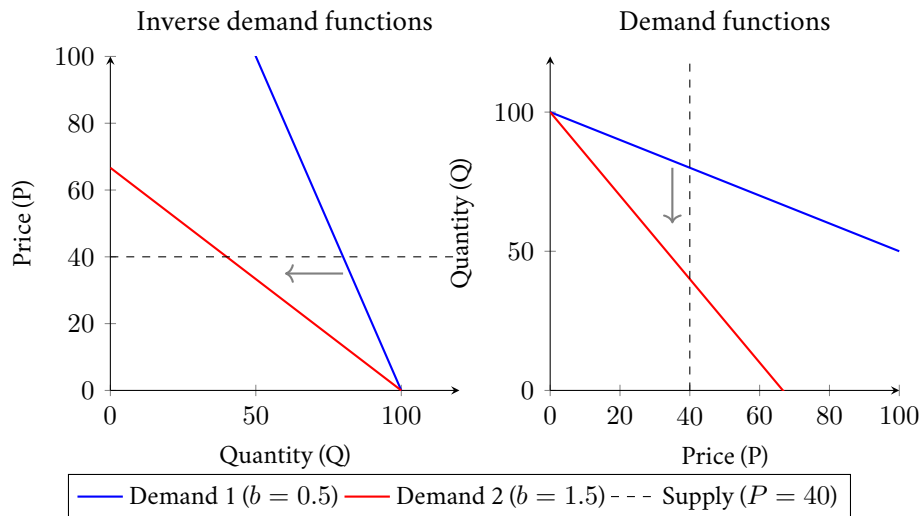


Figure 4.3: Demand and inverse demand with a fixed supply curve (dashed line). We imagine a shift in the steepness of demand, from a steep, inelastic inverse demand (blue curve) to a flatter, elastic inverse demand (red curve). Note that the slopes move in opposite directions in the demand function figure (right panel).

4.3.3 Demand functions and utility functions

We sometimes may want to link demand functions, which are derived from a measure of consumers, to their microfoundations. This is where we come back to the notion of *utility*. In principle, we would like to start with a *utility function* (possibly including some kind of heterogeneity) for a set of N agents (possibly very large). Then, we could try to derive a *demand function* from these individuals. This would allow us to justify the use of a specific demand function, whose parameters could then either be estimated or simulated.

Unfortunately, it's not so simple. Even if every individual has rational, well-behaved *utility functions* (e.g., Cobb-Douglas), the overall *demand function* can behave unpredictably. For example, demand can *increase* when price *increases* if income shifts to consumers who strongly prefer that good. This result is formalized in the **Sonnenschein-Mantel-Debreu theorem**: demand functions can have almost any shape, even if individual utility is well-behaved (Kirman 1992). As a result, when

we talk about demand, we will often need to either assume that there is a representative consumer, or we will use some kind of convenient demand function.

This being said, we can look at the link from (individual) utility to (aggregated) demand. Commonly used utility functions, from which we can derive Marshallian demand curves, are reported in Table 4.1.

Table 4.1: Type of utility functions. Technically, a quasi-linear utility function states that the function of q_1 must be concave. We use log for its simplicity. The implicit logic of the log-linear function is that q_2 captures all alternative goods $-i$. Linear functions are rarely used, since they can yield negative quantities and prices.

Name	Utility function	Marshallian demand (q_1)
Cobb-Douglas	$\prod q_i^{\alpha_i}$	$q_i = \alpha_i \cdot \frac{I}{p_i}$
CES	$[\sum \alpha_i q_i^{\rho}]^{1/\rho}$	$q_i = \frac{\alpha_i^{\sigma} p_i^{1-\sigma}}{\sum \alpha_j^{\sigma} p_j^{1-\sigma}} \cdot \frac{I}{p_i}$
Quasilinear	$\log(q_1) + q_2$	$q_1 = \frac{p_2}{p_1}$
Linear	$aq_1 - \frac{b}{2}q_1^2 + cq_2$	$(\frac{a}{b}) - (\frac{c}{b} \frac{p_1}{p_2})$

See Figure 4.4 to see how these functions differ from each other graphically. One way in which they differ is in the constraints they impose (or not) on elasticity, a notion we return to below.

To make things concrete, let's us use a specific utility function: a Cobb-Douglas utility function:

$$u(q_1, q_2) = q_1^{\alpha} q_2^{1-\alpha}, \quad 0 < \alpha < 1.$$

The parameter α tells us the share of income that goes toward buying q_1 .

Solving the utility maximization problem gives:

$$q_1^* = \alpha \frac{I}{p_1}, \quad q_2^* = (1 - \alpha) \frac{I}{p_2}$$

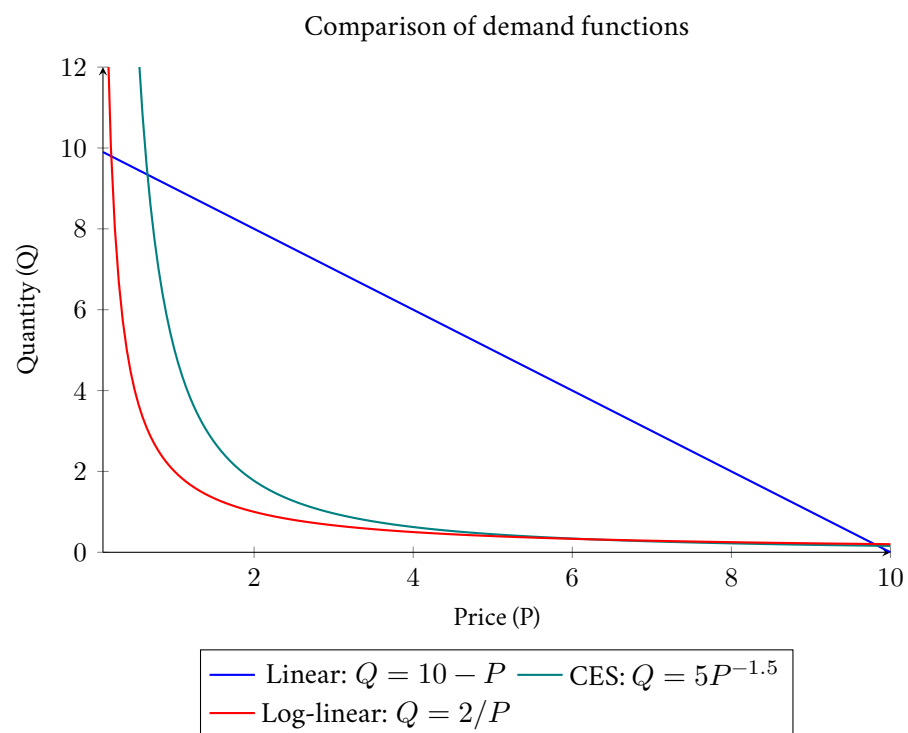


Figure 4.4: Different function forms of demand, depending on various utility functions (linear, constant-elasticity, and log-linear).

These are the Marshallian demand functions: they show how the quantity demanded depends on prices and income.² Note that α represents the consumer's taste for good 1. Note also that here, the consumer always spends a fixed share of income on each good. These equilibrium quantities make sense: q_1 increases in taste (α) and income (I) and decrease in price (p_1).

4.3.4 Elasticity price of demand

Elasticity of demand measures the responsiveness of quantity demanded to price:

$$\varepsilon := - \frac{\% \text{ change in } Q \text{ demanded}}{\% \text{ change in price}} = - \frac{\partial Q}{\partial P} \frac{P}{Q}$$

An elasticity of 1.2 means a 1% price increase reduces quantity demanded by 1.2%.

Elasticity is related to, but not the same as, the steepness of the demand curve, and generally varies along the curve.³ Steep inverse demand curves correspond to **inelastic** demand ($\varepsilon < 1$): price changes have little effect on quantity — typical for necessities. Flat curves correspond to **elastic** demand ($\varepsilon > 1$): consumers respond strongly to price changes, as when easy substitutes exist (e.g., switching from EVs to gasoline cars). The choice of demand functional form (linear, constant elasticity, Cobb-Douglas) has implications: linear demand gives varying ε ; constant-elasticity forms fix it.

Economists typically use the (arbitrary) criterion of 1 as a cutoff to assess whether demand is elastic or not:

- $\varepsilon < 1$: inelastic demand
- $\varepsilon > 1$: elastic demand

Note that non-price factors (e.g., gender, sociodemographics, etc.) can also affect β . We say that consumers have **heterogeneous preferences**. Such heterogeneity can be systematic and vary by groups. For

²We have to assume that prices are strictly different from zero for this expression to work.

³Flat inverse demand curves imply high elasticity (approaching infinity); vertical curves imply zero elasticity. Nonlinear curves have more complex profiles.

instance, a market with more homeowners might have stronger demand for rooftop solar panels (versus a market with more renters, who might not be as interested in this technology). Suppose that we can write the demand as a linear function (for simplicity):

Systematic: $Q_d = \alpha + \beta p_d + \gamma \text{Nbr homeowners} + \lambda p_d \cdot \text{Nbr homeowners}$.

In that case, the marginal effect of price is:

$$\frac{\partial Q_d}{\partial p_d} = \beta + \lambda \text{Nbr homeowners}.$$

Or heterogeneity could be just reflecting the fundamental randomness and variety in consumer demand. Using subscript i for consumers we could write:

Random: $Q_d = \alpha_i + \beta_i P_d$ where $\beta_i \sim \mathcal{D}$,

for some distribution $\mathcal{D}$ (e.g., normal) that could take other variables as inputs (age, gender, etc.). Aside from reminding us that people are complicated, heterogeneity has implications for how we can measure demand. More on this later.

What, then, determines elasticity and β ? There are several factors that are relevant:

- **Tastes.** If people really like a good, demand will be inelastic. Tastes can be stochastic and/or correlated with other variables (age, gender, geography, etc.).
- **Substitutes.** If people can easily switch to another good that offers the same services, then demand will be elastic.
- **Time horizon.** In the short run, consumers may find it hard to identify substitutes, making short-term demand inelastic.

Consider a rural population that commutes to work: initially, car demand is inelastic (no alternatives). If local authorities create public transit, demand becomes more elastic as an outside option emerges.

Elasticity matters for both business and policy. A firm selling wind turbines needs to know whether a price increase will significantly depress sales. A policymaker imposing a **Pigouvian tax** needs to know whether demand is elastic enough to reduce harmful consumption — for inelastic demand, only very large taxes will be effective.

4.4 Demand with network externalities

The models so far treat demand as independent individual decisions. But for some goods, demand is interdependent: **network externalities** arise when someone's valuation depends on how many others are using the same good (Easley and Kleinberg 2010). Examples include social media (Twitter's value depends on who is on it), EVs (charging infrastructure grows with the user base) (Springel 2021), electricity conservation (peer effects) (Allcott 2011), and solar adoption (O'Shaughnessy, Grayson, and Barbose 2023). These all share a strong coordination problem structure.

4.4.1 A model of network externalities

To see how this works, let's first write down a baseline model without network externalities.⁴ We have a continuum of consumers who must decide whether to buy a single good (e.g., a solar rooftop panel). We index the set of consumers by $x \in [0, 1]$, which we sort from most to least interested in this good. As a result, x can be interpreted as a share. Define $r(x)$ as the willingness to pay (or *reserve price*) for this good of consumer x . $r(x)$ is assumed to be continuous and strictly decreasing (since we ordered potential customers starting with those most interested).

The supply side is given by a flat market price p , a constant. This means that the demand cannot change the price (consumers are price-takers). At p , all individuals x such that $r(x) > p$ buy the good (these are people who value the good more than it costs). Define the share of consumers buying the good as x^* . This standard market is illustrated in Figure 4.5.

So far, this is in line with what we already saw.

Let us now add network externalities. Assume that willingness to pay becomes $r(x)f(z)$, where:

- Reserve price $r(x)$: like before (taste, etc.)
- Network effect $f(z)$: benefit to x of a share z of consumers using this good

⁴This model builds heavily on Easley and Kleinberg (2010).

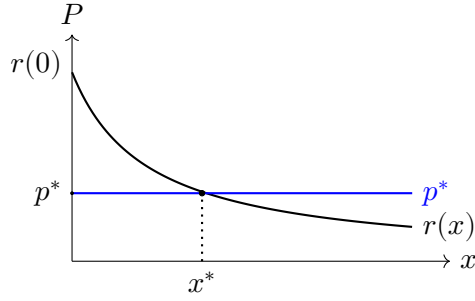


Figure 4.5: Market without network externalities. Since each agent x buys at most one unit of the good, the x -axis corresponds to quantity (or, more precisely, the share of potential consumers who buy the good). The supply side is a flat price p^* .

The function $f(z)$ captures the network effect: it says how a potential consumer's utility changes as a function of other users z . We assume that $f(z)$ is increasing in z (more users, the more value the good has), continuous, and that $f(0) = 0$. If nobody uses it, then the good has no value. For instance, if nobody has a phone, the value to you of having one is nil. Then, the consumer buys the good if:

$$r(x)f(z) \geq p^*.$$

Assuming that we know z and $p^* > 0$, the question is: **what will the equilibrium x^* be?** We have two cases. The first case is $z = 0$: then $r(x)f(0) = 0$ (by assumption). The good isn't sold, since nobody sees value in it.

What is $z > 0$? If Consumer $\tilde{x}$ buys the good, then all $x < \tilde{x}$ buy it too (since x is ordered from most to least interested). Thus: buyers will be the set $[0, z]$ (all x below z). And thus: the equilibrium level of demand will be $p^* = r(z)f(z)$.

In practice, the result will depend on $r(\cdot)$ and $f(\cdot)$. Let us pick two concrete functions and see what the model yields:

- Reserve price: $r(x) = 1 - x$ (i.e., individual 0 is willing to pay 1 unit of currency, individual 0.1 is willing to pay 0.9, etc.)
- Network: $f(z) = z$

Thus, we have a demand curve, such that:

$$r(z)f(z) = z(1 - z).$$

The demand curve is quadratic now as in Figure 4.6, instead of being strictly declining like in Figure 4.5. As a result, there are several feasible equilibria (assuming that $p \leq \max r(z)f(z)$):

$$\begin{aligned} &(p^*, z') \\ &(p^*, z'') \\ &(0, 0) \end{aligned}$$

But note: these equilibria are not all equally stable to small disturbances. One equilibrium is **unstable** and two are **stable**. To see this, think about what would happen to this market if we generate small departures from any z' or z'' :

- If $z < z'$: there will be a downward spiral to 0 (the network too weak, too few users)
- If $z > z'$ but $z < z''$: there will be an upward spiral to z'' (growing 'hype' or 'viral' dynamic)
- If $z > z''$: there will be a downward spiral back to z'' (the last few potential customers are really not interested)

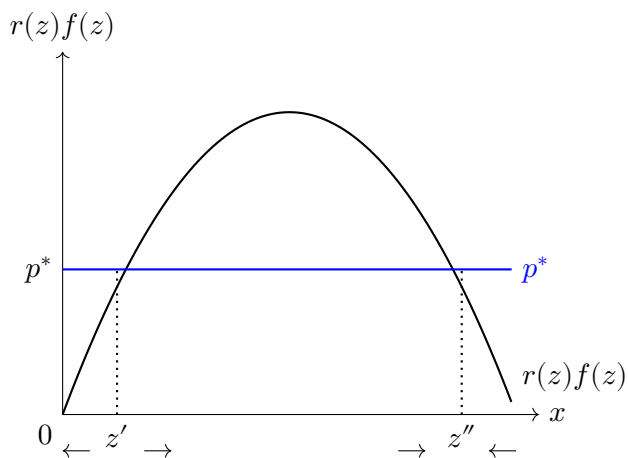


Figure 4.6: Multiple equilibria as a result of network externalities. Arrows show dynamics in case of small departures from z' and from z'' .

Thus, the $(0, 0)$ and (p^*, z'') equilibria are *stable*: a small shock does not affect them. In contrast, (p^*, z') is *unstable*: a small departure shifts consumers to a different equilibrium. What is critical here is that z' is

a **tipping point**. This has immediate policy implications: if good is desirable, the policymaker must try to push demand above z' . Otherwise, there is a risk of stagnation and collapse.

4.4.2 Policy implications

Goods with network externalities raise several key policy implications.

First, policymakers can try to manage equilibria. In “conventional” markets, the key is to correct for market failures, e.g. via Pigouvian pricing. Networks, in contrast, can have multiple equilibria (Morris and Shin 2001). This creates an opportunity for policymakers who can try to direct consumers toward a ‘superior’ equilibrium (however defined). Policies (such as taxes or subsidies) can generate *virtuous* or *vicious* cycles (Krueger 1993). Note that *calibration* of policies becomes especially important. For instance, if you start at $(0, 0)$, a weak policy intervention will be wasted, since once it stops, the market will drift back to $(0, 0)$.

Second, network externalities can create a strong **first-mover advantage**. Consider two goods, A and B . Let’s suppose that B is better from an environmental perspective, but it was marketed one year after good A . If good A is able to reach an equilibrium like z'' in Figure 4.6, it will be extremely hard to dislodge it. This is a **coordination problem**. Once consumers coordinated on a good, absent any impetus to change *en masse*, there is no reason to expect that a new good that starts at $(0, 0)$ will be able to jump out of its trap. This has several implications. For instance, it means that policymakers must be extremely careful with technologies that seem appealing “bridge” technologies but could instead lock in. Examples of first-mover advantages range from social media (Twitter) to transportation (ICE versus EV) to energy infrastructure.

4.5 Demand for green technology, goods, and services

4.5.1 Tech adoption: individual and social acceptance

At this stage, we can start using demand curves as analytical tools to think about potential policy interventions. Some implications are self evident, but regardless, they help organize thoughts. For instance, we can distinguish between shifts *along* the demand curve and shifts *of* the demand curve:

- In general, a positive supply-side shock will increase quantity and reduce price (i.e., move along the downward-sloping demand curve). One exception, which we did not discuss, are **Veblen goods**, in which higher prices makes the good more prestigious; for such goods, the slope of demand goes upward.
- If a lot of consumers start to care more about the environment, then the demand curve for green goods should shift upward (i.e., we don't move along the demand curve; it's the curve that moves). The market is then anticipated to respond to this higher demand by producing more of such goods. In contrast, 'brown' goods should experience a decline. The idea that consumers value the environment is hotly debated. In finance, analysts wonder about the existence of a "greenium," that is, a green premium for green assets (Larcker and Watts 2020). In other words: are investors willing to pay more for such an assets? At the time of writing, empirical evidence is mixed (Lau et al. 2022). Similarly, there has been a number of studies to assess whether consumers are willing to pay more for green (decarbonized) electricity, with mixed results as well (Roe et al. 2001; Borchers, Duke, and Parsons 2007; Sundt and Rehdanz 2015).

A well-defined demand curve is a useful analytical tool to prioritize policies and interventions. For instance, it helps us find out how to strengthen demand for EV most effectively. However, a demand curve tells us something about an aggregate set of actors; it doesn't tell us much about the adoption decision itself at the *individual* level. By "individual," we mean people, households, firms, or any other agent.

There has been considerable work to understand the logic of technology adoption by firms in particular. The reason goes back to growth

models we looked at earlier: it is hard to explain variation in income at the macro level (country). Access to technology probably explains a substantial share of the cross-national variation in output (Comin and Hobijn 2004). As a result, people have been very interested in understanding what drives technology adoption by firms at the micro level, especially in the agricultural sector (Griliches 1957; Besley and Case 1993; Foster and Rosenzweig 2010). If technology helps increase a firm's output, then presumably firms will be on the lookout for new ideas and constantly try to innovate (in-house) or adopt technologies developed elsewhere.

Yet this doesn't have to be the case. New technologies are sometimes slow to be adopted. There are several potential reasons for this:

- A technology might generate benefits that aren't **excludable**. That's the problem of public goods. For instance, a firm might be reluctant to adopt lightbulbs that consume less electricity since it won't take into account the climatic benefits of doing so.
- There is **supermodularity** (or **complementarity**) problems (Milgrom and Roberts 1990): adding a technology to a firm that is otherwise operating well can suddenly change the returns on other inputs. Put another way: a firm might lose from only adding a single tech; to gain, it might need to adopt the tech *and* implement other changes at the same time. That's for instance what might have slowed down AT&T in adopting automatized phone transfers (see Box).
- The technology is **inappropriate** (Moscona and Sastry 2025). Innovations fit certain environments better than others. For instance, a new type of farming equipment might be more effective in a highly motorized country. The returns on a technology might be too low to justify its adoption.⁵

⁵The difference between complementarity and inappropriateness can be hard to see. If your technology is fundamentally inadapted to a particular context, then the problem is probably one of appropriateness. If the technology could be effective if some other variable changes, then it suggests that the problem is one of complementarity/supermodularity.

Adopting a new technology: the case of phone operators

A case of slow tech adoption comes from AT&T, which served from 1913 onward as a *de facto* monopoly for phone network services in the United States. In a fascinating study, Feigenbaum and Gross (2024b) document how slow its transition from human to mechanical phone switching has been.

Here's the backstory: for a long time, phone calls required a person to connect the caller to the callee. This intermediary was the phone operator and their main task was to do 'call switching'. This system was fairly complex and labor-intensive. Some operators took calls, some operators then called the destination, some operators were here to help customers find numbers, and some operators were here to troubleshoot. AT&T used this system until it gradually replaced it with a mechanical switching technology. What is remarkable is that a mechanical switching technology had been developed in the 1880/1890s. Yet the last human-based operators were only discontinued in 1978.

Why did this take so long? This is especially puzzling since manual switching becomes very quickly complex. As Feigenbaum and Gross (2024b) note, if you have 45,000 customers, the number of potential connections is greater than 1 billion. In addition, getting qualified operators to handle these calls became increasingly expensive. So this should pave the way for rapid automation. Yet it took decades for AT&T to become entirely automatized. Why?

Feigenbaum and Gross (2024b) argue that the problem was organizational: AT&T was entirely organized around operators. They weren't just switching calls. They were also troubleshooting, they were monitoring the duration of calls (essential for billing), and they served as customer service. Thus, replacing operators by machines would maybe help increase efficiency of phone switching, but it would create issues across several other parts of the firm – including getting revenues via phone bills! Worse, automatized switching didn't fit well with AT&T's infrastructure. For instance, when long distance calls were made, the customer would say where they wanted to call ("Pittsburgh Pennsylvania") followed by a four-digit number of the person they wanted to call. Since automatized switching required a rotary dial, it could only handle numbers. Yet another barrier! And finally, customers too

would have to adapt, which wasn't a given.

What about the impact of this technology on jobs? Phone switching was labor intensive and thus employed a lot of people (especially women). In a related study, Feigenbaum and Gross (2024a) show that the *aggregate* effect was nil: overall employment was not reduced in areas that adopted automated switching. New jobs at both medium and low skills (e.g., clerks) picked up the slack. However, the *micro* effect was uneven. People who were phone operators who lost their job often went into lower-paying jobs and sometimes exited the labor market altogether.

At the individual level too, considerable research has been conducted to understand what drives people to adopt – or not – a new technology. For instance, Aklin et al. (2018) analyze what drives the adoption of solar microgrids (see Box). Adoption is typically shaped by the features of the decision-maker and their environment. Here, a particularly important idea is the notion of **social acceptance** of new technologies (Musall and Kuik 2011; Jobert, Laborne, and Mimler 2007; Aklin, Cheng, and Urpelainen 2018; Wüstenhagen, Wolsink, and Bürer 2007). Whether a new technology is adopted isn't solely a function of its price. Broadly, we can identify two sets of problems that might appear.

1. Technologies, even clean ones, have negative (economic) externalities on some constituencies. For instance, a wind turbine might reduce the value of a house located nearby (e.g., worse view, noise). This encourages affected stakeholders to mobilize against it – a phenomenon that is sometimes called **NIMBYism** (*not in my backyard*: even if you are in favor of something, you want it somewhere else). If the legal and political system is friendly to these agents, then technology deployment can be slow. In Switzerland, for instance, there were only 12 wind turbines in operation as of 2022 and about 40 as of 2026!⁶
2. Technologies may be seen as undesirable by the community for non-pecuniary reasons. For instance, communities might disagree with siting of new energy infrastructure and feel like the selection process was not fair nor inclusive. New energy infrastructure, for instance, often generates such responses

⁶Wind energy, SFOE, <https://www.bfe.admin.ch/bfe/en/home/supply/renewable-energy/wind-energy.html/#:~:text=There%20are%20currently%20almost%2040,total%20output%20of%2037.2%20megawatts>.

(Wüstenhagen, Wolsink, and Bürer 2007). In a similar vein, Bullard (2018) developed the notion of **environmental racism**: polluting industries are often located near poor and marginalized communities.

Demand for solar microgrids in India

Aklin et al. (2018) study the determinants of adoption of solar microgrids in the state of Uttar Pradesh in India. They implemented a randomized controlled trial with a local solar company called Mera Gao Power (MGP). The main study was designed to estimate the effect of these microgrids on energy consumption (Aklin et al. 2017). These data allowed the research team to study who, among those presented with Mera Gao Power's products, would acquire such a microgrid. Aklin et al. (2018) then estimates the effect of a range of household and respondent features on this decision.

The main results are reported in Figure 4.7 and summarized as:

Richer households are indeed more likely to adopt solar power, and at highly statistically significant levels ($p < 0.05$ throughout all models). Using monthly household expenditures and savings as measures for disposable income, we find that doubling these two variables increases the odds (ratio of success to failure) of technology adoption (holding all variables at their means) by 129–159% for household expenditures and by 19–23% for increases in savings [...]. While we do not find evidence for a robust association between household business and MGP service—most certainly due to little variation in business ownership, with only 8.4% running their own business in our pre-treatment survey—there is strong empirical support that more entrepreneurially minded households are more likely to sign up for solar microgrids. When the entrepreneurial index goes up by one unit, the odds of adoption increase by 19–26%.

Logistic panel regressions: Technology adoption				
	Full sample			
	(1) Model	(2) Model	(3) Model	(4) Model
HH expenditures (rupees/month, log)	2.59*** (0.77)	2.41*** (0.74)	2.29*** (0.71)	2.61** (1.21)
HH savings (rupees/month, log)	1.19*** (0.06)	1.19*** (0.06)	1.19*** (0.05)	1.22*** (0.07)
HH in debt	0.73 (0.33)	0.71 (0.32)	0.70 (0.33)	0.83 (0.44)
Entrepreneurial spirit		1.19*** (0.08)	1.23*** (0.08)	1.33*** (0.12)
Business owner		1.13 (0.75)	1.09 (0.73)	0.82 (0.71)
Risk aversion (experiment)		0.88 (0.12)	0.85 (0.11)	0.95 (0.13)
Trust in people (index)			0.74 (0.17)	0.65* (0.16)
Trust in firms (index)			0.80 (0.17)	0.74 (0.18)
Kerosene spending (PDS, rupees/month, log)				0.93 (0.15)
Kerosene spending (private, rupees/month, log)				1.27 (0.21)
HH electrification				1.28 (2.03)
HH size (#)				0.88 (0.09)
Age of HH head				0.99 (0.02)
School years of HH head				0.87** (0.05)
Scheduled caste				1.53 (2.28)
Backward caste				9.26** (8.87)
Lighting satisfaction				0.21 (0.35)
Solar power decreases lighting cost				4.21** (2.68)
Observations	1530	1530	1530	1530
Habitations	49	49	49	49

Dependent variable: MGP adoption.
Standard errors in parentheses and clustered by habitation.
All explanatory variables are measured pre-treatment.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 4.7: Determinants of adoption of a solar microgrid. Estimates from Aklin et al. (2018, 44), Table 8. The dependent variable equals 1 if the household acquired the solar microgrid and zero otherwise.

4.5.2 Demand policy

Demand-side policies aim to shift the demand curve — upward for green goods, downward for brown ones — and may also change its elasticity. A simple diagnostic: if a policy aiming to increase quantities causes both price and quantity to rise, it is demand-side; if price falls while quantity rises, it is supply-side.

The first family of policies is *informative* in nature, but the mechanisms through which they operate ranges from rational to emotional.

Factual information. Providing information to consumers can reduce uncertainty and therefore strengthen demand for a green good. For instance, we know that individuals have “range-phobia” when considering electric vehicles. Thus, providing ways to publicize longer ranges of EVs or publicizing the number of charging stations can help

tackle this. One way in which information can be provided is by **labeling** goods. See Note 24.

i Note 24: Fair trade labeling

Hainmueller, Hiscox, and Sequeira (2015) implement two experiments in 26 stores of a grocery chain located in the northeast of the US in 2008 and 2009. First, they randomized whether two popular coffees (French Roast and Coffee Blend) were labeled “Fair Trade” or not, and examined how it affected their sales. They find that this label significantly increased demand (Figure 4.8):

We find that sales of the two most popular coffees rose by almost 10% when they carried a Fair Trade label as compared to a generic placebo label.

TABLE 4.—EFFECT OF FAIR TRADE LABEL ON SALES OF TEST COFFEES

Model	(1)	(2)	(3)
Coffee	FR Regular and		
Dependent Variable	Coffee Blend	FR Regular	Coffee Blend
	Log Sales	Log Sales	Log Sales
Fair Trade label	0.079 (0.033)	0.072 (0.042)	0.122 (0.052)
Constant	5.281 (0.046)	4.853 (0.076)	3.861 (0.117)
Store FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Observations	208	208	207

Models 1–3 display regression coefficients with robust clustered standard errors in parentheses. The unit of analysis is a store week. The dependent variable in the regressions is the logged weekly dollar sales of the test coffees, FR Regular and CB. Model 1 refers to the combined sales of both test coffees; Models 2 and 3 refer to sales of the FR Regular and the CB, respectively. The independent variable is a treatment indicator coded as 1 for the four weeks in which the Fair Trade label was placed on the test coffees and 0 for the four weeks when the generic label was placed on the test coffees. The design is a fully balanced crossover experiment so each of the 26 stores is observed for four weeks under each condition (eight weeks in total). All models include a full set of store and week fixed effects.

Figure 4.8: Effect of labeling on weekly coffee sales. *FR* stands for French Roast, one of the two types of coffees treated with the Fair Trade label (the other being Coffee Blend). Results from Hainmueller, Hiscox, and Sequeira (2015, 250), Table 4.

Second, they conducted a price experiment. They increased the price of both types of coffee by \$1 to see how demand would

change (that is, they looked at the elasticity price of demand) once the “Fair Trade” label was assigned. They find:

Demand for the higher-priced coffee was less elastic: sales of the Fair Trade-labeled coffee remained fairly steady when its price was raised by 8%. Demand for the lower-priced coffee was more elastic: a 9% increase in its price led to a 30% decline in sales as buyers switched to low-priced unlabeled alternatives.

People who consume expensive coffee (French Roast) were willing to pay even more for it when it was labeled Fair Trade without reducing consumption. People who bought the cheaper coffee (Coffee Blend) switched to other types of coffee instead.

Norms. Policymakers can provide information to create norms (“littering is bad”). Here, policy operates via its effect on utility and disutility from various types of behavior. Norms affect how individuals “feel” about consumption decisions. Transgressing a norm imposes a utility penalty that may make certain types of behaviors less desirable (e.g., littering is bad). Likewise, there might be benefits associated to complying with norms.

Policies that try to modify the demand side have been applied to domains such as water (Ferraro, Miranda, and Price 2011) and electricity consumption (Allcott 2011). We discuss studies evaluating the impact of these policies in two Note 25 and Note 26.

i Note 25: Norms and water consumption

In principle, one would expect *factual* information to affect how people behave. In practice, *norms* seem just as important. Ferraro, Miranda, and Price (2011) implement a field experiment on water use in Atlanta, Georgia (USA). They sent letters to a water utility’s customers. These letters randomized treatments and examined if it changed their behavior in the short- and long-term. The treatments included:

- A control group (no message).
- *T1*: Technical (factual) advice.
- *T2*: Technical advice and an appeal to prosocial prefer-

ences.

- T3: An appeal to prosocial preferences that included a social comparison to other households (“your neighbor’s average consumption last month was... You consumed more than x percent of your neighbors.”).

They summarize their findings (p.319):

While appeals to prosocial preferences and social comparisons affect short-term patterns of water use, only messages augmented with social comparisons have a lasting impact on water demand. Within a year of treatment assignment, we are unable to detect meaningful differences in use across households who received an appeal to prosocial preferences and counterparts in the control group. In contrast, impacts from the social comparison treatment can be detected more than two years after the message was sent.

The letter was sent in May 2007. The effect of T3 was a reduction of about -1,739 gallons of water (~6,600 liters) by household in the summer of 2007, and -350 gallons during the summer of 2009 (1,300 liters).

i Note 26: Norms and electricity consumption

Allcott (2011) worked with OPOWER, a company that promotes energy conservation, to find ways to reduce electricity consumption. He implemented a randomized controlled trial in which customers were randomly sent letters. These letters sometimes included comparisons between their power consumption and that of their neighbors.

Allcott summarizes his findings as follows:

The program provides additional evidence that non-price interventions can substantially and cost effectively change consumer behavior: the effect is equivalent to that of a short-run electricity price increase of 11 to 20%, and the cost effectiveness compares favorably to that of traditional energy conservation programs.

What is striking is that the reduction in consumption was done, presumably, without much loss of welfare. Since these reductions were voluntary, the only effect is likely to be the psychological impact of feeling bad compared to one's neighbors. This would not be the case if policymakers had instead increased electricity prices via a Pigouvian tax.

What is also interesting is that this intervention could have backfired (a so-called "boomerang effect"): if low-consumption households realize that they are performing better than their neighbors, wouldn't they increase their consumption levels? The answer is no:

Perhaps because the treatment included descriptive social norms, effects are heterogeneous: households in the highest decile of pre-treatment consumption decrease usage by 6.3%, while consumption by the lowest decile decreases by only 0.3%.

Thus, everyone reduced consumption of electricity, regardless of whether they were above or below the average user.

Framing/priming. Policymakers can use informational campaigns to raise the saliency of topics (Kahneman and Tversky 1984; Chong and Druckman 2007). In contrast to norms, we don't focus on a deep change in preferences. Instead, we try to increase the importance of some factors over others. Alternatively, framing can be done by describing a situation in various ways. Hong, Hossain, and List (2015) for instance implements an experiment in which rewards are sometimes described as *rewards* and sometimes as *punishment*: if people fear embarrassment more, then they might respond more strongly to the latter.

Typically, framing policy is done by sending messages (via ads, speeches, policy manifestos, etc.) that emphasize certain aspects of problems or of technologies. In the case of environmental policy, it is sometimes believed that one can shift preferences by emphasizing the fact that climate policy is about national security. The effectiveness of such framing policies is often questioned. A study on solar PV for instance found limited effects on people's preferences (Wolske et al. 2018). And even if a policymaker finds an effective frame, opponents can neutralize them with counter-frames (Aklin and Urpelainen

2013a).

Vouchers. Another type of policy is less direct: it is about loosening the constraints faced by consumers. This could be achieved, for instance, by increasing people's income. Of course, increasing people's income isn't easy (otherwise it would have been done already). But you can think of a simpler version, which is to hand out *vouchers* or similar instruments to consumers. For instance, by giving people a voucher to buy electric bikes, the policymaker actually strengthens demand.

Note: to be a strictly demand-side policy, vouchers (or any policies) must not affecting production costs and the supply side more generally.

Tax credits. Consumers are incentivized to buy by compensating part or all of their expenses via lower taxes. See Note 27 for an example. What is appealing with tax credits is that you receive (part of) something (e.g., an EV), while still benefitting from whatever public goods are produced with other people's taxes. For instance, if you can get a tax credit by adding solar panels to your rooftop, you benefit from this capital investment in your house. The money diverted from taxes benefits you. And this comes at (almost) no cost, since you still benefit from services like firefighters, law enforcement, etc. ("almost" because total tax revenues go down and therefore the level of services could suffer).

 Tip 27: US electric vehicle tax credit

The US federal government has attempted to support the deployment of electric vehicles via tax credits. These have changed occasionally. At the time of writing, the rule was the following [\[link\]](#):

- You buy an EV or a fuel cell vehicle from a licensed dealer for \$25,000 or less.
- You get 30% of the bill back as tax credit. For instance, if you paid \$20,000, you receive a credit of \$6,000. (In practice, the credit was capped at \$4,000.)
- When comes the time to pay your federal income taxes, suppose you owe \$10,000. You can remove \$6,000 from that amount. The taxes you owe is \$4,000.
- If you owed less than your credit, your taxes go to \$0 (that is, you aren't given money back by the government).

Additional readings

Students interested in more complex models of demand should check out microeconomics textbooks such as Kreps (2020), Mas-Colell, Whinston, and Green (1995), Jehle and Reny (2001), Kreps (2013), or Rubinstein (2022). On social networks, see Jackson (2010) and Easley and Kleinberg (2010).

5 Supply-side problems

This version: September 28, 2026

Summary

New technologies must be imagined, produced, and then scaled up. There are many things that can go wrong on the way.

5.1 Emergence of new technologies and goods

This chapter explores the way in which the **supply** side of the economy behaves. It will blend microeconomic perspectives (e.g., models of firm behavior), evolutionary economics, management, and public policy.

The starting point is a crude distinction between the different stages of development of new technologies and goods. Figure 5.1 splits down the path from fundamental research (the emergence of ideas) to its completion (complete market penetration of a technology or good).

If (and that is a big ‘if’) a good succeeds, then it will generally experience a trajectory as in Figure 5.2. After successful research and development stages, the product will be commercialized among early adopters (‘trendsetters’). It will then become popular among the general population, marking a rapid growth phase. Finally, adoption of this good will slow down as it reaches the last remaining segments of the market.

Note that neither the S-shape nor eventual success is guaranteed. Simplifying, failure can happen in two ways:

1. The path is an inverted-U: the deployment of the product collapses after initial liftoff.

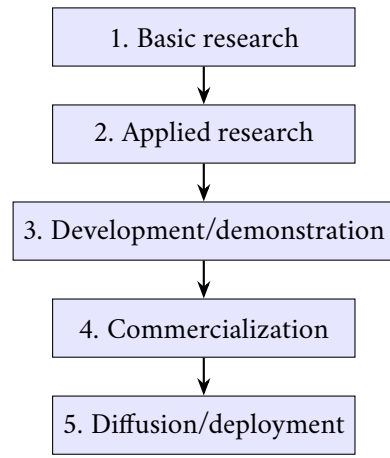


Figure 5.1: Stages of (green) product development.

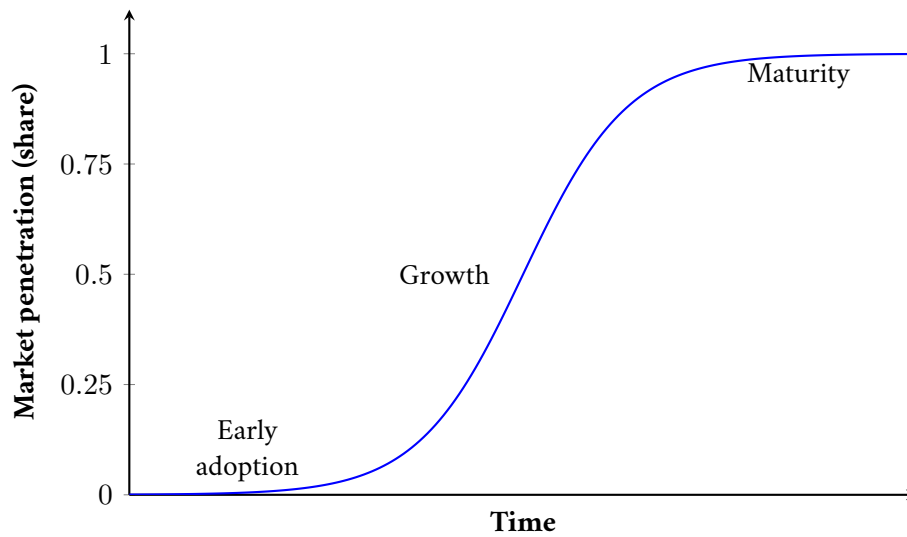


Figure 5.2: S-curve of product adoption.

2. The path is S-shaped, but maxes out at a low level of market penetration.

The causes of failures are many. They may be technical, economic, or demand-side (e.g., poor product-market fit). Most relevant here are **collective action failures**, where public policy can help. Table 5.1 summarizes the key challenges at each stage and corresponding policy instruments.

Table 5.1: Stages of product development.

Stage	Key failures	Policy instruments
1. Basic research	Public good	Public R&D, universities
2. Applied research	Knowledge spillovers, uncertainty, time horizon	Mission-oriented R&D
3. Demonstration	“Valley of death”, capital constraints, coordination failures	ARPA; innovation funds
4. Commercialization	Externalities, information asymmetries	Carbon pricing, tax credits
5. Diffusion	Path dependence, network effects, policy commitment	Feed-in tariffs

5.2 Basic and applied research

Where do innovations come from? In a narrow sense, most innovations come from forms of research and development (R&D). Recall the Solow-Swan model: the parameter A dictates how the law of diminishing returns can be loosened. Thus, the question is to understand what drives higher A .

- Curiosity/serendipity/randomness of evolutionary processes (J. S. Metcalfe 1994)
- Endogenous growth theory (Romer 1990, 1994; Grossman and Helpman 1991) (but see Charles I. Jones (1995))

5.2.1 Innovation and learning

Learning and innovation are crucial for welfare. In Solow-Swan and endogenous growth models, output growth comes from technological change; replacing polluting technologies with cleaner ones requires the same. Innovation refers to (1) the development of new goods and services, or (2) changes in production technology for existing ones. Tablets were mostly new products; electric cars delivered an existing service (mobility) via a new technology.

Learning refers more specifically to improvements in producing a given good — identifying more efficient processes, cutting costs, reducing waste. Since learning rarely reverses, technologies follow a *learning curve*: costs fall as deployment accumulates, driven by economies of scale and experience. For solar cells, this pattern is called *Swanson's law*, after SunPower founder Richard Swanson (Herche 2017).¹ Learning occurs through doing (Kenneth Joseph Arrow 1971) and through imitation (Strang 2010) — observing competitors and applying their insights.

What, then, are the impediments to learning and innovation? We focus on three critical problems:

1. Positive externalities and spillover of knowledge: learning and innovations are public goods.
2. Coordination failures: learning and innovations require sometimes coordinated efforts when developing novel products
3. Internal politics and hostility to organizational change

i Note 29: Jevons' paradox and the rebound effect

The idea that innovation can help us reduce our environmental footprint is at the core of the idea of sustainable development (Robert M. Solow 1991). This, however, may vary by context. One problem is **Jevons' paradox**, which brings in the role of elasticity of demand (which we saw earlier).

The idea is the following: suppose that you find a way to produce a good in a way that requires fewer inputs. This should be good from an environmental standpoint, right? Not necessarily:

¹Swanson's law echoes *Moore's law*: the log of transistor count in integrated circuits grew linearly in time (G. E. Moore 1965).

this innovation may also decrease the cost of production. This shifts the (inverse) supply curve down, reducing the price *and* increasing quantity. If demand is highly elastic (the inverse demand is flat), the new equilibrium quantity could be much higher and erase the gains made by technological efficiency gains.

This connects to the problem of **rebound**, which is often discussed in energy economics. Suppose you find a way to reduce waste in the use of fuel for an engine. If consumption of said fuel doesn't decline by a comparable proportion, then part of the gains was erased by increased demand. But that's not the only mechanism creating a rebound. Gillingham, Rapson, and Wagner (2016) list four ways in which a rebound effect can be created:

- The shift along the demand curve caused by efficiency gains, as per Jevons' paradox.
- The rightward shift of the supply curve reduces the equilibrium price. As a consumer, you can spend the saved money on other goods that will need energy and possibly create externalities.
- Even if innovations lead to a decline energy needs in a large market (e.g., the US), it will increase consumption in other countries.
- Technological innovations can trigger local growth via geographic spillover effects.

One crude way to compute the rebound effect is:

$$\text{Rebound} = \frac{\Delta \text{Gain (pp)} - \Delta \text{Consumption (pp)}}{\Delta \text{Gain (pp)}}.$$

For instance, if you design an engine that uses 10% less kerosene, but demand for kerosene only decline by 2%, we would have:

$$\text{Rebound} = \frac{10 - 2}{10} = 0.8.$$

- A *Rebound* of 0 means that technological gains are entirely passed onto lower demand.
- A *Rebound* score of 1 means that all gains are erased.
- A *Rebound* greater than 1 corresponds to Jevons' paradox (also known as the 'backfire' effect).

What does the data say? Two studies review the state of knowledge regarding the existence of a rebound connected to energy efficiency gains (Gillingham et al. 2013; Gillingham, Rapson, and Wagner 2016). Gillingham, Rapson, and Wagner (2016, 476) find:

Macroeconomic models estimate total combined rebound effects to be in the range of 20–60%

Not trivial, but far from suggesting a backfire effect. Thus, energy efficiency measures are not undermining themselves.

5.2.2 Positive externalities and spillovers

The first problem in innovation is connected to **positive externalities**. A firm that innovates may see part of its contribution spill over to other firms. In turn, since the returns to research are lower than they ought to be, the firm might decide to invest fewer resources into research. This is a classical public good problem (Nelson 1959; Kenneth Joseph Arrow 1971; Griliches 1991).

To show this, we build a simple model of research (or R&D) spillovers, inspired by d’Aspremont and Jacquemin (1988) and Katsoutacos and Ulph (1998). We have two identical firms. Both firms must decide how much to invest in R&D, which in turn affects its capital and therefore its output. The key is that R&D is partly a public good. This means that some of the benefits from R&D spill over to other firms.

Let’s write it down. A firm i produces capital K_i according to its own investments in research R_i plus a fraction of the investment of the other firm(s), scaled by a *spillover* parameter ϕ .² In other words, a firm automatically “receives” learnings made by other firms to improve its own accumulation of capital.³ For instance, imagine that firm 2 discovers how to produce a solar cell that is more robust to weather deterioration, and suppose that firm 1 learns how it did it, then it can improve its own processes.

²This parameter roughly corresponds to β in d’Aspremont and Jacquemin (1988) and to σ in Katsoutacos and Ulph (1998).

³We deviate here from d’Aspremont and Jacquemin (1988), who model R&D as reducing per-unit production costs. But the overall logic of spillovers remains about the same.

We have:

$$\begin{aligned} K_1 &= R_1 + \phi R_2 \\ K_2 &= R_2 + \phi R_1 \\ 0 < \phi < 1 \quad &\text{where } \phi \text{ is the spillover parameter.} \end{aligned}$$

We now want to know how this affects a firm's level of investments in research R_i . This depends on the effect that spillovers have on a firm's profits.

We thus examine our firms' profit functions. A firm's output is assumed to follow a Cobb-Douglas production function. For simplicity, we say that a firm's only input is capital, which is augmented by technology A and whose effect is scaled by α :

$$Y_i = AK_i^\alpha, \quad \text{where } A > 0, 0 < \alpha < 1, i = 1, 2$$

Profits π are defined as a function of capital, minus the costs of research c_R times the amount of research R (e.g., the salaries of R&D scientists and the cost of labs):

$$\pi_i = AK_i^\alpha - c_R R_i.$$

Firms are assumed to be profit maximizers. This is a simple, unconstrained optimization problem. Firm 1's first-order condition is:

$$\begin{aligned} \frac{\partial \pi_1}{\partial R_1} &= A\alpha(K_1)^{\alpha-1} - c_R = 0 \\ &= A\alpha(R_1 + \phi R_2)^{\alpha-1} - c_R \end{aligned}$$

And we have the same for firm 2, since they are assumed to be identical. In a symmetric equilibrium $R_1 = R_2 = R$:

$$c_R = A\alpha[(1 + \phi)R]^{\alpha-1}$$

We can solve the private equilibrium investment in research R^{private} :

$$\begin{aligned} R^{\text{private}} &= \left[\frac{c_R}{A\alpha(1 + \phi)^{\alpha-1}} \right]^{\frac{1}{\alpha-1}} \\ &= \left[\frac{A\alpha(1 + \phi)^{\alpha-1}}{c_R} \right]^{\frac{1}{1-\alpha}} \end{aligned}$$

That's for the selfish firm. But what would a benevolent dictator or social planner do (i.e., someone who wants the best for society)? A utilitarian social planner would like to optimize the value for society, that is, benefits net of costs:

$$W = Y_1 + Y_2 - c_R(R_1 + R_2).$$

The marginal social benefit of R_1 is:

$$\frac{\partial W}{\partial R_1} = A\alpha(R_1 + \phi R_2)^{\alpha-1} + A\alpha\phi(R_2 + \phi R_1)^{\alpha-1}$$

In the symmetric case $R_1 = R_2 = R$, this simplifies to:

$$\frac{\partial W}{\partial R_1} = A\alpha[(1 + \phi)R]^{\alpha-1}(1 + \phi)$$

The planner's first order condition is:

$$c_R = A\alpha(1 + \phi)^\alpha R^{\alpha-1}$$

We can now solve for the *social* (in contrast to individual) optimum level of research R^{soc} :

$$\begin{aligned} R^{\text{soc}} &= \left[\frac{c_R}{A\alpha(1 + \phi)^\alpha} \right]^{\frac{1}{\alpha-1}} \\ &= \left[\frac{A\alpha(1 + \phi)^\alpha}{c_R} \right]^{\frac{1}{1-\alpha}} \end{aligned}$$

We can compare the private the first order conditions for the firm and for a social planner:

- Private FOC: $A\alpha(1 + \phi)^{\alpha-1}$
- Social FOC: $A\alpha(1 + \phi)^\alpha$

Since $(1 + \phi) > 1$, the social marginal benefit is higher. To see why this implies a higher equilibrium R : because $\alpha < 1$, we have $\alpha - 1 < 0$, so $R^{\alpha-1}$ is a *decreasing* function of R . The FOC sets c_R equal to [coefficient] $\cdot R^{\alpha-1}$, so a larger coefficient can only be satisfied by a

larger R (since only a larger R will push the decreasing term $R^{\alpha-1}$ down enough to restore equality). Therefore:

$$R^{\text{social}} > R^{\text{private}}$$

In other words: **private investments in research will be lower than what is socially optimal when there are research spillovers** (positive externalities).

What are the **policy implications** of this model? There are at least two:

1. As long as $\phi > 0$, there is a theoretical justification for involving the social planner in research. The social planner *internalizes* research externalities, and thus isn't affected by leakage caused by them. In practice, this can explain why research – and fundamental research in particular, which is extremely vulnerable to externalities – is often done in public institutions, such as universities or national research labs. For instance, the National Renewable Energy Laboratory (NREL) is funded by the US government to conduct research on renewable energy technology.
2. The state can also try to reduce ϕ and push it closer to zero. One way to do this is by strengthening **intellectual property** laws (Stiglitz 1999; de Rassenfosse and Potterie 2009). Instruments such as **patents** *privatize* the benefits of research to the agent (firm) that conducted it.

i Note 29: Patents: dynamic incentives and static impact

Are patents desirable? That has been a long debate tracing back to Kenneth Joseph Arrow (1962) and Tirole (1988).

The problem is the following. On the one hand, patents reduce spillovers ($\phi \rightarrow 0$), thereby increasing private R&D investments: you increase the incentives to do R&D by making sure that you get its rents. On the other hand, patents also slow down knowledge diffusion, which would help the economy at large.

Imagine, for instance, that you discover an incredible new solar technology. Patents make sure you get a reward. It is this reward that, *ex ante*, encouraged you to devote resources to R&D: you put money into research because you expected getting some return from it.

However, this patent that benefits you will also slow down the

deployment of this piece of technology. There are many firms that might be able to produce the same solar technology cheaply, and maybe even improve on it. Yet they might be discouraged by the licensing fees that you will ask for it (your rent).

There is therefore a balance to be found between providing incentives. Should I lean toward protecting the innovative firm? Or toward accelerating the spread of the technology? Lerner (2009) reviews the literature and finds limited evidence that longer patent protection increases innovation. If that's true, then you could possibly tilt toward lower IP protection.

The role of population

Since Malthus (1888) (at least), people have tended to worry about excessive population growth and its impact on the environment. Malthus' idea was that natural resources (e.g., food) can grow at best linearly, whereas population dynamics tend to grow exponentially. Thus, at some point, societies hit a limit. Modern versions of this take are to be found in Hardin (1968), D. H. Meadows et al. (1972), or J. L. Simon (1998).

Be this as it may, the study of innovation economics suggests that population growth may have a *beneficial* effect on technology. One mechanism, discussed in Kremer (1993, 681), is that more people means more brains to generate new ideas:

the cost of inventing a new technology is independent of the number of people who use it. Thus, holding constant the share of resources devoted to research, an increase in population leads to an increase in technological change.

Another study from the same time similarly builds a model in which population growth is the driver of inventions and, by extension, output growth (Charles I. Jones 1995). This model de-emphasizes the importance of R&D: over the long term, what matters is population growth.

Thus, assuming that the demand for green and sustainable innovation increases over time, more people means more of such innovations.

5.3 Development and demonstration to commercialization

5.3.1 Building out new technologies

The *demonstration* stage is when a product exits the laboratory and faces real consumers. It must establish technological viability, estimate costs, and assess market potential. For sustainable technologies, this phase requires sustained policy support — typically by *creating a market*. Governments generate demand at prices sufficient for firms to refine technology, learn by doing, reduce costs, and build production capacity.

Let us illustrate this with the emergence of renewable energy technologies for the power sector (Aklin and Urpelainen 2018). After supporting R&D for many years, as wind turbines and later solar panels became technologically viable (in the sense that they could be connected to the grid without safety concerns), a new type of public support was needed: renewable energy companies needed a market of buyers to scale up production.

In practice, states used two policy instruments to do so (Cory, Couture, and Kreyzik 2009; Smith and Urpelainen 2014).

- **Feed-in tariffs.** The owners of renewable energy installations could connect them to the grid and get paid for the power that they contributed at a fixed price over a given number of years (often ten years). If they consumed some of it themselves (e.g., if it is a houseowner who consumes part of it), then net-metering is used to calculate the net contribution to the grid.
- **Renewable portfolio standards (RPS).** Under an RPS, utilities are required to generate a preset percentage of electricity from renewable energy sources. For instance, utilities might have been told to generate 30% of their power by 2020.

Such policies can effectively increase the volume of new technologies in operation by creating demand *ex nihilo*. Assuming that firms learn by producing more of these new technologies, then they will gradually become more cost-competitive. We will enter the *commercialization* stage of the process.

Can technologies improve too fast?

In principle, a consumer (whether a household or a firm) will be interested in acquiring a new technology if it gives it some kind of advantage. For instance, a firm might want to digitalize to reduce costs and improve efficiency. Thus, technological innovation gradually spreads beyond early adopters.

Could fast technological improvements undermine itself? Rosenberg (1976) examines this question. His point is the following: if, as a firm (or a consumer, for that matter), you live at a time when future innovations are predictable (you expect that technology will improve in the next few years), then you might delay the acquisition of this technology. If you buy it now, it will shortly be obsolete. Thus, you should wait. But then, paradoxically, demand might slow down precisely when innovation speeds up.

It is easy to see how this could apply to sustainable technology. For instance, if you expect solar cells to decline in cost due to some innovation, you are going to delay investments in new panels. Likewise, if you think that EV will have a greater range within a few years, you will wait to buy one.

This, in turn, could slow down innovation. If firms don't receive orders, then they might cut down on investments in development and commercialization. Now, this is a theoretical result. Who knows if it affects the deployment of green technology in real life? A few studies raise this point, suggesting that investment delays may occur in domains where investments are locked-in for the long term (e.g., power sector) (Jaffe, Newell, and Stavins 2004).

5.3.2 Valley of Death

Innovators working on R&D and product development must generally have a long time horizon. A new product often takes several years to be commercially viable. This is especially the case in highly regulated domains with 'red tape' (administrative regulations), such as drugs or potentially dangerous technologies (e.g., next-generation nuclear power plants).

This creates a danger zone for innovations, especially in that space in-between fundamental research and commercialization. Fundamental

research can be oriented to the long-term, thanks to universities and research centers (assuming they are funded accordingly). And commercialization takes care of itself, if the product is successful. In between exists a **Valley of Death** for innovation (Ellwood, Williams, and Egan 2022), that is, a high-risk period when innovations may fail in their attempt to become economically and commercially competitive.

In this critical phase, innovations often still need some form of public support. However, such support may be lacking. As noted, universities plan funding over decades. But similar support doesn't exist, typically, for products on the cusp of success or failure.

5.3.3 Uncertainty and political credibility

There exist several reasons for the existence of a Valley of Death, including access to finance, the need to transform the firm's strategy toward commercialization, and so forth (Ellwood, Williams, and Egan 2022). Here we concentrate on the specific role of public policy and the threat of **political failures**.

One problem to get through the Valley of Death is the lack of **credible commitment** by policymakers, also referred to as the **dynamic inconsistency** or **time inconsistency** problem (Kydland and Prescott 1977; Harstad 2020).

The basic idea can be illustrated with the following example.

- Suppose that an agent (say, the government) promises to implement a strategy π at time t over the time horizon $t, t + 1, \dots, T$. For instance: a subsidy program to stimulate the deployment of solar panels.
- Suppose, then, that another agent (say, a solar company) decides to invest in new production capacity at time $t + 1$, which, given the policy π , becomes profitable. So far, so good.
- But next, let's say that at time $t + 2$, the government figures out: "the firm has built out its production capacity, which is what we wanted. Why don't we now use the cash for something else than subsidies?"
- If the firm, at time t , realizes that the government is likely to renege later, then it will *not* respond to the policy incentive. This is because the government's promise is not credible: it cannot

guarantee that the promises it makes today will be kept in the future.

Policy uncertainty thus deepens the Valley of Death: investors who cannot trust that support will last until competitiveness is reached have little reason to proceed. Feed-in tariffs illustrate the risk — in practice, states often renegotiated contracts unilaterally when costs proved higher than expected (Ritzenhofen and Spinler 2016).

5.4 Commercialization and diffusion: externality problems

We now focus on technologies that have reached maturity — viable, and comparable in quality to existing alternatives. The questions are:

1. Why do firms produce goods in a way that overuses (natural) inputs?
2. Why do firms use technologies that generate negative externalities?
3. Why don't firms transition to the use and production of green(er) technologies?

To examine this, we must dig a bit deeper in the logic of firms.

5.4.1 Firm behavior 101

We start by assuming a reasonably well-functioning market. This typically means that:

- There is a large number of competing firms (no monopoly or oligopoly)
- Agents have sufficient information about themselves and others
- Transaction costs are low
- Property rights are well defined (no theft, no externalities)

Of course, these are assumptions that will be weakened when sustainability problems emerge. But first, let's start with this benchmark. We would then expect the market for a given good to look like Figure 5.5. As before, the demand curve would likely be downward sloping (lower prices equate demand for more units of that good).

We now add a supply curve. It is constructed on the basis of a continuum of firms. Each solve an optimization problem: how to maximize profit. We use this as a first approximation of firm behavior. To do so, the firm decides how much to produce, given the price of the good. This reflects the notion that firms don't set prices in competitive market: markets do. For instance, a solar panel producer doesn't set the price of panels; it is set on the market for solar panels.

The optimization problem for firm i , then, is to maximize profit π , defined as total revenues (price times quantity produced by i) net of production costs:

$$\max_{Q_i \in \mathbb{R}_+} \pi_i = PQ_i - f_i(Q_i).$$

The cost function $f(\cdot)$ is defined at the firm level. Costs depend on inputs — labor, capital, and natural inputs (L, K, N) — needed to produce goods. It's reasonably safe to assume that $f'(Q) > 0$: the marginal cost of production rises with output, since additional inputs (e.g., land, raw materials) become progressively more expensive. For instance, when you start a firm, you will hire the 10 cheapest employees (holding their ability constant). Then, as you need to add more staff, you will need to increase wages to attract them.

To maximize profit, we take the derivative of π with respect to Q_i and set it to zero. This yields the result that a firm produces up to the point where marginal cost (i.e., the derivative of the cost function $f'(Q)$) equals market price, which we call the *equilibrium quantity* Q_i^* :

$$Q_i^* \text{ such that } f'_i(Q_i^*) = P$$

Thus, a firm selects a quantity Q^* up to the point at which the marginal cost of production equals the market price for the good. For instance, if the market price for a new car is CHF 30,000, then the car manufacturer will produce cars until the extra car it produces reaches that cost (the earlier cars will have been cheaper to produce and generate surplus).

The supply curve is then made of many firms that must to decide whether to produce, given a certain price level. Or, put another way, for each price, you compute the sum of quantity that firms are willing to produce. This curve is upward sloping. At low prices, only the most efficient firms can profitably supply. These are those with lower marginal costs (since the price is the same for everyone), which can

come from using superior technology or having better management. As prices rise, more firms find it worthwhile to enter this market, even those with higher costs. Aggregate supply therefore slopes upward. See Note 30 for another example coming from electricity markets.

i Note 30: Electricity supply

The electricity market has several peculiarities. One is that demand must be met by supply at any given time point, otherwise the grid collapses. As a result, this market is typically highly organized and regulated. In many countries, we have a system in which *generators* produce electricity that they sell to *utilities*, which buy it on behalf of their customers.

In the US, this market is organized by *Independent System Operators* (ISOs) or *Regional Transmission Organizations* (RTOs). There exist several markets to coordinate the supply of electricity. The main one is the *day-ahead* market. Its exact design varies, but roughly:

- At Day $t-7$ to Day $t-1$: generators prepare bids (supply; e.g., “we can provide 300 MWh at EUR 30/MWh”) and utilities forecast their demand (based on past data, expected weather, etc.).
- On Day $t-1$ (“day-ahead market”):
 - The ISO/RTO estimates total demand.
 - The ISO/RTO ranks bids from cheapest to most expensive. This is called the **merit order**.
 - The market clears at the price (*spot price*) at which the last unit of MW is sold (i.e., everyone gets the same price).
- On Day t : the ISO/RTO refines demand forecasts and taps into the *balancing market* to fill the gap between day-ahead scheduled production and actual consumption. See Figure 5.3 for real-life data from CAISO (California).

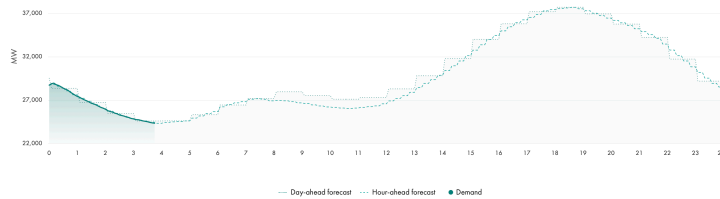


Figure 5.3: Day-ahead and hour-ahead forecast, compared to actual consumption. Source: CAISO [link](#)

The merit order can be interpreted as a supply curve. There is a unique equilibrium price. Given the design of this auction, each generator has an incentive to provide a bid that is as low as possible, in order to ensure that their electricity will get bought. Figure 5.4 illustrates what the merit order could look like.

The merit order also shows why renewables such as wind and solar are successful: since their marginal cost is close to zero (they need no expensive fuel and they don't need staff to turn them on and off), their owners can submit bids at very low prices. They will get the price that marginal generator (often, but not always natural gas) needs to cover their own cost.

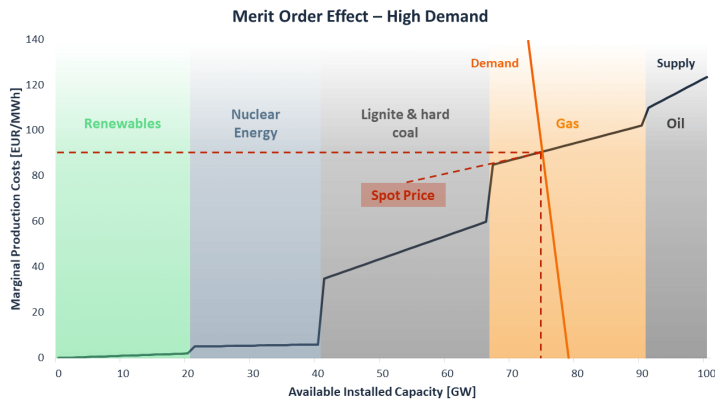


Figure 5.4: Illustration of the merit order (supply) curve (black) in the power market. Source: RVI [link](#)

Let us now assume that we have well-defined demand and supply curves, as in Figure 5.5. The market clears at E . This generates *surplus* for consumers and producers:

- **Consumer surplus.** This is the area under the demand curve above the market price. This corresponds to the volume (surface) of quantity demanded that consumers would have been willing to get at a higher price. Thus, for these consumers, the market price is lower than what they would have been willing to pay, and therefore represents a 'bonus'. For instance, if I had been willing to pay CHF 20,000 for an electric car, and I found one at CHF 15,000, I would have a surplus of 5,000.
- **Producer surplus.** This is the area above the supply curve and under the market price. This corresponds to the volume of sales producers would have been willing to make at a lower price. For instance, if I can produce solar modules for 30 cent per Watt, and if the market is at 50 cent per Watt, I pocket a surplus of 20 cent.
- **(Utilitarian) social welfare:** consumer surplus plus producer surplus (minus social costs, if any, such as negative externalities).

From a (utilitarian) social welfare perspective, what we want is to maximize such surplus. And this happens at Q^* , P^* . If we imposed a limit on quantities at $Q' < Q^*$, part of the surplus would disappear. If we forced quantities to be larger than the market equilibrium, $Q' > Q^*$, then someone would lose (firms would lose money if they can't increase the price, and if they can increase the price then consumers wouldn't be interested).

This is the state of the market at the macro level.

We can use this definition of a market to formulate predictions about policies. For instance, imagine that the price is set at a level below P^* . What will happen? Well, the price is low, so consumers will want a lot of this good. The quantity they want is going to be larger than Q^* . However, at this low price, few firms want to produce (most won't find it profitable to do so). As a result, quantity produced will be lower than Q^* . Thus, we will have a poorly functioning market: too much demand, not enough supply.

Could such a situation happen in real life? Yes! See Note 31.

i Note 31: Affordable train

A policy that often comes up in public debates is the cost of public transportation. The argument is the following: internal combustion cars generate negative side effects (congestion, pollution,

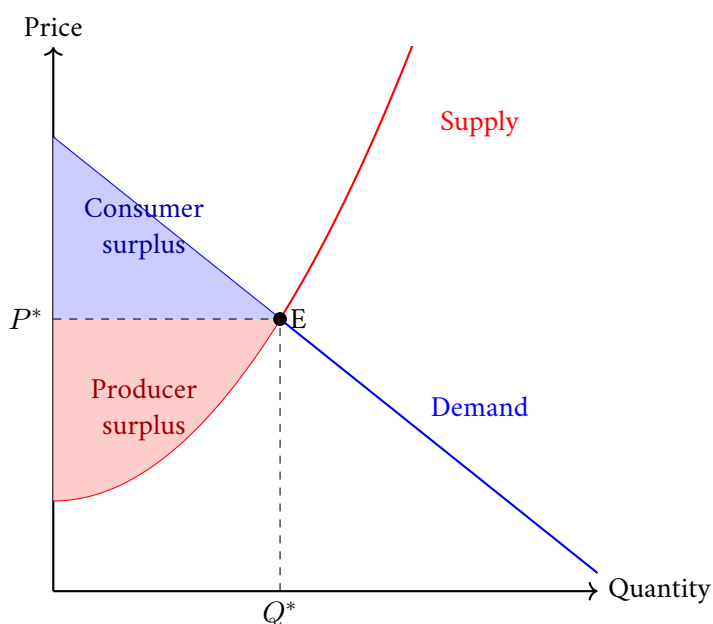


Figure 5.5: Market supply, demand, and surpluses.

greenhouse gas emissions). Thus, the state should encourage a switch from cars to public transportation.

This seems reasonable, but how? Germany experimented with the following policy: create a very cheap train ticket to make the relative price of train (versus cars) much more appealing. During the summer of 2022, domestic train tickets were limited to EUR 9. That's very cheap!

In Figure 5.5, this corresponds to imposing a cap on price at EUR 9. Thus, we can conjecture that demand will increase, but supply will be under tremendous pressure.

What happened? Liebensteiner et al. (2026) completed a statistical analysis of consumer responses to these cheap tickets. The result:

"We find a strong increase in rail travel with modest substitution away from cars. Train trips rise by around 35%, while car traffic falls by 1–5%, with the smallest traffic reductions during commuting times and the largest increases in rail travel for leisure purposes on

weekends and to touristic destinations. Moreover, the train demand surge materially worsens service quality via elevated train delays, which even spill over to long-distance trains.”

Thus, people used the cheap tickets for leisure. However, they continued using their car, especially for work. Why? One can imagine that taking the train for work depends on availability (is there a train nearby?), convenience (is the train coming at the right time?), and reliability (is the train punctual?) of the train. Cheaper tickets don't directly affect availability and convenience, and it turns out it made reliability worse. As a result, commuters continued to take the car.

And the, the big question then is: will low-price tickets undermine availability and convenience? Forcing Deutsche Bahn (DB, the German train operator) to issue cheap tickets not only means more users, but less revenues for DB. Will DB have to cut on investments? Or reduce service? If so, quality of trains will go down. Or will the government compensate DB to make up for the difference between service demanded and services that can be supplied with a EUR 9 price? If so, what other expenses will need to be cut?

5.4.2 Property rights and inputs

So where do sustainability issues come into play? Here, we return to the issue of **property rights**.

Suppose that the firm needs some scarce minerals as input (e.g., cobalt). In a well-functioning market, these would appear in the firm's calculus as an input cost, which in turn affects its production level (as we saw, the firm produces up to the moment its marginal cost equates market price). In Figure 5.6, this corresponds to the intersection of the marginal cost curve $f'(Q)$ (in red; upward sloping, as per our assumption) and the price line (inelastic demand). The firm produces quantity E_f of goods. At the macro level (top graph), this corresponds to a market equilibrium at E_m .

But markets often don't function so well. Suppose now that the firm finds a free deposit of cobalt. Its cost per unit of good produced will

go down. This corresponds to a shift of the marginal cost curve to the right (at a given price, you can produce more goods). In Figure 5.6, this corresponds to a shift from E' to E (at the firm level on the bottom and at the macro level on the top). In other words: an over cheap input has two connected implications:

1. The firm will use more natural inputs than what a well-functioning market would yield.
2. The firm will produce more goods than what a well-functioning market would produce.

These two implications suggest that **under-priced natural resources cause an unsustainable level of consumption**.

Why are natural resources underpriced? When firms can freely extract from their environment — i.e., no property rights exist over those resources — they face zero cost and exclude them from production decisions.

Resources may also be underpriced when acquired through non-transparent means — for instance, when a firm bribes a corrupt official to obtain natural resources at a fraction of their market value (Fredriksson and Svensson 2003; Fredriksson, List, and Millimet 2003). A price was paid, but not a fair or efficient one.

5.4.3 Externalities in production

Let us now consider the problem of negative externalities. The setup is the following. We have two goods, one is dirty d and one is clean c . For instance, we can think of an electric car versus a ICE car, and we assume that electricity is produced without greenhouse gases.

First, consumers have an inverse demand (willingness to pay) for each good:

$$p_i = P_i(q_i), \quad i \in \{c, d\},$$

with $P(\cdot)$ being how much people are willing to pay for quantity q_i , with $P'_i(\cdot) < 0$. For simplicity, let's assume that we have a linear demand function:

$$q_i = \alpha - \beta p_i \implies P_i(q) = \frac{\alpha - q}{\beta}.$$

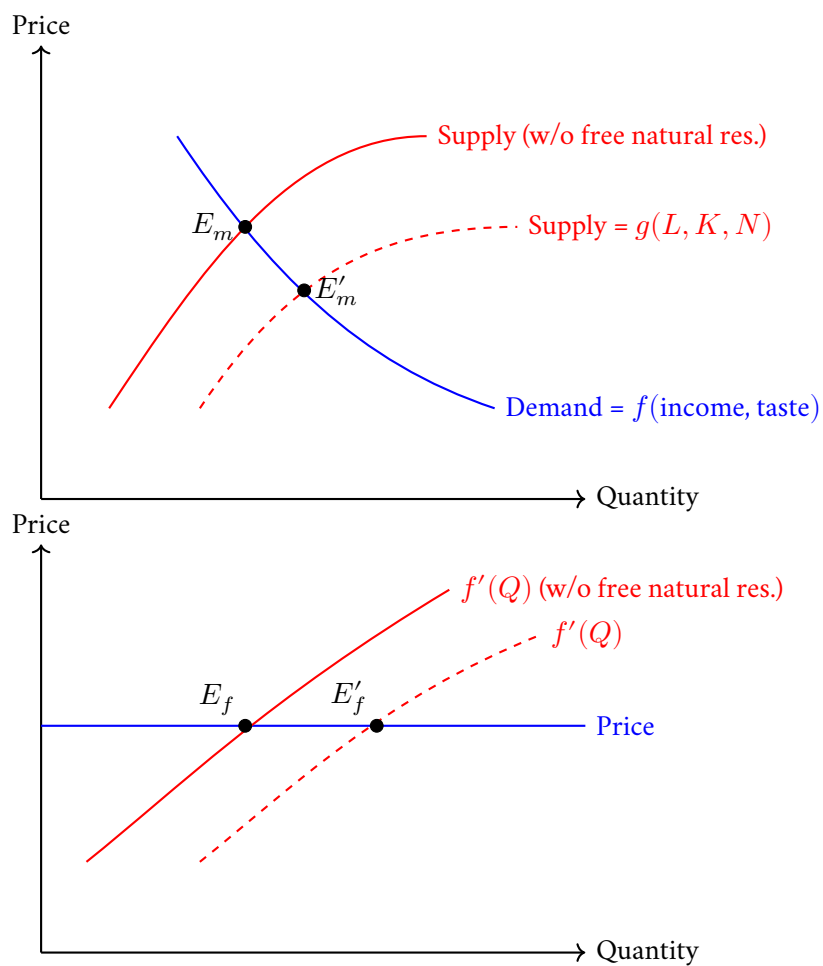


Figure 5.6: Macro (top) and micro (bottom) markets. Supply-side in red. Demand-side in blue. The dashed line corresponds to the market equilibrium when the natural inputs are free.

As before, firms are price-takers with constant private marginal costs MC_c and MC_d . We assume that firms are optimizers, which gives us the earlier equilibrium condition: firms produce up to the point where their (private) marginal cost equates the price:

$$p_c^{\text{priv}} = MC_c, \quad p_d^{\text{priv}} = MC_d.$$

What is new is that we have an *externality*, namely a social ‘damage’ caused by the dirty good. The externality is given by a function $E(q_d)$, with marginal damage $E'(q_d) > 0$. To keep things simple, we will assume that there is a constant per-unit damage $E(q_d) = e q_d$, so that $E'(q_d) = e$. This would not apply in non-linear systems where damage can explode once a tipping point is reached.

Now, consumer surplus for good i when quantity q_i is consumed at price p_i is:

$$CS_i(q_i) = \int_0^{q_i} P_i(q) dq - p_i q_i,$$

which for the linear case reduces to a triangle:

$$CS = \frac{1}{2}(P_{\max} - p) q.$$

Let us now look at the market equilibrium. The private equilibrium without any government intervention is:

$$q_i^{\text{priv}} = D_i(p_i^{\text{priv}}) = D_i(MC_i).$$

We can compute social welfare, which is the sum of consumer and producer surpluses, minus the social cost of the externalities. Note that the producer surplus is zero, given our assumption that the supply curve is flat (i.e., marginal costs are constant)

$$W^{\text{priv}} = CS_c(q_c^{\text{priv}}) + CS_d(q_d^{\text{priv}}) - E(q_d^{\text{priv}}).$$

Is this what a social planner (or benevolent dictator) would do? A social planner chooses q_c, q_d to maximize:

$$W = CS_c(q_c) + CS_d(q_d) - MC_c q_c - MC_d q_d - E(q_d).$$

- First-order condition for the dirty good:

$$\frac{dW}{dq_d} = P_d(q_d) - MC_d - E'(q_d) = 0.$$

Thus, the social optimum q_d^* satisfies

$$P_d(q_d^*) = MC_d + E'(q_d^*).$$

We can compare the welfare without intervention and what would be socially optimal:

$$W^{\text{priv}} = CS_c(q_c^{\text{priv}}) + CS_d(q_d^{\text{priv}}) - E(q_d^{\text{priv}}).$$

and:

$$W^{\text{soc}} = CS_c(q_c^*) + CS_d(q_d^*) - MC_c q_c^* - MC_d q_d^* - E(q_d^*).$$

The welfare gain from moving the dirty-good quantity from the market level to the social optimum is

$$\Delta W = W^{\text{soc}} - W^{\text{priv}} = [CS_d(q_d^*) - CS_d(q_d^{\text{priv}})] - MC_d(q_d^* - q_d^{\text{priv}}) - [E(q_d^*) - E(q_d^{\text{priv}})].$$

Or, equivalently:

$$\Delta W = \int_{q_d^{\text{priv}}}^{q_d^*} [P_d(q) - MC_d - E'(q)] dq$$

We can interpret this as follows: ΔW is the area between the marginal private benefit $P_d(q)$ and the social marginal cost $MC_d + E'(q)$ between the two quantities. If the private market over-consumes ($q_d^{\text{priv}} > q_d^*$), reducing q_d to q_d^* yields the positive welfare gain equal to that area.

5.4.4 Pigouvian taxes and the internalization of externalities

We thus have two related problems: (1) underpriced natural inputs, and (2) underpriced goods that generate negative externalities. The standard economist's remedy for (1) is to create property rights over resources, making them costly and shifting the supply curve left. For (2), the key policy response is to *internalize* externalities — forcing producers to face the costs they impose on others. This is the idea behind a **Pigouvian tax**, named after Alfred Pigou (1877–1959) (Pigou 1920): raise production costs, the market clears at a higher price and lower

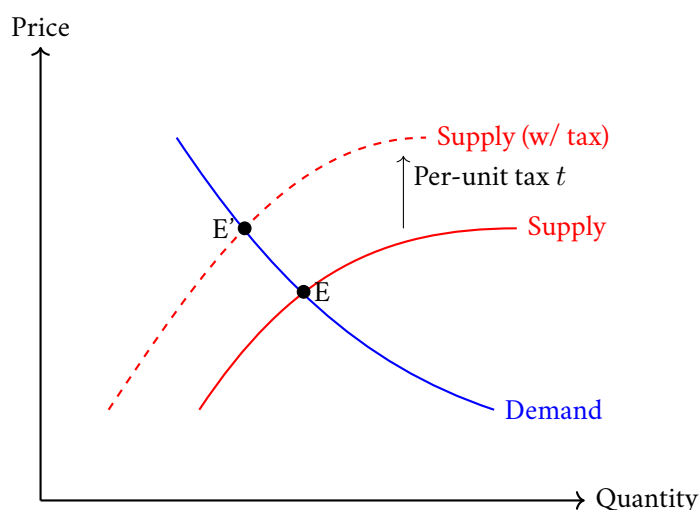


Figure 5.7: The effect of a Pigouvian tax.

quantity, and externalities (if proportional to quantity) fall accordingly. See Figure 5.7.

A well-calibrated Pigouvian tax achieves a first-best outcome with a relatively simple instrument.⁴ Pigouvian taxes work through two channels:

1. Reduce demand via higher prices.
2. Encourage substitution — toward alternative goods or new technologies.

On channel (2): if a carbon tax of \$300/ton makes coal uneconomical, and a carbon capture technology costs \$280/ton, the firm has an incentive to adopt it. The tax thus spurs innovation. Evidence for this effect has been found in studies of the EU ETS and Sweden (Calel and Dechezleprêtre 2016; aus dem Moore, Brehm, and Gruhl 2025).

⁴Administrative capacity is still needed to identify the social optimum, measure externalities, implement the tax, and enforce compliance. Baumol (1972, 316) was skeptical about our ability to find an optimum: “we do not know how to estimate the magnitudes of the social costs, the data needed to implement the Pigouvian tax-subsidy proposals. For example, a very substantial portion of the cost of pollution is psychic; and even if we knew how to evaluate the psychic cost to some one individual we seem to have little hope of dealing with effects so widely diffused through the population.” Baumol’s alternative was to pick some limit (e.g., level of air pollution deemed acceptable) and then tax the behavior causing it to meet it. This is similar to Pigouvian taxes, but without the reference to a social optimum.

Tax versus subsidies

Policymakers are often encouraged to reduce the price of (sustainable) technologies. As the price goes down, we move along the demand curve and the quantity demanded goes up. This raises the question: should you stimulate green technologies or discourage unsustainable technologies? A few things to keep in mind in such a debate:

Who funds (green) subsidies (if anyone)? Is the state imposing a price or are they compensating producers? If they are imposing a price without compensation, then producers see their revenues decline and as a result investments will go down too. In the long-term, this threatens the viability of the production of this good. For instance, Indian politicians had a habit of forcing utilities to decrease electricity prices before elections without compensating them (essentially via threats). This was popular among voters but made utilities miss out on revenue, contributing to the poor electric infrastructure (Min and Golden 2014; Min 2015; Aklin et al. 2016).

In Geneva (Switzerland), public transports were made free for certain groups of people (free for under-25 years old and 50% off for retirees). The cost of this policy was expected to be about CHF 31m, but this proved to underestimate how popular this policy would be. An extra CHF 20m was needed, yet the parliament refused to fund the renewal of this policy, leaving the *Transports Publics Genevois* with a considerable shortfall ([link](#)).

The problem of picking (green) winners. When we have an externalities problem, the culprit is generally clear (at least in terms of technologies). For instance, we know that fossil fuels-based energy systems are the main source of human-generated greenhouse gas emissions. But the solution is often less clear. For instance: should a government promote solar? Nuclear? Wind? Geothermal? States operate under imperfect information. It is not clear whether they know better which technologies are the ones that will survive. This is why policies that are *technology-neutral* are generally preferred. But it's not always easy to implement them.

Political calculus. It is generally easier to *give* subsidies than to *take* money (via taxes) (Olson 1965; Stigler 1971; Weingast, Shepsle, and Johnsen 1981). The reason is that the negative im-

pact of Pigouvian taxes is highly concentrated (e.g., coal plants) whereas its benefits are diffused over the entire population. The negative impact of subsidies, however, is much more spread out, since it is divided over the entire tax base and its benefits can be concentrated, depending on how narrowly it is targeted. As a result, there is generally less mobilization against subsidies.

5.4.5 Setting a Pigouvian tax

To set a Pigouvian tax, you need the following information: (a) a supply function, (b) a demand function, and (c) an externality function. The latter can, in practice, be challenging. How do you quantify all the impacts of an externality? These estimates have, mathematically, a significant impact on the level of Pigouvian taxes you must set to solve the problem. See Note 32.

The step-by-step design of an Pigouvian tax, t^* , is the following. We will need 3 equations:

1. Cost of supply: $C = f(Q)$, with $f'(Q) \geq 0$
2. Demand: $P = g(Q)$, with $g'(Q) \leq 0$
3. Externality: $E = h(Q)$, with $h'(Q) \geq 0$

Step 1 (optional). Start by solving for market equilibrium Q^e :

- Marginal *private* cost = price
- $\partial C / \partial Q = P$
- Solve for Q^e

Step 2. Next, solve for *socially* optimal quantity Q^* :

- Marginal *social* cost = price
- $\partial C / \partial Q + \partial E / \partial Q = P$
- Solve for Q^*

Step 3. Equate the price to the marginal *private* cost *plus* the Pigouvian tax at Q^* and solve for optimal tax rate t^* :

- $P = \partial C / \partial Q + \partial t / \partial Q$
- For a flat per-unit tax: $P = \partial C / \partial Q + t$
- Plug in Q^* and solve for t^* .

i Note 32: Example of a Pigouvian tax

Suppose we have the following setup:

- Market price (demand function) $f(Q)$: $P = 100 - Q$
- Private cost $g(Q)$: $C = 10Q + Q^2$
- Externality $h(Q)$: $E = 10Q + 0.5Q^2$

Let us find the optimal Pigouvian tax rate, which we will assume is a flat, per-unit tax.

1. **Private equilibrium:** find Q that equates marginal *private* cost to the market price (Q^e)

$$\begin{aligned} P &= \underbrace{g'(Q)}_{\text{marginal private cost}} \\ 100 - Q &= 10 + 2Q \\ Q^e &= 30 \end{aligned}$$

2. **Social optimum:** find Q that equates marginal *social* cost to the market price (Q^*)

$$\begin{aligned} P &= \underbrace{g'(Q) + h'(Q)}_{\text{marginal social cost}} \\ 100 - Q &= \underbrace{(10 + 2Q)}_{\text{marginal private cost}} + \underbrace{(10 + Q)}_{\text{marginal externality}} \\ Q^* &= 20 \end{aligned}$$

Thus the market produces 30 units of the good, but once we include externalities, we actually would prefer 20 units to be produced. The Pigouvian tax must therefore increase the price of the good until quantity is reduced to 20.

3. **Find Pigouvian tax:** add a tax to increase private marginal cost

$$\begin{aligned} P &= g'(Q) + t \\ 100 - Q &= (10 + 2Q) + t \\ t &= 90 - 3Q \\ t^* &= 30 \quad \text{by plugging in } Q^* \end{aligned}$$

Thus, the Pigouvian tax rate per unit is 30. To see that this moves the equilibrium quantity of goods produced to 20, we can plug it in the private equilibrium:

$$\begin{aligned}100 - Q &= (10 + 2Q) + 30 \\ Q &= 20\end{aligned}$$

5.4.6 Pigouvian taxes in practice

The mechanism underpinning a Pigouvian tax is that, by making a technology (or behavior) more expensive, we can capitalize on the fact that demand-curves are downward sloping (in general) to reduce the “quantity” of that technology. This is therefore a *pricing* mechanism. Such a policy is ubiquitous. We briefly give a few examples here.

Tobacco. We can think of smoking as generating negative externalities.

Climate. Carbon pricing is a Pigouvian strategy. There are two main types of pricing policies. The first is a *carbon tax* (or more generally a greenhouse tax). The tax is set per unit (e.g., ton) of CO₂. Externalities then come from the effect that the source of carbon has on welfare, e.g., via lower agricultural output, poorer worker productivity, or health dangers. This is captured by the *social cost of carbon* (see earlier chapter). Carbon taxes can take various forms, such as a tax on gasoline fuel. The estimates of such policies suggest large benefits (see Box), but these are politically contentious (see Box).

The second pricing strategy is a *cap-and-trade*. This is not technically a tax, but it corresponds to the pricing logic underpinning Pigouvian policies. Here, the policymaker reduces emissions by setting a total amount of emissions allowed per unit of time (say, in a year). Carbon permits must then be bought, normally on a carbon market, by firms that wish to emit. In this approach, quantity is fixed and the price varies. A typical example of a cap-and-trade system is the European Union Emissions Trading System (EU ETS) (see Note 33).

Carbon pricing, despite political setbacks, is a popular tool. The World Bank built a dashboard that lists jurisdictions that have or are planning

to implement it ([link](#)). At the time of writing, there were 37 cap-and-trade in place and 43 carbon taxes across 55 countries and 44 subnational jurisdictions.

Does carbon pricing work? Evidence is mixed (Aldy and Stavins 2012). A meta review by J. F. Green (2021) suggests that its effect on emissions is generally limited, with a range of 0 to -2%. Another meta analysis by Döbbeling-Hildebrandt et al. (2024) is considerably more optimistic, stating that 17 out of 21 carbon pricing schemes reduced emissions and that the range of reduction is estimated to be between -4% and -15%.

Note 33: The EU ETS

The EU ETS was established by the European Union in the aftermath of the ratification of the Kyoto Protocol (Ellerman, Convery, and De Perthuis 2010). It was launched in 2005 (Phase I) and is now in its fourth trading period (Phase IV, 2021-2030). The design of the EU ETS has varied over time. Currently, it includes all EU member states, EFTA, and Northern Ireland. It covers carbon dioxide (CO₂), nitrous oxide (N₂O) and perfluorocarbons (PFCs) from a defined set of sectors (electricity and heat generation, energy-intensive industries, aviation within the EEA, maritime transport, aluminium, etc.). This corresponds to about 10,000 installations. See *Scope of the EU ETS* at the EU ([link](#)).

In practice, the ETS roughly works as follows:

1. The EU Commission sets an annual emission target (the *cap* of cap-and-trade). Generally, the cap declines year to year by about 4% to meet broader greenhouse gas reduction goals. The cap is translated into a number of allowances (called EUA). One EUA gives the right to emit one ton of CO₂.
2. The annual cap is divided into daily volumes that are auctioned off on the *primary market* by the European Energy Exchange. The calendar for auctions is available online ([link](#)). Only a small number of firms participate to these auctions ([link](#)).
3. Most of the action happens in the *secondary market* (the *trade* part of cap-and-trade), where thousands of firms may buy and sell permits. These are mostly traded on the European Energy Exchange and on the Intercontinental Ex-

change.

Firms must monitor their carbon emissions year-long. Then, by March 31, they must report the total emissions they generated in the previous calendar year. The report is verified. By September 30, they *surrender* the necessary number of EUA to the EU, which then ‘burns’ these EUAs (that is, the EU doesn’t trade them back). If they don’t have a sufficient large number of EUA, they will face penalties for non-compliance (currently set at EUR 100/ton; [link](#)).

After a difficult start, the ETS is now often heralded as a success. Bayer and Aklin (2020) find that it reduced emissions by about 3.8% in covered sectors. Cael and Dechezleprêtre (2016) find that it also triggered innovation – a desired side effect of Pigouvian taxes.

But this was not easy. The EU made policy design mistakes early on. In its first phase (which covered 2005-2007), the EU did not let EUA be valid for the second phase; as a result, as the end of Phase I approached, the price rapidly collapsed.

Another criticism was that member states gave many allowances for free to critical (or powerful) sectors and firms. Excessive free EUAs undermines their price and therefore reduces the ETS’ impact. Why did states do so? Partly, this is done to protect some industries from *carbon leakage*: if the price of production in the EU goes up by too much, some firms might just move to other jurisdictions and produce their goods there. This would weaken the European economy with no environmental benefit.

But set aside the leakage problem for a second. Can you think of this on a political level? What could the political calculus be from the EU’s perspective?

Congestion pricing. You can think of traffic as a tragedy of the commons: I get the benefits of driving while I externalize the negative effects onto others. This can be particularly challenging for urban areas: people living in the suburbs must come into the city (e.g., for work or tourism), profit from the benefits of a private vehicle (flexibility, maybe speed) but impose an externality on other cars (slower traffic for other commuters, air quality for local inhabitants). Newbery (1990) finds that the journey speed in London was slower in 1990 than in 1900 – before cars became widespread! One way to address this is by imposing a

price on entering the region in question (such as a city). This is the idea behind a *congestion price*, which is a geographically-defined Pigouvian tax. It is typically implemented as a flat fee imposed to certain types of vehicles. Depending on the design, it could affect all private vehicles (to encourage a switch to public transportation), or it could affect only internal-combustion engines (to encourage a switch either to electric vehicles or to public transportation). Congestion charging can also be done on specific roads (e.g., highway tolls) rather than specific cities. In practice, there is a rich variety of congestion pricing policy design (Parry 2009).

Urban congestion pricing has been implemented in several cities, including London, Milan, New York, Singapore, or Stockholm (Leape 2006; Eliasson et al. 2009; Lehe 2019; Cook et al. 2025). Empirical studies on their effect are still relatively scarce (Parry 2009). A recent attempt to use congestion pricing in New York City, which included a fee to enter its central business district, finds that it reduced emissions from vehicles and increased traffic speed, both consistent with a decline in the number of cars circulating (Cook et al. 2025).

What about building more roads? Wouldn't this reduce traffic jams? Two things:

1. There are other externalities we care about aside from slower trips, such as air pollution. This still requires a reduction in the number of cars.
2. Adding roads can mobilize more drivers. It can also lead to Braess' paradox. See Note 34.

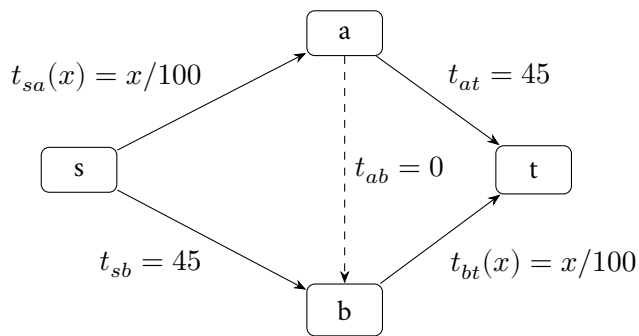


Figure 5.8: Braess' paradox from start s to terminus t , via a and b . Adapted from Easley and Kleinberg (2010).

i Note 34: Braess' paradox

Braess' paradox shows that adding roads can sometimes worsen the time it takes to complete a trip (Braess 1968). This is paradoxical because our intuition tells us that increasing the supply of roads should make things faster. Yet we can show that this is not necessarily the case, even without having to say anything about additional roads adding more drivers.

Let's go over an example, summarized in Figure 5.8 (using numbers from Easley and Kleinberg (2010)). We have 4,000 drivers and 3 paths ($s-a-t$, $s-b-t$, and $s-a-b-t$), with $s-a-b-t$ initially closed.

Before adding the shortcut

1. $s \rightarrow a \rightarrow t$: travel time = $x/100 + 45$ minutes
2. $s \rightarrow b \rightarrow t$: travel time = $45 + x/100$ minutes

The Wardrop equilibrium (a type of Nash equilibrium) states that all users have the same travel time. This is obtained when drivers split equally on both roads, such as 2,000 take the top route and 2,000 take the bottom one, for a trip time of 65 minutes. To verify this, check whether 1 driver would deviate (which is not the case: travel time for that person would increase to 65.01).

After adding the shortcut $a \rightarrow b$

Let's assume we are at the previous Wardrop equilibrium. Now, it makes sense for the marginal driver on the top route to deviate to the new path $s-a-b-t$. Doing so makes the trip go from 65 minutes to 40.01 minutes, a considerable gain.⁵ But this is not an equilibrium yet, since the next driver will also do so. Time will then gradually increase.

- All the 2000 drivers from the top route will take the new route: at that moment, their trip will be 60 minutes.⁶
- The bottom drivers now spend 85 minutes because of the newcomers on the $b-t$ segment.⁷ Since going via a instead of b is better for them, they will deviate. Everybody now goes $s-a-b-t$, for an average time of 80 minutes for everyone.

- This is an equilibrium: if you deviate individually, you would spend 85 minutes on the s - b - t route.

But note: we went from 65 to 80 minutes. Thus, adding a free road made traffic slower.

💡 Solving Braess' paradox

Many car drivers use an app on their phone to go from A to B, such as Google Maps. The way these apps do this, in a nutshell, is to identify a set of paths from A to B. Then, a cost is computed for each path. This cost generally includes duration (most people want to go faster rather than slower) as well as other constraints (e.g., low emissions, no tolls, etc.). The paths are then ranked based on these costs and reported on your screen.

This, as we saw, can generate a paradox, whereby individually rational decisions can make trips longer for everyone because everyone receives the same information (assuming they start and end at the same location and make the request at the same time). Arora et al. (2026) explored whether overall speed could increase if some app users received slightly suboptimal recommendations. The way this was done is by adding an extra cost to each segment of the path from A to B when that segment was congested, changing the order of the suggested routes. Arora et al. (2026) tested this idea in ten large US cities and found:

“we observe a city-average 2% increase in vehicle speeds on the intervened segments, along with improved travel times 0.7% and potential annual reductions exceeding 1,000 tons of CO₂-equivalent emissions per city in the majority of studied locations.”

⁷The first driver to deviate would spend $2000/100 + 2001/100 = 40.01$ minutes.

⁷These 2,000 drivers take s - a and then all 4,000 drivers take b - t , such that $2000/100 + 4000/100$ minutes.

⁷ $45 + 4000/100 = 85$ minutes.

💡 The political challenges of a Pigouvian tax

You might recall the concept of elasticity of demand:

$$\eta = - \frac{\Delta Q(\text{percent})}{\Delta P(\text{percent})}$$

If $\eta > 1$, then demand is elastic; if $\eta < 1$ then it is inelastic.

Now, if demand is elastic (flat demand curve), then:

- People don't need this good (easy substitutes)
- Pigouvian tax can be small and yet have an effect

However, if demand is inelastic (steep demand curve)

- People need this good
- Pigouvian tax will need to be large to have an effect

The latter raises a difficult political problem: taxes tend to be unpopular and thus politically difficult to implement. Thus, one may conjecture that they are most likely to succeed (both politically and in reducing externalities) where demand is reasonably elastic. This would indicate that they are most effective when substitutes are readily available.

A good example is the French Yellow Vest movement, which started after an attempt by the French government to increase a fuel tax. This might be fine in urban areas, where people don't use cars much. But it has a large cost for people in rural areas that are poorly served by public transportation. In such cases, there is no way for people to 'escape' the tax (e.g., by taking the train instead of the car); as a result, their private welfare declines considerably. The public did not respond well to this (Pigouvian) tax and so the Yellow Vest movement started.

What is interesting is that the French government wanted to accompany its fuel tax increase with tax credits for lower income households, thereby reducing its impact. But the French Constitutional Court rejected this policy, leaving only a tax increase.

There are additional challenges when establishing a Pigouvian tax. One is that it requires a lot of knowledge. But estimating all these parameters can be very difficult. For instance, they might differ across regions, depending on the preferences and income levels of local populations. And they might change over time, as new technologies come in.

Measuring avoided externalities: the case of fuel taxes

It is hard to estimate externalities, since they can be multidimensional. For instance, the externalities from cars with internal combustion engines could include: air pollution, climate change, time lost in traffic, noise pollution, etc. As a result, it is also hard to estimate how much a Pigouvian policy helped. Some estimates suggest that these benefits can be large. A recent study by Basaglia, Behr, and Drupp (2025) estimates the externalities avoided by a (Pigouvian) fuel tax in Europe. They proceeded as follows:

1. They estimate the effect of the tax on CO₂, PM2.5, and NO_x emissions at the regional level.
2. They use estimates of the (social) cost of each pollutant in terms of public health (p.28). These are (in 2010 euros):
 - EUR 80/t of CO₂
 - EUR 122,800 to 364,100/t of PM2.5 (rural versus urban)
 - EUR 15,400/t of NO_x.
3. You can then compute the value of avoided emissions by multiplying the volume of reduced emissions times these social cost estimates.

They summarize their findings (p.17):

First, between 1999 and 2009, mean annual emission reduction amounted to 0.37 metric tonnes of CO₂ per capita, which cumulatively sums up to 334,970,340 tonnes of CO₂ [...]. Second, we find that, on average, 0.17 kg of per capita PM2.5 less were emitted each year in comparison to a scenario with no eco-tax, which amounts to total PM2.5 savings of around 158,432 tons. Finally, our SDID estimation suggests that, after the eco-tax, per capita NO_x emissions were lower by 2.2 kg, on average, with a cumulative reduction in NO_x of 2,030,644 tonnes.

Furthermore (p.29):

The average estimates from our preferred specification suggest benefits from reduced carbon and pollution costs due to the eco-tax of 112 billion euros [...]. Our baseline [synthetic difference-in-difference] results thus suggest that the eco-tax has reduced external damages by 120 Euros per capita on average per year, amounting to around 0.7 percent of the average purchasing power per-capita in the year 2005.

5.5 Coordination and sustainable technologies

Public goods and externalities are not the only impediments to sustainable technologies. A third type of problem is **coordination failures**: situations where agents have aligned interests but fail to converge on the best strategy. A classic example is which side of the road to drive on — everyone benefits from a common choice, but getting there requires coordination.

5.5.1 The nature of coordination problems

We can illustrate this problem in a simple game theoretic setting. Suppose we have two firms, 1 and 2. Each can implement one of two strategies, L and R . In the benchmark case, we assume that they both want to do use the same strategy but they don't care which one. This is akin to the case of driving: it doesn't matter whether we drive left or right – what matters is that we all drive on the same side of the road.

This can be presented as a 2x2 matrix, as in the table below. Player 1 selects the row and Player 2 selects the column. Payoffs associated to each outcome are listed in the cells. By convention, the first number is the payoff for Player 1 and the second is the payoff to Player 2.

		Player 2	
		L	R
Player 1	L	1, 1	0, 0
	R	0, 0	1, 1

Coordination problem (benchmark). Game with two players, 1, 2 and two strategies, L , R . This yields four outcomes. Numbers in the cells indicate ordinal payoffs.

Payoffs are here *ordinal*: players rank each of the potential four outcomes. In this specific case, we assume that both players prefer coordination ($\{LL\}$ or $\{RR\}$) over non-coordination ($\{LR\}$ or $\{RL\}$), and both sets of outcomes are equivalent. We thus have $u_i(LL) = u_i(RR) > u_i(LR) = u_i(RL)$.

A **Nash equilibrium** in pure strategies is a set of strategies that yields an outcome from which no player wishes to deviate *unilaterally*.⁸ In a coordination game such as the one above, we have two Nash equilibria: $\{LL\}$ and $\{RR\}$. To verify this, note that if we are at $\{LL\}$, and Player 1 deviates from L to R to generate $\{RL\}$, then their payoff declines ($0 < 1$).

What could go wrong here? In principle, nothing: if both players can communicate, they can simply toss a coin and select one of the equilibria. They both want the same and are indifferent between the two equilibria. This is why selecting the side on which cars drive is non-controversial. Failure (that is, reaching $\{LR\}$ or $\{RL\}$) only happens if the two players cannot communicate, or if they have doubts over the other player's preferences.

Coordinating (music) pitches

Composers share music through sheet music. This is a standardized way to tell musicians how to play songs. But how do you set a pitch? Until the 19th century, without electric instruments, it wasn't straightforward to establish a common pitch. This meant that pitches varied from city to city and region to region. A coordination problem!

This problem was solved gradually over a period of about 100 years (Haynes 2002). First, France established a country-wide standard in 1859, with 'A' set at 435 Hz (called the *diapason normal*). In 1885, several central and southern European countries adopted it as well in a joint agreement. This agreement was then included in the Versailles Treaty of 1919, Section II, Art. 282 (!).

⁸By "pure" strategies, we mean that we exclude mixed strategies in which the player places probabilities over all strategies.

Twenty years later, a conference in London (1939) proposed a new standard, with A set at 440 Hz (called the *concert pitch*). This standard was confirmed after World War II by the International Organization for Standardization (1955).

The next example of a coordination problem still has two equilibria (table below). However, both players now want the same equilibrium (here: {LL}, since $u_i(LL) > u_i(RR)$). We say that {LL} is *Pareto superior* to {RR} (and to {LR} and {RL}).⁹ Absent other indications, it is hard to expect the players to fail to coordinate on LL. Only misperceptions about other actors appears to create a threat, but presumably this can be dissipated quickly.

		Player 2	
		L	R
Player 1	L	2, 2	0, 0
	R	0, 0	1, 1

Coordination problem with varying equilibria. Game with two players, 1, 2 and two strategies, L, R. This yields four outcomes. Numbers in the cells indicate ordinal payoffs.

Batteries versus fuel cell electric vehicles

Decarbonizing the transportation sector requires a shift away from cars (or other vehicles) away from internal combustion engines. Two technologies appear particularly likely to succeed: battery electric vehicles (or BEV) and fuel cell electric vehicles (FCEV). For a long time, it wasn't clear which of the two technologies would succeed. Furthermore, initially at least, it wasn't clear whether one technology would eventually be better. In fact, until around 2008, fuel cell EVs were perceived to be more promising, at least for long-range transportation.

From a sectoral perspective, this represents a coordination problem. Producers and consumers, presumably, would like to coordinate one one technology rather than the other. Otherwise, you

⁹Note that this requires no statement about interpersonal comparison of utility. Here, each individual player has preferences that make them prefer {LL}. You could multiply each player's number by 100 or 10,000 and nothing would change.

may have to maintain two parallel set of infrastructures. Dugoua and Dumas (2024) document how the coordination problem was solved. It was *not* solved by an external actor, such as the state, picking an outcome. Instead, it was an exogenous technological shocks (which the Dugoua and Dumas trace via patents, R&D and other data sources) that made batteries suddenly more appealing. Innovations spread rapidly, as car manufacturers could gradually learn from hybrid vehicles (which have both internal combustion engines and batteries), making the transition smoother. These signals were picked upon by producers, who shifted their attention to electric EVs rather than fuel cell EVs.

The last example of coordination problem raises thornier problems. Consider the table below. We still have two equilibria. They still agree that “landing” on equilibrium is better than missing it ($\{LL\}$ and $\{RR\}$ are superior to $\{LR\}$ and $\{RL\}$). However, Player 1 and Player 2 don’t agree on which one is preferable. Player 1 would like $\{LL\}$ and Player 2 would like $\{RR\}$.

		Player 2	
		L	R
Player 1	L	2, 1	0, 0
	R	0, 0	1, 2

Coordination problem with conflicting equilibria. Game with two players, 1, 2 and two strategies, L, R. This yields four outcomes. Numbers in the cells indicate ordinal payoffs.

Absent further information, we don’t know what will happen. We would still anticipate the players to select a Nash equilibrium, but we don’t know which one. If we loosen the game a little bit, we can imagine situations in which Player 1 and 2 are engaged in a fight over the equilibrium to select. This could, in real life, create situations in which non-equilibrium strategies are used. For instance, Player 1 could play *L* and Player 2 could play *R* with the intention of holding a hard line until the other blinks. In the long run, we still expect a Nash equilibrium, but in the short run, we may experience what seems like ‘irrational’ events happening. Box *Coordination and EV plugs battles* has an example.

This does not exhaust the range of coordination games. Another one is referred to as the **Stag-Hare** game. Its inspiration comes from a story by Jean-Jacques Rousseau (1755). Imagine two hunters. They hunt together but are physically apart, without means to communicate. At some point, they face a decision: go after a stag (lots of food but hard to catch) or go after a hare (less food but easy to catch). This can be modeled as below.

		Hunter 2	
		Stag	Hare
Hunter 1	Stag	3, 3	0, 2
	Hare	2, 0	1, 1

Coordination problem with conflicting equilibria. Game with two players, 1, 2 and two strategies, Stag and Hare. This yields four outcomes. Numbers in the cells indicate ordinal payoffs.

This game is subtly different from the second coordination game above. There are two equilibria, one of which is Pareto-superior to the other. So far, no difference. However, introduce another element in the game: suppose that Player 2 is uncertain about Player 1's action and is risk averse. Player 2 notes that if she goes after the stag, she will get her best payoff (3) if and only if Player 1 also does this. Otherwise, she gets her worst payoff (0). In our story, this could be starvation and thus pretty bad. If, however, she goes after the hare, she is guaranteed a payoff of 1. Maybe that's good enough for her. This game thus raises the role of safety and risk preferences.

The problem of uncertainty is often an important one. Instead of hunters, think of firms that must invest *jointly* to achieve some breakthrough. Now, imagine that such firms are worried that their partners are not entirely reliable. For instance, it may be that Firm 2 will soon have a new CEO whose priorities are unknown. Should Firm 1 invest (e.g., in R&D) without knowing if Firm 2 will continue to hold its side of the bargain?

We now have a general sense of what coordination problems consist of. Of course, there is a richer set of models that capture coordination problems, such as global games (Morris and Shin 2001). For now, this gives us a sufficiently large set of analytical frameworks to play with.

5.5.2 Coordination failures: innovation

New products are bundles of new technologies. A laptop requires simultaneous progress on motherboards, batteries, screens, and more. A firm specializing in screen materials faces a familiar question: invest in R&D without knowing whether the battery producer will do the same? Any complex product depends on a web of complementary technologies developed by actors with different goals and constraints. This partly explains why innovations often emerge from individuals who value discovery for its own sake — they don't wait for the full ecosystem to align.

Another type of innovation failure comes from the complexity inherent to ambitious ideas. Whereas the public good game captures the challenge that is inherent to *marginal* changes to products, creating new technologies altogether adds a layer of difficulty.

We can think of this as a **coordination problem** with multiple equilibria. Let us go over a model that captures its spirit. The model is inspired by threshold and *weakest link* public good games (Hirshleifer 1983; Palfrey and Rosenthal 1984; Varian 1994; Azoulay et al. 2019).

We have N identical agents (firms, labs, etc.). Each agent chooses either to *invest* in an ambitious project (action 1) or *not* (action 0). The cost of investing for agent i is $c > 0$. There is a large potential social benefit $B > 0$ (e.g., commercially viable fusion energy) that is realized if the project succeeds. We assume that this benefit is a public good accruing to all if the project succeeds. Success, in turn, depends on aggregate effort.

Here's how the game is structured. Each agent chooses $a_i \in \{0, 1\}$ (0 = not invest, 1 = invest). Let $S = \sum_{j=1}^N a_j$ be the number of investors.

Now, we assume that the project succeeds if, and only if, the number of investors passes some threshold k .

$$\text{Project succeeds} \iff S \geq k,$$

for some threshold integer $1 \leq k \leq N$. That is, we need k firms/labs/teams to invest in order to make the project viable.

We can write down the payoff to agent i as:

$$u_i(a_i, a_{-i}) = B \cdot \mathbb{I}\{S \geq k\} - c \cdot a_i,$$

where $\mathbb{I}$ is an indicator that the condition is met.

What will happen? We can list i 's *best responses* to what other firms are doing. For every set of actions by all other actors (denoted $-i$), we identify what i should do. There are three scenarios:

- If $S_{-i} = \sum_{j \neq i} a_j \geq k$, then investing ($a_i = 1$) is dominated (i.e., it is inferior to at least one other strategy). The project succeeds without i anyway, and $a_i = 0$ still yields payoff B . No need to spend resources.
- If $S_{-i} = k - 1$, then agent i is *pivotal*: choosing $a_i = 1$ gives $B - c$, choosing $a_i = 0$ gives 0. Agent i invests as long as benefits are larger than its (private) cost, i.e., $B \geq c$.
- If $S_{-i} \leq k - 2$, a single agent cannot reach the threshold alone. Investing simply means that you spend and pay $-c$ without any benefit.

Thus, this game has multiple equilibria:

1. **No-investment equilibrium:** $a_i = 0$ for all i . Nobody invests and nobody has an incentive to change their mind alone.
2. **Coordination equilibria:** any profile with $S = k$. In that case, nobody on the team has an incentive to jump ship (the benefits would be lost), nor does anyone off the team want to join (since they would only pay without any additional benefit).

Which equilibrium obtains depends on agents' expectations.

What would a social planner want? The planner compares total social benefit to total cost. If success occurs when $S \geq k$, planner finds it desirable to reach the threshold when:

$$B \geq kc \quad (\text{social benefit exceeds total cost of } k \text{ investors}).$$

If B is large relative to kc , the planner would like to induce the k investors — but private incentives may fail because each agent fears being the only investor.

What are the policy implications? We see that this large-scale project might fail because individual agents are worried to contribute to it (and thus spend resources) even though it might be ineffective. Without public intervention, this means that some of the time, we end up in an equilibrium where no action is taken. Thus, one of the state's role is to ensure that such concerns are alleviated. This could mean:

- Provide guaranteed compensation for the first k teams. Thus, firms don't have to worry about reaching the threshold, since they get something out of it. This requires the compensation to be greater than c .
- Create private benefits. The project's value is B , but firms could be given some extra, private benefits. For instance, if the project generates new intellectual property, participating firms might receive privileged use of these new technologies.
- Organize the project under public resources. You build national labs and universities and task them with achieving this goal.

Examples include a range of **moonshot** initiatives, such as the race for the moon in the 1960s or ARPA (DARPA) (Azoulay et al. 2019).

5.5.3 Coordination problems: commercialization stage

Coordination problems arise at several stages of commercial development for sustainable technologies.

Technological standards. Suppose your product is now entering its growth phase. The product is nearly mature from a technological perspective. But suppose that it has network properties. The demand-side would like to see a standardized product, but the market is not there yet. Phone consumers don't want seventeen different types of charging stations. Closer to us, EV buyers would like to see a unified set of charging stations. See Box.

Likewise, the supply-side would like to see unified demand for the product. For instance, suppose that you are developing wagons for a maglev, a levitating train. You would like all countries and cities that buy from you to want the same specifications (e.g., width, wattage) to reduce your costs. You don't necessarily care what these specifications are, but having City A ask for something different from City B is not helpful. Since sustainability is often related to technologies and services that have a strong network features, standard problems often slow down their deployment.

💡 Coordination and EV plugs battles

Coordination have plagued the spread of electric vehicles in several ways. One relates to plugs and recharging. Initially, each EV producer designed their own plugs. This is inconvenient for users. If you stop on the highway, you want to be able to charge regardless of the brand of your EV. In contrast, ICE cars have standardized filling tubes (with different tubes for different fuels to avoid blowing up your car by accident).

Eventually, EV manufacturers will therefore want to converge on a single plug technology. The benefits from accrued demand should be worth it. But which standard will be chosen? Each producer would like their technology to be chosen. This allows them to avoid having to retrofit their production line. And they maybe have other benefits attached to it.

This is indeed what happened. In the US, one producer after another converged to Tesla's standards. All except one: Stellantis, which resisted for a long time. We can model this as in the table below. Tesla preferred the SAE J3400 standard, whereas Stellantis preferred CCS. For a while, consumers were trapped in a world in which this market was off equilibrium at {SAE; CCS}. Maybe from a consumer's perspective, which standard would win didn't matter. But they certainly wanted one to win.

		Stellantis	
		SAE J3400	CCS
Tesla	SAE J3400	2, 1	0, 0
	CCS	0, 0	1, 2

Coordination problem with conflicting equilibria. Game with two players, Tesla and Stellantis, and two strategies, SAE J3400 and CCS. Numbers in the cells indicate ordinal payoffs.

And who won? Tesla. In 2024, Stellantis announced that it would use Tesla's charging plug ([The Verge](#)). One year later, Honda and Acura marketed a new EV charging adapter to allow their customers to use one of the 23,000 Tesla Supercharger stations ([The Verge](#)).

Industrial development. Some transitions require system-wide coordination. Decarbonizing the power sector illustrates this: adding a

few wind turbines is straightforward, but building a grid around intermittent renewables is not. A renewables-heavy system needs storage — batteries, hydrogen — supplied by actors not currently in the market. Storage firms won't invest unless solar deployment is certain; solar deployment is uncertain without storage. This is a coordination problem with multiple equilibria. Government intervention is warranted when a superior equilibrium exists but cannot be reached privately. The barrier typically lies in one of:

- Coordinating other actors (e.g., suppliers) is too costly?
- Uncertainty regarding how credible the commitment of other actors are?
- Payoffs are too uncertain?

When coordination fails and precludes industrial development, states like to undertake **industrial policy**. Juhász, Lane, and Rodrik (2023, 216) define industrial policy as:

We define industrial policies as those government policies that explicitly target the transformation of the structure of economic activity in pursuit of some public goal. The goal is typically to stimulate innovation, productivity, and economic growth, but it could also be to promote climate transition, good jobs, lagging regions, exports, or import substitution. Since industrial policy targets structural change, a key characteristic is the exercise of choice and discretion by the public authorities, as in “We promote X but not Y,” though the latter part of this statement is typically left implicit.

Industrial policy has become common to speed up the development of sustainable technologies. This includes industrial policies in domains such as the green sector generally (Rodrik 2014) and the renewable energy sector in particular (Gerarden, Reguant, and Xu 2025). See *Box*.

Examples of industrial policies

Policies akin to *industrial policy* have been around for a long time (Juhász and Steinwender 2024). After an apparent decline in the 1980-90s (at least in the West), states have started to use them again in a range of settings, including sustainability:

- The Inflation Reduction Act and the CHIPS Act under US

President Joe Biden (Bistline and Wolfram 2025). Includes several instruments to help energy communities and promote the clean energy transition (notably via batteries).

- The European Green Deal (Fetting 2020).
- China's solar industrial policy (Zhi et al. 2014), including the "*Tenth Five-Year Plan*", which calls for an increase in the production of solar, wind, and geothermal energy.

Each of these programs have an "industrial policy" flavor, in that they coordinate large-scale economic transformations.

The consequence(s) of industrial policy

In recent years, several studies have suggested that industrial policy can be successful. (We set aside the problem of defining what qualifies as industrial policy.) For instance, Mazzucato (2011) argues that states have played a successful role in the innovation processes of several new technologies. Even when the goal is not industrial policy per se, compatible policies such as protectionism have occasionally generated domestic development, as France showed during the Napoleonic wars (Juhász 2018).

Yet there are several reasons to be skeptical of such claims. One is a problem of *design* of such studies: the select cases that are possibly known because they were successful. For instance, consider the study by Juhász (2018) who looks at regions of France that were more or less protected from trade during the 1803-1815 period. She shows that regions that were protected increased production capacity in the textile sector (thanks to mechanized cotton spinning). The causal link is rooted in the notion of **infant industry**: some industries need temporary protection to overcome internal coordination problems (and possibly other market failures) to blossom.

Let's take Juhász (2018) for granted. The question is: what about all the other cases where trade protections were implemented and industry didn't grow? The concern is a form of **selection on the dependent variable**: you study cases based on the outcome. Maybe trade barriers in Country XYZ had the opposite effect, but, for some reason (e.g., lack of data or obscurity of the case), we don't know about it.

Same for Mazzucato (2011): what about technologies promoted by the state but that didn't work out? An example that many observers use is France's infatuation with the 'minitel' (a precursor to the Internet). The French government pushed hard for it (Mailand and Driscoll 2017), but it never took off outside of France, and is believed to have slowed down France's transition to the internet. If we say that the state was good for Technology A, then shouldn't we also flag when it is bad for Technology B?

Consider today's debates about renewable energy. Several states have used industrial policy to prop up domestic production (e.g., the United States with solar PV). These policies can be very expensive and generate negative side effects in the long term. So we need to exert some caution when assessing whether industrial policy (or any policy, for that matter) is desirable.

The critiques of industrial policy

The modern case for industrial policy, as presented in Juhász, Lane, and Rodrik (2023), is pretty straightforward:

1. It solves externalities. For instance: we need IP protection against R&D spillovers.
2. It solves coordination failures. For instance: it only makes sense to sell computers if the grid is reliable.
3. It provides 'activity-specific public inputs'. For instance: the state must provide certain infrastructure.

But what about the negative case? Juhász, Lane, and Rodrik (2023) flag two main problems.

- **Information failure.** Identifying these failures and calibrating solutions is very difficult. Suppose you're elected head of state tomorrow. How do you know whether to invest in broadband for all or not? Is it going to unlock growth somehow? And how do you know which technology to select? Should you go for solar or wind? What if you accidentally slow down an even better technology (the problem of *picking winners*) (Krugman 1983). The strength of markets is precisely to aggregate decentralized information (e.g., about which technologies are promising and which ones aren't).

- **Political failure** (Juhász, Lane, and Rodrik 2023). Should we assume that governments and civil servants have preferences that are aligned with the public's? And can they always manage resources optimally? Clearly, that's not always the case (Keech and Munger 2015). Industrial policy concentrates a lot of resources into the hands of a small number of people. It's easy to imagine some engaging in rent-seeking and corruption (Rowley, Tollison, and Tullock 1988). We sometimes talk of *government failure* (Le Grand 1991; Tullock, Brady, and Seldon 2002).

Additional readings

Aside from the resources mentioned in the chapter on demand, see also Goulder and Parry (2008) on environmental policy instruments. For textbook treatments on public economics, see Hindriks and Myles (2013).

Part III

Policies and Institutions

6 Policy design

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Summary

This chapter investigates the problem of **policy design** — how to construct interventions that achieve their goals. We treat policymaking as a technical optimization problem, examining the Tinbergen framework, how policy operates in simple versus complex systems, the risks of adverse side effects, and strategies such as policy portfolios and sequencing. We also examine economic and bureaucratic constraints on implementation. Throughout, we assume that policymakers have the political space to act; the next chapter examines what happens when that assumption fails.

6.1 Policymaking process

A very simplified way to describe the policymaking process is presented in Figure 6.1. The first stage (*problem identification*) roughly corresponds to what we discussed (Chapter 2 and 3). We then spent time to understand what causes problems in the specific context of market economies (Chapter 4 and 5). We tackled here and there the type of policies that may address the market failures we identified. This was a soft introduction to the second stage (*policy design*).

That's only part of the story. What follows next is a process that leads to the decision to *adopt* a policy. In many countries, this is done by *parliament* (Congress, Assemblée Nationale, House of Commons, Conseil National and Conseil d'Etat, etc.). Once a policy is adopted, then civil servants (the bureaucracy) must translate it into laws, regulations, and conduct the monitoring and punishment needed for the policy to have

bite. Finally, policies are evaluated (by citizens, by politicians, by governments, etc.), which will at a later stage lead to the formulation of new problems.

What we will do next is to review systematically how policy should be designed (stage 2). Then, in the next chapter, we discuss the **political constraints** that limit the kind of policies that can be implemented, with a focus on policy adoption (stage 3) and bureaucratic implementation (stage 4).

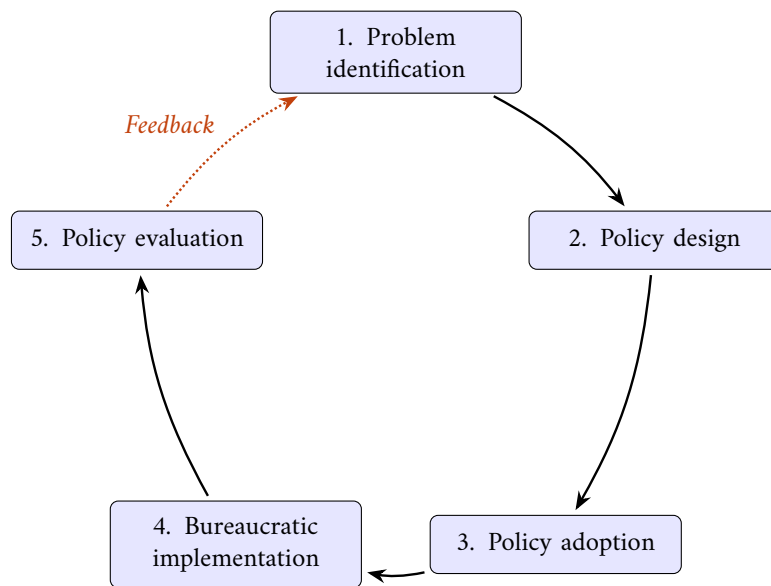


Figure 6.1: The policy cycle.

6.2 A catalogue of policies

Let us briefly recall the different types of interventions that we have seen so far. To organize them, let's divide public policies between those helping the *onset* of new technologies and those helping their *diffusion*. Across the board, we focus primarily on the market failure framework: what kind of interventions are needed to help markets function properly. Note: this catalogue is not exhaustive by any means!

6.2.1 Technology onset

We assume that new technologies is fundamentally a function of **human capital** (our collective brainpower and knowledge) and of returns to that capital (G. S. Becker 2009). New technologies emerge in general when human capital is high, and from markets in particular when firms have a sufficient return from them.

This raises the following problem:

- Who is going to invest into people to increase their human capital (*inputs*)?
- Who is going to invest into research (*outputs*)?

This problem arises because of knowledge spillovers. As a firm, why would I educate employees if they will leave me after? And as a firm, why would I invest into research if the output (a new technology) is a public good?

In both cases, we have a *public good problem*. But we can also have a *coordination problem*. Knowledge often exhibits strong complementarities (or supermodularity). For instance, GPS were made feasible (and precise) thanks to the combination of atomic clocks and relativity. New ideas are often born from combining other ideas in novel ways (S. Page 2008).

In many cases (but not universally), what we observe is a two-pronged strategy by States:

1. Human capital (input): provided by the State via public schools. The State funds and provides education from the primary to the tertiary level. Private schools exist and can be particularly important in some countries (e.g., United Kingdom) and at some levels (e.g., universities in the United States).
2. Research (output): (a) direct public and private provision and (b) indirect regulation. *Direct initiatives* include national research labs, like the Paul Scherrer Institute (nuclear research in Switzerland), the Joint Research Centre (general research for the European Union), the National Renewable Energy Laboratory (NREL), or the Manhattan Project are public solutions to the joint public good and the coordination problem. *Indirect regulation* helps make private labs (i.e., R&D departments) more

valuable. This includes protecting intellectual property (e.g., via subsidies for R&D, grants, patents).

6.2.2 Technology diffusion

Here, we have a much broader set of policies that may be useful depending on the exact source of market imperfection.

First, we have demand-side policies.

Knowledge. Consumers don't know about the benefits (or costs) of a technology. *Information and awareness campaigns* can reduce information gaps. Sometimes, these are combined with *behavior nudges* (attempts to use behavioral science to change people's actions). For instance, households may not realize that their water consumption is very high; adding information about their consumption levels compared to their neighbors can induce them to moderate their own consumption.

Cost. A new benign technology is initially expensive, at least until firms learn to reduce costs. *Consumer credits* can increase the affordability of new technologies. For instance tax credits can encourage consumers to build solar panels on their rooftops (e.g., *Residential Clean Energy Credit* in the United States).

Second, we have supply-side policies.

Negative externalities. Some undesirable technologies produce negative externalities that can be internalized.

- (1) Policies that target the price of externalities. *Pigouvian taxes* make them less appealing to consumers by increasing prices (e.g., carbon tax), letting actual quantity of the externality vary.
- (2) Policies that target the quantity of externalities. These include *permit trading systems* in which actors must buy permits that they can trade; quantities are fixed, while prices fluctuate (e.g., European Union Emissions Trading System).
- (3) Finally, another approach is to use *standards* to reduce the prevalence of such technologies. For instance, car emissions standards prevent some kind of vehicles from being used. Likewise, some countries impose limits on when (polluting) cars can be used (e.g., based on plate numbers).

Positive externalities. Some desirable technologies are not compensated for their broader beneficial effects. *Pigouvian subsidies* can help them capture some of these rents.

International externalities. Some externalities cross borders (in one or in both directions). International negotiations produce *treaties* that can address the problem. Such agreements can explicitly list the kind of interventions that culprits will undertake. Some agreements will also include compensatory measures (e.g., foreign aid).

In the absence of international treaty, states might try to impose regulation using their market power. We talk of a *California effect* when a market is so large that its regulations are adopted by others to preserve access to that market. For instance, the EU imposes tariffs on goods produced in countries that don't have a carbon tax or carbon trading system comparable to the EU ETS; this is called the Carbon Border Adjustment Mechanism (CBAM).

Coordination failures. Some goods and technologies depend on complementary goods and services. For instance, a factory needs electricity and water. In some cases, these complementary inputs are geographically important (e.g., Silicon Valley). *Industrial policies* are tools to coordinate public and private investments to break these deadlocks.

Uncertainty. Firms may hesitate to invest in new technologies given uncertainty about future prices. For instance, a utility might be unwilling to build solar PV farms without knowing if the price of electricity will be sufficiently high. In the power sector, a *feed-in tariff* (FIT) guarantees a certain price to producers. Similarly, a *Contract for Difference* (CfD) guarantee a fixed price to electricity producers. If the market price is below that fixed price, the state pays the difference; if the market price is above, the producer gives the surplus back to consumers.

Common goods. States sometimes manage common goods (rival but non-excludable). For instance, many States run natural parks. This can lead to overuse.

- One policy is to make the good *excludable*. For instance, impose an entry price for visitors of a public park or a pool. There is a range of related policies that can help avoid the problem (E. Ostrom 1990), including limiting inputs, outputs (how much is extracted), and other collective agreements.

- Another policy is to *privatize*, i.e., assign (segments of) the good to private operators. This is often done via *auctions*. For instance: airwaves are often assigned to radios and TV channels via auctions. Similarly, new plants to supply electricity are also auctioned off by the State.¹

Market structures. Markets operate well, in general, when no actor has excessive market power. In practice, this means that there are no firms with monopolistic (single seller) or monopsonistic (single buyer) power. Otherwise, they could use that power to make goods excessive expensive (monopolist) or cheap (monopsonist). Policies addressing these problems include *anti-cartel* measures (breaking up firms), *nationalization* (taking over the monopolist and then cutting down prices),

Problem to be solved	Policy	Example
Underinvestment in human capital (knowledge spillovers; public-good problem)	Public education	Public schools
Coordination failure in research	Research centers	Paul Scherrer Institute; NREL; Manhattan Project
Appropriability of R&D benefits (externality)	Intellectual property, R&D subsidies, etc.	Patent protection for inventions
Imperfect consumer information about a technology's benefits/costs	Information & awareness campaigns	Showing households their water use vs. neighbours'
Affordability of technology	Consumer/tax credits	Residential Clean Energy Credit
Negative externality	Pigouvian tax	Tax on polluting / high-emission vehicles

¹The power sector isn't really a common good, but you can think of it as such if you think that the state only wants a certain level of capacity in place.

Problem to be solved	Policy	Example
Negative externality (command-and-control)	Standards	Car emission standards
Negative externality, acting directly on quantity	Quantity restrictions	Plate-based limits on when cars can drive
Uncompensated positive externality	Pigouvian subsidy	Subsidy for education
Coordination failure across complementary goods/inputs	Industrial policy	Innovation Parks
Price uncertainty	Feed-in tariff/Contract for Difference	CfD
Overuse of a common good (rival, non-excludable)	Common good exclusion	Entry price for a public park or pool
Allocating a common good to operators	Auctions	Auctioning airwaves to radio/TV; auctioning new power plants

6.3 Policy goals and criteria

In one of the first chapter of these notes, we had discussed families of social and ethical goals. As we now think about policy design, let's briefly revisit this point.

First, what is the **goal** of public policies? We can think of two potential answer:

1. Enforce (deontological) norms
2. Improve social welfare (consequentialism)

Of course, not all goals neatly fit in these two categories. Some libertarians would probably say that the goal of public policy is to enforce

property rights, but this could be done on deontological or on consequentialist grounds. Furthermore, saying that your policy is to have no policy is hard to include in this typology.

If you pick (2), then you have another step to complete: what is your social welfare function?

- Utilitarian? If so, how do you aggregate utilities? And how do you measure it?
- Egalitarian? Of what kind?
- Rawlsian?
- etc.

You then will select policies that meet these goals. Some will be about providing public goods (education, etc.), reducing public bads (reduce pollution), solving information asymmetries, etc.

Now, cutting across these perspectives, we must also think about the **criteria** to evaluate policies *ex post*. What criteria would you use, after the facts, to assess whether a policy *A* was the right choice over *B*, *C*, ...? Again, we can revisit some of our earlier concepts.

- Effectiveness: a policy is successful if it achieves its goal.
- (Pareto) Efficient: a policy is successful if it achieves its goal at the lowest cost.

These two criteria may seem common sense, but they can raise tricky problems. For instance, should you, as the policymaker, take into account popular backlash when computing the probability of success of Policy *A*? Perhaps, but that's not typically done when looking for efficient policies. Conventional models of welfare generally assume that policymakers can neatly implement their ideas. In practice, the *politics* of policy can make things extremely messy. That's why we sometimes refer to strategies that are optimal on paper as **first best**, with the caveat that we may sometimes have to rely on **second best** policies instead. What we do for the rest of this chapter is twofolds: (1) how to design policies in general ('on paper'), and (2) how to design robust policies that are implementable.

💡 Taxing windows

To do stuff, states need money. The main source of income of states are taxes (with some exceptions, like petro-states). Yet (most) people don't like to get taxes, all things equal. They want the benefits of taxes (e.g., public goods), but they would rather free-ride and enjoy the benefits of the money that they would keep. As a result, historically, we observe a constant struggle between the state (which looks for new sources of funds) and people (who try to hide their savings).

An example comes from England (Oates and Schwab 2015). In 1696, William III was in dire need of money. There was a war with France (the Nine Years' War) and England itself had just gone through the Glorious Revolution (when William replaced James II). This requires a lot of cash: you must pay your allies and your army. Otherwise, your head would be at risk, since the aristocracy and the army could decide it best to do another revolution.

The King thus decided to impose a tax on housing, which was politically easier to do. It's also an asset that's hard to hide (unlike gold that you can stash; in French real estate is called "*immobilier*", i.e., unmovable). But how do you do so in a way that's implementable by tax collectors? At the time, you didn't have registries with surfaces of houses, so you can't use the size of the house if you need to do it fast. Nor did you have a market price for housing that you could use to guess the value of buildings (which is how real estate taxes are often done in the United States today). The solution: take a proxy that is easy to observe. What the King did is to use the *number of windows*. If you have lots of windows, you are likely to have a big house. And you won't to use a ruler to find out how big the house is, nor argue its value. Oates and Schwab (2015, 164) explain it:

the King levied a new tax consisting originally of a flat rate of 2 shillings upon each house and an additional charge of 4 shillings upon houses with between 10 and 20 windows and 8 shillings upon houses with more than 20 windows...

Now, there are two things that are interesting. First, once a tax is implemented, it's hard to get rid of it. The state is now using these

resources, and if you don't cut spending, there is little incentive for the state to eliminate this tax. That's why taxes tend to be asymmetric: they are easier to increase than to decrease (with some exceptions, of course). In this case, the tax lasted a long time. Oates and Schwab (2015, 164):

The tax was intended to be a temporary levy, but it was restructured and increased several times. In the end, the window tax lasted in various forms for over 150 years; as we noted above, it was not repealed until 1851.

Second, taxation (or other policies) never happen in *ceteris paribus* (holding everything constant). People have agency and can decide to respond strategically to state action. Can you guess what they did? Well, if you get taxed by number of windows, why not reduce the number of windows? In cities, it was often the landlord who paid the tax. Landlords would then promptly board off windows to reduce their fiscal burden. This, however, had secondary consequences on public health. Oates and Schwab (2015, 165)

... the renter, not the landlord, paid the tax. However, large tenement buildings in the cities, each with several apartments, were an exception. They were charged as single residences with the tax liability resting on the landlord. This led to especially wretched conditions for the poor in the cities, as landlords blocked up windows and constructed tenements without adequate light and ventilation...

England was by no means the only place to have such taxes. France had an “*impôt sur les portes et fenêtres*”. See wikipedia for references ([link](#)). The question then is: was this efficient? Probably not. The tax could be gamed and it created considerable long-term issues (think of health impact). But we have to put this in its historical context. Without a well-trained bureaucracy, lots of data, and an effective law enforcement agency, it was hard to implement reasonably efficient forms of taxation. This might be the case of a *second best* (or maybe even *third best*...).



Figure 6.2: Strategic responses to taxation. Source: wikipedia.

6.4 The nature of policymaking

Policymaking is an intervention by a public authority to modify the state of the world in a particular direction. We treat it here as a consequentialist optimization problem — the goal is a state of the world that maximizes social welfare, possibly subject to constraints. We leave aside which specific objective society chooses (see earlier chapters on ethics) and focus on how to achieve it.

Tinbergen (1956) offers a useful taxonomy of policy interventions by level:

1. **Foundations.** These are the “fundamental elements in the organization of human society” (p.4). Think of these as institutions that run our (public) lives: state institutions (education, security, etc.), religions, and so forth. Foundations are transformed by **reforms**.
2. **Structures.** These are the “details or economic and social organization” (p.4). We are one step below foundations. Structures include the way institutions are designed (e.g., division of education into primary, secondary, and tertiary school) and how agents are set up (teachers’ union, parent associations, etc.). Structures are modified by **qualitative policies**.
3. **Instruments.** Parameters that characterize the economy. For instance, this could be the legal number of students per classroom, tuition fees, etc. Instruments are defined by **quantitative policies**. These policies can be *fixed* (“set a cap at 24 students per classroom”) or *flexible* (“maximize average PISA score”).

This division is illustrated in Figure 6.3. It is *directional*: a reform can change foundations, structures, and instruments; however, a quantitative policy cannot generally change a foundation (Tinbergen 1956, 7).

Means	Foundations	Structure	Instruments
Interventions	Reforms	Qualitative policy	Quantitative policy <i>Flexible vs fixed</i>
Frequency	Rare	Occasional	Frequent
Example	Health insurance	Set of taxes	Max premium

Figure 6.3: The levels of policymaking in Tinbergen (1956). *Means* describes the environment in which the policymaker operate. *Interventions* label the type of actions available to the policymaker. *Frequency* indicates how often each intervention is typically implemented.

While Tinbergen doesn’t make the distinction, we will classify *reforms* as a type of *institutional intervention* and not as being part of the *policies* family of interventions. The latter will be reserved for qualitative and quantitative policies. The reason is that we understand *policy* to be an intervention at the margin rather than a profound reorganization of the rules of society.

6.5 Policymaking in simple systems

Simple problems are well-defined, with no nonlinear responses, no tipping points, and no feedback dynamics. Policy choice reduces to selecting an instrument that (a) achieves a given goal at (b) the lowest cost.

The goal itself must be specified. The most common approach is to identify a market or collective action failure — a negative externality, an undersupplied public good — and define success as correcting it. Other goals are legitimate too: concerns about distributional inequity may warrant redistribution even when no market failure is present.

A guiding principle for instrument choice in this setting is the **Tinbergen Rule** (Tinbergen 1952, 1956; Kenneth J. Arrow 1958). In a nutshell,

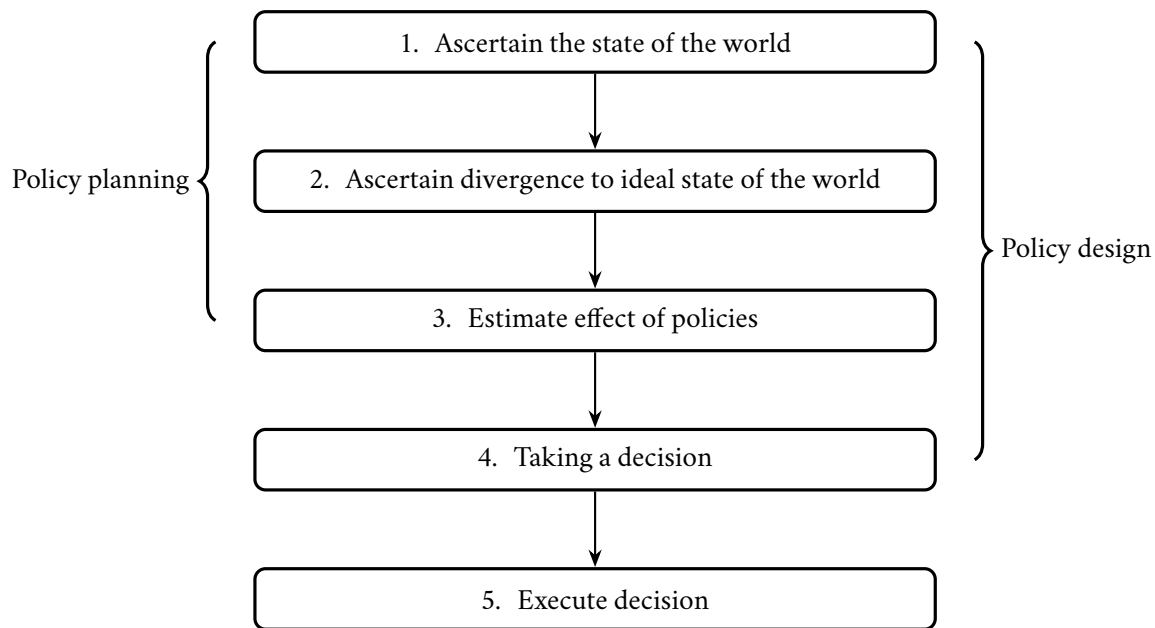


Figure 6.4: Stages of policymaking according to Tinbergen (1956).

Tinbergen argued that a policymaker will need at least n distinct policy instruments to achieve n distinct policy targets. Tinbergen's point is that there is generally a lower bound on the number of policy instruments needed when you have multiple goals. There is also the converse point. If you have n goals and only $n - k$ policy instruments, then you are likely to face situations in which you won't be able to achieve all of your goals (targets) and will need to compromise. For instance, imagine that you have two goals: (1) universal electricity access and (2) decarbonization, and you have a single instrument (say, building gas power plants), then you can achieve (1) or (2) but not both.

Regardless, the key point is that goals and how to achieve them is at the heart of policymaking. A core theme of this book is to move beyond the *benign dictator* model. Policymaking is embedded in a political environment that imposes constraints: the best-in-theory policy may prove deeply unpopular in practice. The contrast is illustrated in Figure 6.5: the top panel shows the technocratic ideal; the bottom shows the political reality.

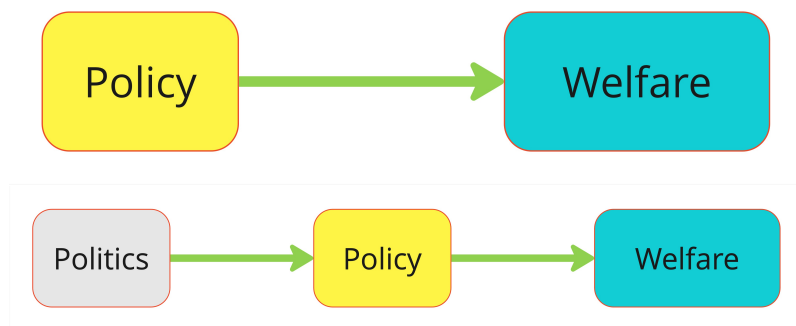


Figure 6.5: Policy process, from a technocratic (top) to a political problem (bottom).

Additional readings

The literature on public choice took off in the 1950s, notably with Kenneth J. Arrow (1951), Buchanan and Tullock (1962), and Olson (1965). A good textbook (now a bit dated) is Mueller (2003). See also Austen-Smith and Banks (2000) and Austen-Smith and Banks (2005). For textbooks in political economy, see Drazen (2000) and Persson and Tabellini (2002).

7 Politics of public policy

This version: September 28, 2026

Summary

This chapter investigates the problem of the **politics** behind policies: what explains their observed design and what kind of issues may arise that make policies depart from what you might think is adequate.

7.1 Policymaking process redux

We had introduced a stylized model of policymaking in Figure 6.1. That figure showed that policy typically follows stages from its design to its implementation.

These stages are not trivial. A social problem is not solved solely by finding the right policy (and technology). What matters is to understand the kind of constraints that the policymaking process can impose on the type of policies that are feasible.

7.2 Policy adoption

7.2.1 Legislative constraints

In many (democratic) countries, policies are rooted in **laws** and other regulatory instruments. The body that writes and votes on laws is the parliament or the legislature. Thus, if you are a minister, you must make sure that the parliament will indeed adopt the law you want.

There are a few things that could go wrong in the legislative process. First, the legislative process could be stopped by **veto players** (Tsebelis

2002). A veto player in an agent that cannot design policies but that can *block* them. For instance, in some countries, whoever runs the legislature has the power to block bills (laws) even if they would in principle have a majority.

Second, even if a bill reaches parliament, you must still obtain a majority to get it through. Here comes another key result: the **median voter theorem** (Downs 1957). The idea is the following. Suppose that you have k members of parliament who must vote (yes/no) on a bill. Suppose that the only thing that matters is how aligned the bill is to their ideology. We can think of ideology as a latent variable defined as a single dimension. For instance, you can think of each legislator being positioned somewhere on a line that represents ideological views from left- to right-wing. The median voter theorem then states that the winning bill will be located at the point corresponding to the ideology of the median legislator.

Third, legislators (or other voters) must typically pick an outcome from a set of potential outcomes. In some places, there are agents that have **agenda-setting** powers (Black et al. 1968; McKelvey 1976). These actors can influence which potential outcomes will be voted on, and how votes are sequenced. A skilled politician can therefore organize a set of votes such that the outcome is good for them, but not necessarily for you.

Tip 1: Median voter theorem and agenda setting

Imagine a committee with five members. They are all defined by a single variable: their ideology, which is a real number. Let's go through an example in which their ideological preferences (sometimes called "ideal point") are:

$$\{-2, 0, 3, 7, 8\}$$

Thus, the most left-wing legislator has an ideology score of -2, the second most left-wing has a score of 0, etc. Let's also assume that they have symmetrical preferences centered on their ideal point. So Legislator #2 prefers a law located at 0 the most and is indifferent between a law that is scored as -1 and another as +1 (but both are worse than a law located at, say, 0.2). For instance, think of this dimension as the stringency of a carbon tax, where -2 is a very strong carbon tax and 8 is nearly zero.

Suppose now that you want to get a bill through. Legislators will go through a series of pairwise contests between bills and must choose which one they will support. Each contest is decided by simple majority. Suppose that anyone can bring up a bill for vote at any time.

Where will the final bill be located on this left-right axis?

Let's suppose that the first version of the law is located at 7.5. But then, the left-wing legislators could say: "wait a second, let's draft another bill situated at 7.1." The reason is clear: if these five members have a choice between a bill located at 7.1 and 7.5, four legislators would vote for the 7.1. Only the most right-wing legislator will pick 7.5 (because 7.5 is closer to 8 than 7.1 is).

But these left-wing legislators can do even better: why not pick a law at 2.5? The two most left-wing legislators (#1 and #2) will like it: 2.5 is much closer to -2 and 0 than 7.5 is. Even Legislator #3 will also prefer 2.5 (half a point away from their ideology) to 7.5 (which is 4.5 units away). Thus, they would have a majority of 3 votes in favor, and 2 against.

This isn't the end of the story. Now the right-wing legislators can find a better law (from their perspective) that could gather a majority. Why not pick a bill situated at 3.2? Legislator #4 and #5 clearly prefer this. And Legislator #3 prefers it too (3.2 is closer to 3 than 2.5).

We can continue like this to finally realize that we will converge to the position of Legislator #3, who is the median voter. The reason is that any winning bill must include him/her. This gives that person a lot of power and to get the final bill to be exactly what they want.

Now, consider a variant of this situation in which Legislator #4 is the **agenda setter** (Black et al. 1968). This means that Legislator #4 happens to have the right to pick the bills that will be considered or not. If #4 has complete discretion over the bills, then what should they do? They can bring up two bills such that the first one is their ideal point (7) and the second is a competitor that cannot win (many values would fit, such as 7.2 or -8). This "false" choice ensures that their preferred outcome arises. That's why agenda setting positions are coveted.

7.2.2 Electoral constraints

7.3 Policy implementation

Suppose a policymaker has well-defined preferences and a coherent policy design that was adopted by parliament (or whoever is in charge). What real-world constraints could still prevent her from succeeding? We focus on two:

- Economic constraints
- Bureaucratic constraints

Political constraints — how power is distributed, how preferences aggregate, and how policies generate political backlash — are examined in the next chapter.

7.3.1 Economic constraints

The state engages in two primary sorts of activities:

1. Production of (possibly public) goods.
2. Transfers from one group of agents to another.

Either of these activities (outputs) requires resources (input). Inputs include human capital (e.g., teachers, tax collectors). But they especially require money. That's why taxation is so important: without taxation, what resources can a state use to fund its activities?

Thus, states tax. They tax all kinds of things, including:

- Income
- Wealth
- Transactions (value-added tax, or VAT)
- Inheritance
- Imports (taxes on imported goods are called tariffs)

Yet how much revenues they can extract is not self-evident. Take income: could a government just tax everyone's full income? No. That's partly the idea behind the **Laffer curve**. The logic is the following. Start from a world with no taxes. Increasing taxes to 1% (e.g., of income) will generate state revenues. So the effect is positive. However, imagine that you start at a 99% tax rate. Will increasing the tax rate to

One answer is looting and extortion, which is what many agents have done in history, notably via wars.

100% increase revenues? No: why would anyone work if the marginal income you make is taken by the state? If my next hour of work doesn't yield any payoff for me (as a worker), why would I not instead do something else, like spend time with my family? Thus, at high levels of taxes, more taxes have a negative effect on public finances. This implies that the relationship must be an inverted-U of some kind, with a maximum somewhere.

Note. The max of the Laffer curve maximizes state tax revenues. But it doesn't mean that this is optimal in a social welfare sense. To make a welfare claim, we would have to say that each additional tax revenue is used better by the state than by the person who earned it. This may or may not be the case.

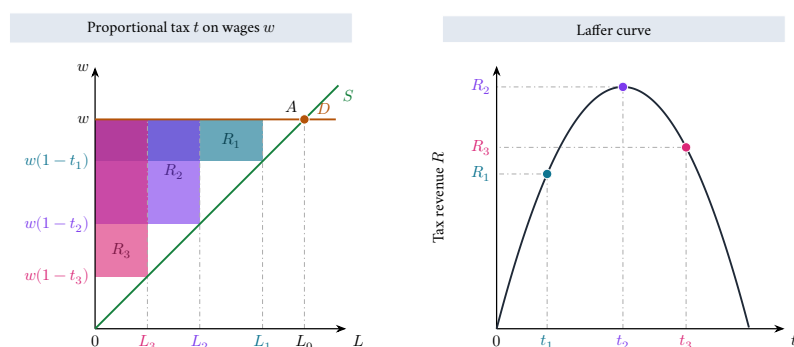


Figure 7.1: Deriving the Laffer curve from the labor market. **Left panel** shows a labor market, where supply is increasing in wages (green line) and demand is assumed to be flat (orange line). Tax rates reduce the amount of labor supplied. Government revenues correspond to the rectangle between gross wages w , net wages $w(1 - t)$, and the new equilibrium. **Right panel** shows how the size of tax revenues (y axis, corresponding to the surface of the rectangles on the right) change as the tax rate (x-axis) increases. The max is achieved at t_2 .

The Laffer curve explains the constraints that happen assuming that *politicians/civil servants are free to pick a tax rate*. Yet this is not a realistic assumption. In many countries, bureaucrats don't have the capacity to extract resources from their society. Researchers talk of **state capacity** to describe the ability (or lack thereof) of public institutions to

design, coordinate, and enforce complex policies. Weak administrative systems can stall transitions through fragmentation, lack of expertise, or poor monitoring even when economic conditions are favorable.

7.3.2 Bureaucratic constraints

Imagine you are a mayor or a prime minister. You want to launch your policy. But the people (say, civil servants) who are supposed to implement may not want to comply. They might disagree with your policy. Or it is too hard given the resources they have access to. Or maybe it is not in their contract and they do not want to put in the effort into doing something new.

If you cannot perfectly monitor your staff's actions, there will always be a residual risk that they are not going to put the level of effort that you expect. This is one type of **principal-agent problem**. Principal-agent problems are situations in which there is asymmetry of information between an agent (e.g., an employee) who knows something that a principal (e.g., a boss) doesn't. This can generate welfare losses. In our case, the principal is a policymaker who is delegating a task (e.g., implement a public policy) to an agent (e.g., a civil servant) (Holmström 1979; Bolton and Dewatripont 2004).

Principal-agent problems come in two main versions. First, the principal ("she") might face different types of agent ("he"), ranging from good ("high type") to bad ("low type") ones. Here, the agent (e.g. the bureaucrat) might be someone who works well or not; only he knows. The agent has private information about himself. The principal doesn't know which one she is dealing with. This corresponds to the problem of **adverse selection**, whereby the principal struggles to identify the type of agents. (That's why human resources are such an important department in larger firms: they are here to screen good from bad recruits.)

Second, the principal can fail to ensure that the agent exerts a sufficient level of efforts. Here, the private information is that the agent's efforts are not fully observable to the principal. For instance, employers are not constantly checking how much work their employees do and thus won't know for sure whether they are slacking or working hard. This is the **moral hazard** problem.

Principal-agent problems are abundant and can help shed light on all kinds of relations:

- An employer (principal) and their employees (agents who complete tasks on the employer's behalf)
- A government (principal) and firms (the government wants firms to exert effort to reduce pollution)
- Voters (principals) and the government (agent who implement what voters want)
- A teacher (principal) and their students ("agents" who study to learn)

 Tip 2: The logic of university education

What is the purpose of a university? Or of a course?

One answer, naturally, is to answer: to educate. We could formalize this by saying that what a university or a course does is to try to maximize the difference between a student's knowledge at the beginning and the end of the program.

But that's probably not quite accurate. If the purpose was only learning, then we wouldn't need grades and exams. We would teach, you would learn, and that's it.

Another way to understand how universities are setup is as a *screening* mechanism. When we look at an incoming class, we (instructors, university administrators) don't know who is a good student and who isn't. And you (students) cannot be expected to tell us truthfully (at least those of you who aren't among the best). Thus, grades are a way to distinguish students and screen between those who are "high types" and "low types." This is just like an adverse selection problem. A GPA of 6.0 (or A+) can then be used by agents (employers, etc.) to learn about the type of the student.

But this immediately raises difficult questions:

- Can a single scalar capture the ability of students? What if a student is outstanding in one domain (math) and poor at another (chemistry)?
- What if grades are measured with error? A student might be sick during an exam. The grade is a noisy signal.
- What if the instructor isn't good at getting signal out of students? An exam filled with trivial (or impossible) questions

won't allow the instructor to distinguish between different types.

- How should grading be benchmarked? In a class with 10 students, it seems unfair to set an average at a particular grade. In a small-n sample, it is perfectly possible to have lots of very good (or very bad) students.
- How can screening be protected against negative feedback? In the United States, a commonly observed problem is that of "grade inflation": the share of A's has increased continuously. There are probably lots of reasons for this, but presumably one of them is that students who worry about being flagged as "low type" (low grades) prefer a system in which such discrimination isn't available. Thus, they will lobby intensively to get better grades, weakening the signaling effect of grades.

The problem of policy implementation often raises moral hazard issues. There are many ways to address them. One is to create a contract that rewards observable performance in order to make sure that effort levels are high. An example comes from an unlikely place: the Ukrainian army. Armies in general are prone to mismanagement. The Ukrainian government created a system in which brigades received credits for successful drone strikes. These credits could then be used to buy more equipment. The use of credits creates a pricing signal. The use of performance bonuses makes sure that officers think carefully what kind of equipment would be most effective (instead of carelessly overordering). See [A Choice of Deadly Drones Is Only a Few Clicks Away for Ukrainian Troops](#), *New York Times*, March 23, 2026.

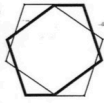
Bureaucratic moral hazard: from public choice to Parkinson's Law

Civil servants (or bureaucrats) are critical actors at several stages of the policymaking process. They routinely pre-write regulations, produce data and feasibility studies, and translate laws into policies.

The question is: are they passive executants or are they active agents? Weber (1922, pt. 3, Chapter 6), a German social scientist, described the bureaucracy as a tool that can rationally implement the desires of the ruler. A modern and effective bureaucracy

comes hand in hand with the modern state that has to tackle a complex set of issues.

Yet this doesn't tell us that bureaucrats actually act as Weber expects them to. There has been considerable research on what civil servants optimize, which can be quite complex (Downs 1965; Moe 1997). One particularly influential research agenda is often dubbed the **public choice** literature which insists on the notion that civil servants may have private goals that depart from collective goals (Tullock, Brady, and Seldon 2002, ch. 5). Niskanen (1968), for instance, presents a model in which an agency's objective function is to maximize its budget, subject to meeting the costs of producing whatever the agency is expected to produce. A letter by Arthur Schlesinger to Raymond Aron – two individuals who were important actors of public policy and international relations in the US and France, respectively – tells a similar story (Figure 7.2). This perspective is why many analysts and policymakers worry a lot about principal-agent relations. A Prime Minister will be told that an agency (e.g., the army) needs a larger budget, but she won't know whether it is for legitimate reasons or because the agency just wishes to grow.



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December 20, 1982

Mr. Raymond Aron
École Des Hautes Études
En Sciences Sociales
54, Boulevard Raspail
75270 Paris
France

My dear friend:

Forgive my delay in answering your letter of some weeks back,
but life here has been hopelessly (and rather uselessly) busy.

I share your dislike of facile assumptions of symmetry between
totalitarian and democratic societies, and I should have
phrased my comments with more precision. What I had in mind
was rather the parallelism derived from the natural history
of institutions. All defense departments have an institutional
inclination to increase their budgets. After all, the professional
duty of generals is to guarantee the safety of their countries,
and their professional instinct is to demand enough to provide
for all contingencies. As old Lord Salisbury once wrote, "If
you believe the doctors, nothing is wholesome; if you believe the
theologians, nothing is innocent; if you believe the soldiers,
nothing is safe." Defense budgets grow when enemies appear
menacing. It seemed to me no more than a truism to say that
the Pentagon and the Soviet Ministry of Defense prosper
symbiotically. My point was limited to that.

I need not say that I agree completely with your concluding
sentence. Do let me know if anything brings you to New York.
It has been too long since we have met, and I have learned much
from you through the years. Best Christmas wishes.

Faithfully yours,

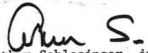

Arthur Schlesinger, Jr.

Figure 7.2: Letter from Arthur Schlesinger to Raymond Aron. Schlesinger was a professor at City University of New York and a speechwriter for John F. Kennedy and for Robert Kennedy. Raymond Aron was a journalist, member of the French Free Forces during World War II, and a professor at the *Ecole pratique des hautes études*. Source unknown.

There are variants on Niskanen's thesis. Tullock (1965) focuses on the career aspirations of civil servants, arguing that they will behave in a way that maximizes their chance of climbing the professional ladder. As in Niskanen, this creates a wedge between private and public interests. A satirical variant on these ideas can be found in Parkinson (1957). *Parkinson's Law* describes how overworked civil servants will first ask for reinforcement,

but they will prefer to hire people below them hierarchically (to avoid creating rivals). And since a single employee would still be a threat, they will want multiple sub-employees who fight between themselves. This creates a continuous growth process.¹

Regardless of the variant of public choice model, the fundamental problem is that the incentives of civil servants aren't aligned with the interests of the public. This connects therefore closely to principal-agent problems in general and moral hazard issues in particular.

Moral hazard problems can sometimes be solved by smartly designing contracts and institutions. China is a case in point (Landry, Lü, and Duan 2018). China's central government had long wanted stronger economic growth to increase its popular legitimacy. To make sure that bureaucrats would do their best to implement pro-growth strategies (instead of, say, engaging in rent seeking), the government linked career promotions to economic performance. And it worked. Landry, Lü, and Duan (2018, 1088) find that "a one standard deviation increase in relative revenue growth [*a measure of economic performance*] enhances prefectural party secretary's promotion probability by around 5 percentage points, which is a 14% increase in average promotion probability."

The public choice approach is by no means the only way to explain the behavior of civil servants. For instance, Dunleavy (1991) argues that what high-level civil servants really want isn't a large budget, but an interesting job. Instead, the thing to remember from the public choice model is the importance of designing incentives and institutions that align civil servants' to yours (V. Ostrom 2008).

Tip 3: Smart implementation of public policy

Implementing policies is costly: you need civil servants who monitor and enforce rules. The cost of doing so might well outweigh the benefits of the rule. A case that can be tricky is tax collection. People can hide their revenue or other taxable sources. Hiring people to check transactions and find out if people cheat is quickly expensive, especially for sophisticated systems. We saw

¹Think of Sir Humphrey Appleby in *Yes Minister* and *Yes Prime Minister*.

that tax collectors use tricks like counting the number of windows as a proxy for hard-to-measure variables (like wealth), but that doesn't always work. Indeed, we also saw that agents respond strategically to that.

One particular case where this can be problematic is tax collection from business transactions in a cash-based society. The problem is the following: a small shop receives a payment for some item. It's tempting not to report the sale and avoid paying taxes (e.g., Value Added Tax or VAT) on it. Since it's cash, it's hard to detect (unlike credit cards, which leave a trace). The buyer doesn't really care: they get their item. You could in principle hire tax collectors, but it would be very expensive, since you'd have to check every shop almost permanently.

Taiwan suffered from this problem, which created real missed resources for the state. In the early 1950s, this could have been tragic, since the island was clashing with China. Here's how the government solved the problem: every receipt was simultaneously a lottery ticket. You buy a bottle of shampoo? You get a lottery ticket on the flip side. Every other months, the lottery yields a number of winners who can pocket up to about CHF300,000 if they win the top prize. This is named the *Uniform Invoice* system. This is a smart design:

- Buyers suddenly want their ticket. Shops cannot hide their sales anymore.
- Each buyer becomes a tax auditor for free. No more scaling problems created by a large number of hard-to-monitor shops.

According to Wikipedia, the income boost was immediate, with an increase in tax revenues by 75% between 1950 and 1951, when the system was launched.

This is a form of **decentralized enforcement**. Decentralized enforcement is very appealing because it can help reduce cheating at a fraction of the cost. That's partly where the fascination with blockchain comes from. But these smart designs aren't necessarily desirable for people. Authoritarian regimes often find out that it's really expensive to check on what their people does. That's why they often try to turn people against each other. They outsource and decentralize monitoring in a way that is destruc-

tive for people.



Figure 7.3: Uniform Invoice. Source: twitter (user: CR1337).

7.4 Policymaking in complex system

Policymaking is considerably harder in *complex* systems (D. H. Meadows et al. 1972; D. Meadows 2008) — those with multiple target variables, feedback effects, and potentially nonlinear responses. We focus on three problems:

1. **Uncertainty:** the consequences of a policy are hard to predict.
2. **Negative side effects:** a policy effective on one target may harm another.
3. **Negative feedback loops:** a policy that works in a static system may trigger responses that undermine it over time.

7.4.1 Uncertainty

Policymakers constantly operate under **incomplete information**: we don't fully understand how physical systems or agents respond to change. Errors are inevitable — not necessarily because decisions are irrational, but because they are made under uncertainty. A decision that is correct given available information can still turn out to be wrong.

We can think of this using the framework of **multi-armed bandits** (Sutton and Barto 2018), in which a policymaker must build an algorithm to find the best policy. The name of the bandit comes from casinos, where gamblers must decide which slot machine is the one that will give them the highest payoff. The gambler can activate arms to try to learn about them (do they pay well?).

Concretely, suppose that the effect of a policy is θ_k (θ could be the impact of the policy on CO2 emissions), and that the policymaker must identify which of the k policies available (e.g., carbon tax, cap-and-trade, emissions standard) is the one with the highest θ . The policymaker doesn't know *ex ante* which one it is. She can *learn* by trying these policies, but she will obtain a noisy piece of evidence. For instance, imagine that instead of obtaining θ , she receives a payoff from a distribution centered around it: $\hat{\theta}_k \sim \mathcal{D}(\theta_k)$. After multiple draws, she will start to have some belief over which of the k available choices has the highest expected payoff.

Learning via sampling (experimenting) should improve the quality of policymaking. In addition, policymakers can try to learn from other contexts (e.g., other countries). This could help them refine their beliefs over θ_k . The notion of **policy diffusion** via learning captures the idea that a policymaker in one area might be able to learn (or copy) what happens elsewhere (Dobbin, Simmons, and Garrett 2007; Gilardi 2010).

Regardless, despite our best efforts, there will always be a residue of uncertainty when we want to forecast the effect of any given public policy. The chance of making errors will be compounded by the fact that, in real life, θ might change, making this a moving target. As a result, we say that policymakers operate under **bounded rationality** (H. A. Simon 1957, 2013). They can learn, they can reduce uncertainty, but doing so is costly. At some point, it might not be worth acquiring more knowledge because it is too costly or time is lacking. This is why policymakers must often launch programs without being entirely sure that it's a good idea.

7.4.2 (Negative) side effects

Negative side effects come in two forms. First, a policy may have a positive direct effect on the main target while simultaneously having

a *direct* negative effect on another. The example in Figure 7.4 comes from clean energy: phasing out coal and nuclear requires large-scale solar and wind, which may be sited in rural areas rich in biodiversity. The result is a risk of mitigating climate change while worsening biodiversity loss.



Figure 7.4: Direct policy side effects. Bottom: a renewable energy policy has a *direct* positive effect on climate stability and a *direct* negative effect on biodiversity (e.g., by building solar farms on land that is rich in biodiversity). In this example, each effect is certain; however, how the tradeoff is handled depends on decision-makers.

💡 The side effects of wind turbines

New technologies that have desirable features on one dimension may have negative effects on a second dimension. How to weight the two is then up to individual and collective decisions (as we saw when we discussed utilitarianism).

Wind turbines, for instance, produce electricity with no carbon emissions. From a climatic perspective, that's great. But opponents have raised a number of concerns about this technology. Some are financial in nature: wind turbines could negatively affect real estate value. We then enter a situation that's similar to a Coasian externality. At the time of writing, studies on this remain scarce. Yet in general, there is limited evidence of an impact of turbines on housing value: neither Lang, Opaluch, and Sfinarolakis (2014) (in Rhode Island) nor Hoen et al. (2015) (US at large) find a statistically significant relation between the two.

Another potential negative impact of wind turbines is that they cause public health issue. @Rottetal2026 test this idea against longitudinal household data from 2011-2023.² They link these data to the construction of wind turbines to see whether these turbines affect a wide range of health outcomes, “including depression, anxiety, sleep disorders, headaches, and use of sleep aids and painkillers.” They find no statistically significant effect, and

the confidence intervals of the effect of turbines are close to zero, suggesting that even if there was an effect, it would likely be very small. This being said, they also note that there are other outcomes for which they have no data and yet that could affect health, such as noise.

And yet: Stokes (2016) finds that voters from Ontario (Canada) who lived near newly established wind turbines ended up punishing incumbents: “precincts with a proposed or operational turbine, the vote share for the governing party declined by 4–10% compared to similar precincts without a turbine.” In the world of politics, such an effect is very large. Thus, it seems hard to rule out some kind of disutility caused by wind turbines to voters.

The side effect of “electrify everything” policies

Electricity, when generated from decarbonized sources, offers a clean way to deliver a range of services: cooking, heating, transportation. As a result, there has been a push toward the “electrify everything” policy approach. This has notably been the case for household services like heating and cooking.

Can the electrification push have side effects? Yes. Davis and Hausman (2022) and Garibay-Rodriguez et al. (2025) make the following point. Suppose that a municipality relies on natural gas for heating. Then, let’s say that a fraction of households transitions toward electricity. This means that the cost of maintaining gas infrastructure is now divided by a smaller base. Such fixed costs are not trivial: they are about half the total cost faced by utilities (Davis and Hausman 2022). Left-behind households have now to pay more. Looking at Massachusetts and the US more generally, Garibay-Rodriguez et al. (2025) find that “[h]ouseholds that do not electrify may bear a 46% increase in gas bills on average over the next 15 years (relative to current costs).”

Here, the policy (electrification) has adverse side effects on households that are not covered by it. Addressing this, in the words of Garibay-Rodriguez et al. (2025), requires better policy *coordination*. Below, we discuss the potential for policy *bundles* to tackle such side effects.

²“Longitudinal” means that the same units (e.g., households) are observed repeatedly (e.g., once a year).

How to weight these tradeoffs is an ethical question: fighting climate change may justify some biodiversity loss. You can't make an omelet without breaking eggs.

Second, a policy may have beneficial *direct* effects while generating *indirect* negative ones (Figure 7.5). A fiscal stimulus that boosts GDP may simultaneously accelerate clean technology deployment (a co-benefit) and increase consumption of meat and electricity (a co-harm). The net effect on climate is uncertain.

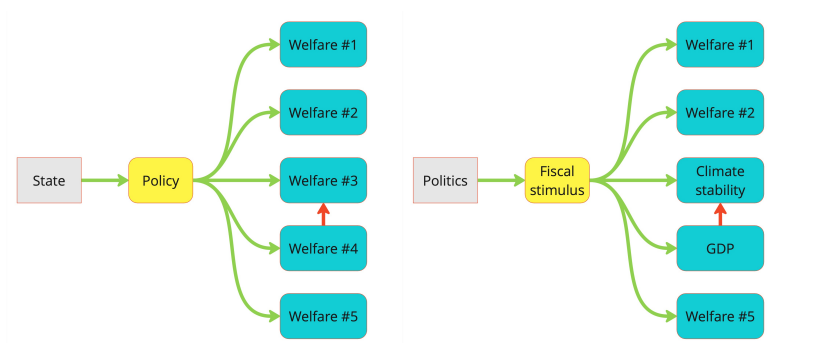


Figure 7.5: Indirect policy side effects. Right: a fiscal stimulus package has a *direct* positive effect on climate stability (e.g., by accelerating the deployment of solar technology); however, it has a negative *indirect* effect on CO₂ by increasing GDP and thus increasing consumption. The net effect on climate stability is uncertain.

💡 Positive side effects

Of course, side effects can also be positive. Hence the expression “killing two birds with one stone.” One example comes from (urban) congestion pricing. Recall that this is a policy in which access to an area with some kinds of vehicles (typically cars) is made costly. This is a form of Pigouvian pricing: instead of being free and therefore leading to overuse, access to a common good is made more expensive to discourage demand. The immediate benefit is fewer vehicles, and thus less pollution, fewer traffic jams and accidents, and faster mobility.

This, in turn, may generate secondary benefits. Less crowded roads give more space to ambulances. Medical staff can more rapidly arrive at the location of an emergency, thereby saving pre-

cious time. Chikish et al. (2026) estimate such a benefit by looking at the implementation of New York City's congestion charge in 2025. To get a causal estimate, they compare areas right above and below Manhattan's 60th Street.³ Vehicles only had to pay a congestion charge when entering the southern part of Manhattan (i.e., going south of 60th St); the streets above did not entail a charge. Thus, the latter still had lots of traffic compared to the streets south of 60th St. Using this analysis, Chikish et al. (2026) find:

we find that the January 2025 implementation of congestion pricing reduced passenger vehicle traffic by roughly 21 percent near the boundary and improved total EMS [Emergency Medical Service] travel times by 63–70 seconds, approximately 5–6 percent.

This is good news. The authors note the implication of this finding for public policy choices:

These findings suggest that cost-benefit analyses of congestion pricing systematically understate net social benefits by omitting emergency response improvements.

But if you were in charge of conducting a cost-benefit analysis, would you have thought about listing this benefit? You might not have thought of it, and even if you had, you might not have had good data to quantify these benefits. This is a limitation of cost-benefit exercises: the list of potential benefits and costs is unknown, thereby creating considerable uncertainty.

7.4.3 Feedback loops

Another problem that plagues public policy is the presence of *negative feedback loops*. The idea is the following. Imagine a dynamic system in which input variables (e.g., a carbon tax) affect the state of the world

³In causal inference, this kind of design is called a *regression discontinuity design*, whereby units on either side of an arbitrary cutoff (here: 60th St) are compared. The idea is that the kind of emergencies occurring on, say, 59th St are likely to be very similar (on average) to emergencies experienced on, say, 62nd St.

(e.g., total carbon emissions) at time t . Such a system includes a feedback loop if the state of the world at time t affects input variables at time $t + 1, t + 2, \dots$ It is a *positive* feedback loop if it monotonically changes the state of the world and it is a *negative* feedback loop if the system returns to its original point. In other words, feedback loops capture the way agents respond to shocks (such as public policies).

 Tip

Swiss authorities forbid publicizing the presence of speed radars. As a result, apps like Waze have turned off the possibility, for drivers, to share information about speed checks they may see. This policy turned out to be vulnerable to feedback effects. Waze users, unable to flag radars, started instead to indicate the presence of *potholes* on the highway. Potholes are very rare (it generally doesn't get cold enough). Thus, this button simply replaced the one tagging radars. In theory, authorities could try to block that too, but it's starting to be arbitrary and inefficient (there are genuine reasons to allow drivers to signal real potholes). It turns out that it's very hard to regulate that kind of issue.

Sidenote: drivers' response suggests that they overcome a *coordination* problem. It's not clear *ex ante* that they would pick one button (potholes) over another. Drivers cannot communicate on that platform. Yet they seem to have found a way to overcome it.

Un «nid-de-poule» sur l'autoroute? C'est peut-être un radar

L'application de navigation Waze a désactivé les signalements de radars en Suisse pour se conformer à la loi. Mais les utilisateurs ont déjà trouvé la parade.

Figure 7.6: Source: 24 Heures, August 29, 2026.

What we are particularly interested in are **political feedback loops** (and we will generally refer to this kind of feedback loops from now on). In recent years, a number of studies have shown that green policies generate negative responses from the public. In the US, the closure of coal mines led to a shift of voters toward Republicans and away from Democrats (Egli, Schmid, and Schmidt 2022). In Canada, the construction of wind turbines undermined local incumbents (Stokes 2016). In Italy (Milan), congestion pricing increased the vote share of the Lega, the far-right party (Colantone et al. 2022). In France, an increase in

fuel tax triggered the Yellow Vest movement (Tatham and Peters 2023). In Europe, higher electricity prices undermined public support for renewable energy (Aklin 2021).

In contrast, policies that trigger *positive* feedback loops are hard to overturn. Policy programs such as Franklin Roosevelt's *New Deal* created entrenched constituencies that allowed key provisions to last several decades. Barak Obama's healthcare program ("Obamacare") initially received considerable pushback, but eventually proved very hard to overturn. In the UK, the popularity of the National Health Service (NHS) means that despite its inefficiencies, it has remained at the core of the British healthcare system since its inception. In the case of energy, support for the renewable energy industry has made the sector politically powerful in countries such as Denmark and Germany (Aklin and Urpelainen 2018). This is a form of **path dependence** (Kline 2001; Pierson 2000). Once you take a given (policy) path, changing becomes increasingly costly with time (Aklin and Urpelainen 2013b). Thus, we face *self-reinforcing equilibria*.

Figure 7.7 provides an illustration of such processes. This is where the importance of embedding policy into a *political* system becomes very important. A purely technocratic approach would ignore the role of political conflicts in shaping public policy in the long term. The idea that policy should "follow science" ignores the fact that policy depends on politics first and foremost. A technocratic government that implements a policy that is not popular among key stakeholders (which can vary by country and include voters, interest groups, the army, etc.) will not last long. Thus, for policies to be *feasible* in the long-term, policy-makers must address negative feedback loops.

Feedback loops can be categorized in two ways (Pierson 1993; Schmid and Lumsden 2023):

By whom they affect:

1. **Mass feedback:** operates through large numbers of agents such as voters and residents.
2. **Elite feedback:** operates through smaller groups closer to policymakers: elected officials, civil servants, interest groups, media.

By their mechanism:

1. **Resource mechanism:** a policy generates resources that actors use to mobilize against it (e.g., fossil fuel profits fund lobbying).

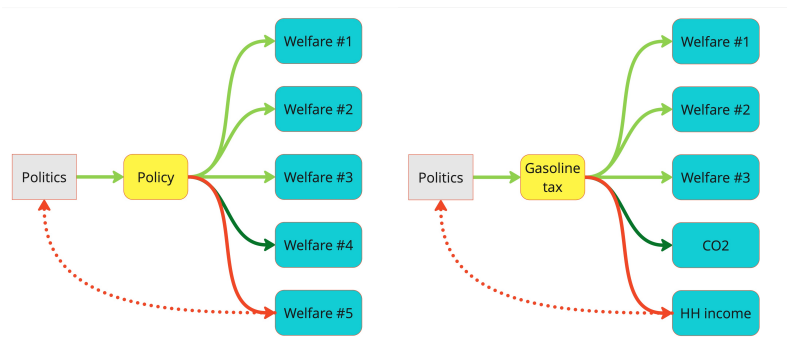


Figure 7.7: Political feedback loops. Right: illustration of a negative feedback loop in which a gasoline tax, applied to an inelastic demand, reduces household income which undermines political support for this policy.

2. **Interpretive mechanism:** a policy generates media coverage or public narratives that shift how policymakers and citizens understand the problem.

Climate policy and negative feedback loops

Climate policy tends to generate significant negative feedback loops (Jordan and Matt 2014). Pigouvian pricing imposes concentrated costs on key groups — fossil fuel companies, heavy industries, consumers with inelastic demand, homeowners affected by wind turbines — creating a structural tendency toward what is sometimes called **carbon lock-in** (Unruh 2000). An effective and durable climate policy must therefore be designed to (a) minimize negative feedback loops and (b) generate positive ones that create constituencies for continued reform (Jordan and Matt 2014).

Understanding feedback loops is essential for strategy. The goal is not merely to pass a policy but to design it so that its political consequences make it harder to reverse.

7.5 Designing effective policy strategies

Two main strategies can help manage negative feedback loops and adverse side effects in complex systems: policy portfolios and policy sequencing.

7.5.1 Policy portfolios

When a single policy generates feedback loops or unwanted side effects, one remedy is to combine multiple instruments into a *policy portfolio* (also referred to as *policy bundles*, *policy programs*, or simply *policy combinations*) (Doremus 2003; Sovacool 2008; Howlett and Rayner 2013; Schaffrin, Sewerin, and Seubert 2014). The logic is illustrated in Figure 7.8. Consider carbon pricing: economists often treat it as a first-best instrument, but it is frequently rejected by the public — particularly when consumers face inelastic demand and lack affordable substitutes, as the French fuel tax episode showed.

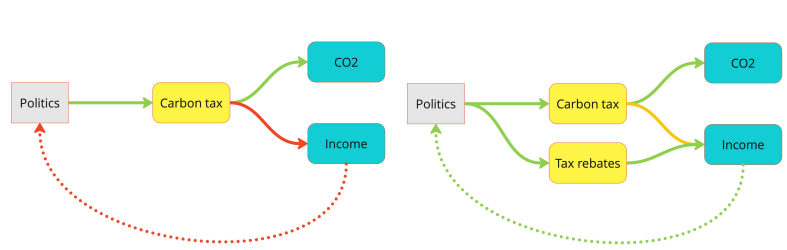


Figure 7.8: Policy portfolio. Combining policies can mitigate undesired side effects and negative feedback loops.

One response is to pair a carbon tax with a tax rebate — redistributing the revenues back to consumers so they retain the Pigouvian price signal without bearing the full distributional burden. This is the logic of the right panel of Figure 7.8. Empirical evidence on whether this works is discussed in the box below.

💡 Can rebates mitigate the unpopularity of energy taxes?

We present here the striking experimental evidence from Fremstad et al. (2022).

William Nordhaus estimates that a carbon tax of approximately

\$230/tCO₂ is needed today to keep global temperature increases below 2.5°C. In contrast, the International Monetary Fund reports that the global average carbon price is only \$2/tCO₂. Consequently, most carbon prices implemented to date have had limited impact on national emissions trajectories.

A major challenge is the lack of popular support for increasing the price of carbon, as numerous international examples show. Voters have rejected carbon pricing initiatives on multiple occasions, including in Switzerland in 2000, 2015, and 2021, and in Washington State in 2016 and 2018. In other cases, new governments have quickly rolled back existing carbon prices, as in Australia in 2013 and the Canadian province of Ontario in 2018. The ‘Gilets Jaunes’ protest movement in France highlighted working-class opposition to carbon taxes, which disproportionately burden the poor.

One possible solution is rebates as a form of policy design that may address the negative feedback loop by increasing popular support for carbon pricing. Rebates typically take the form of equal lump-sum payments, often referred to as carbon dividends or rebates. These measures can create a short-term, tangible economic benefit associated with policy enactment, helping to reshape the politics of carbon pricing. Publics may subsequently mobilize to defend the carbon price from opponents or even advocate for its expansion to maintain these benefits. Rebates may also address the public’s seemingly low willingness to pay to mitigate climate change.

Through a survey experiment, Fremstad et al. (2022) demonstrate that rebates are indeed effective in increasing public support for carbon pricing, even at levels high enough to efficiently mitigate climate change. However, they also find that political messaging can reduce the effect of rebates, leaving such policy designs vulnerable to shifting political dynamics.

Another example comes from food (Fesenfeld et al. 2020).

💡 Can policy packages mitigate the unpopularity of sustainable food policy?

There are increasing attention and interests for policies to regulate and incentivize sustainable food consumption and production. While supply-side measures promote environmentally friendly production methods, demand-side policies targeting consumer behavior are crucial for achieving climate goals, avoiding carbon leakage, and preventing environmental tipping points. However, these demand-side policies often face low public acceptance, which makes their implementation politically challenging.

One strategy to address this issue is policy packaging. Policy packaging refers to the bundling of multiple policy instruments in a single package, which can collectively reduce the risk of public backlash. Fesenfeld et al. (2020) studied the effects of policy packaging in the context of sustainable food consumption, using survey-experimental data from China, Germany, and the United States.

Their findings reveal a trade-off between policy effectiveness and political risk: policies that are highly effective at reducing carbon emissions tend to be costly and can face strong opposition. Importantly, the design of policy packages can mitigate this trade-off:

- Policy packaging mitigates trade-offs: Bundling costly instruments with measures that provide benefits to citizens (e.g., discounts, environmental improvements) increases public support across income groups and ideological constituencies.
- Policy design matters: Effective packages balance medium-level stringency for costly demand-side measures (e.g., higher taxes), strong supply-side regulations (e.g., stricter production standards), and consumer benefits (e.g., discounts for sustainable alternatives).
- Challenges remain: Organized producer interests may resist packages that favor consumers, emphasizing the need to consider both citizen and stakeholder preferences in policy design.

Policy portfolios are not free. Each added instrument risks *policy in-*

coherence (policies working against each other), raises administrative transaction costs, and adds system complexity. A classic illustration of incoherence is shown in Figure 7.9: simultaneously promoting renewables and running a cap-and-trade scheme can backfire. As renewables displace fossil fuels, demand for CO₂ permits falls, permit prices drop, and heavy industry faces weaker incentives to decarbonize — partially offsetting the environmental gains.

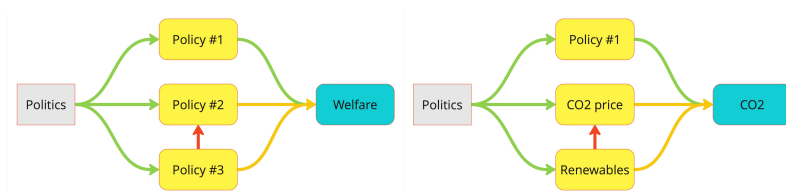


Figure 7.9: Policy incoherence. Right: higher volumes of renewable energy reduces the demand for carbon permits, reducing incentives to switch to cleaner technologies.

7.5.2 Sequential policymaking

Sequential policymaking recognizes that the *order* of implementation matters, not just which instruments are chosen (Meckling et al. 2015; Pahle et al. 2018). A policy that triggers a negative feedback loop when implemented first may succeed if a prior policy has already changed the underlying conditions — as illustrated in Figure 7.10.

Meckling et al. (2015) give a concrete example using carbon pricing. Rather than imposing a carbon tax immediately (and risking political backlash due to high costs), a sequencing strategy first invests in R&D and substitutes — EVs for transportation, renewables for power — allowing their prices to fall via learning curves. By the time the carbon tax arrives, affordable alternatives already exist, and opposition is weaker.

Hale (2020) extends this logic to international institutions, arguing they can play a **catalytic** role by accelerating innovation and lowering the costs of future policies.

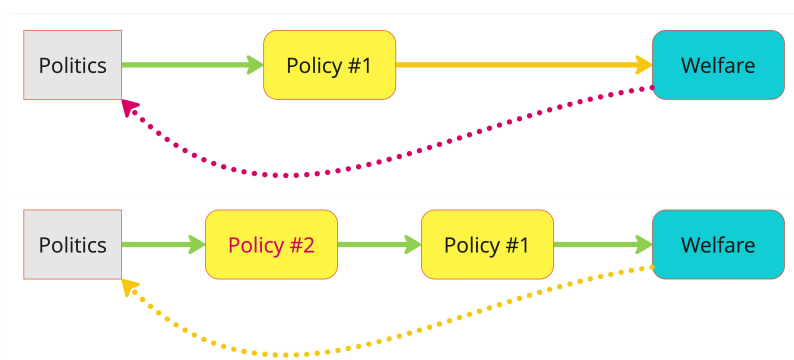


Figure 7.10: Policy sequencing. Top: Policy 1 is ineffective because it generates a negative feedback loop. Bottom: sequencing policies by starting with Policy 2, followed by Policy 1, mitigates negative feedback.

7.6 A checklist for policymaking

- Define *what you want to optimize*: social welfare, distributional equity, or both. Think through intended and unintended consequences.
- Diagnose *why* the optimum isn't being reached (externalities? coordination failures?). Select instruments accordingly, keeping the Tinbergen rule in mind.
- Assess whether you need *proscriptive* or *prescriptive* rules. Prescriptive rules (telling agents what to do) require substantial information.
- Identify the right *level* of intervention: demand-side instruments within existing institutions, or deeper institutional reform.
- Build *coalitions*. Political feasibility depends on securing support and avoiding vetoes from key actors.
- Account for strategic responses. Agents adapt: policies designed in partial equilibrium can be undone by moral hazard and adverse selection (see box below).
- Design for *self-reinforcement*. Policies that create their own political support are more durable.

- Consider the *costs of creating and maintaining* your policy apparatus. Political failures can be as costly as the market failures you are trying to fix.

Throughout this chapter, we have assumed that policymakers can act — that they have the authority and the political space to implement what the analysis recommends. The next chapter drops this assumption. It examines how power is distributed, why collective preferences are inherently difficult to aggregate, and how organized interests and political feedback loops define the true boundaries of the feasible.

Moral hazard and adverse selection

One problem with policymaking is that agents may “game” your interventions (without any malice). Two examples are *moral hazard* and *adverse selection*.

Moral hazard refers to situation in which an intervention changes the way an agent takes risks. The typical example is a seatbelt: drivers may decide to drive faster if they believe they are safer. States might take greater foreign policy risks if they believe that their defenses are impossible to beat.

In sustainability, an example of moral hazard would be a policy that offers financial guarantees to firms that invest in, say, building solar panels. If they have a guaranteed lifeline, these firms might take exaggerated risks that they would not engage in otherwise.

Adverse selection assumes the existing of different *types* of agents. An intervention would then create a selection effect, whereby the ‘worse’ types behave differently from ‘better’ types. The canonical example is optional health insurance: if people aren’t forced to buy one, then sick people will disproportionately subscribe, whereas healthy people won’t. Your health insurance will then not fulfill its missing to pool different types of risk.

Consider for example a case where the government provides loans at a higher cost than the private capital market. Suppose that poorly performing firms cannot get loans from banks. This will create a selection effect, whereby the ‘bad’ firms borrow from the government and the good firms from the private sector. You can imagine what happens when a recession hits: the government-funded firms will fail at a higher rate than the

privately-funded ones.

Interestingly, policymakers can also try to leverage these problems to their advantage. Cicala, Hémous, and Olsen (2022) came up with an intriguing idea for climate policy. They suggest a policy whereby firms can choose between two types of taxes:

1. A tax on verifiable CO₂ emissions. This is a conventional Pigouvian tax.
2. A tax on output based on the average rate of emissions among firms that don't disclose emissions.

Cleaner firms will pick (1) (assuming the tax isn't set too high). But this raises the tax for the other firms: since they pay a tax based on the *average* emissions, and since cleaner firms exit, this increases the average emissions. Thus, gradually, firms will migrate toward Policy (1) as the cost of the tax in Policy 2 increases. Furthermore, by giving firms the choice, it is possible that their political hostility will be weaker than under mandatory Pigouvian pricing.

Additional readings

The literature on public choice took off in the 1950s, notably with Kenneth J. Arrow (1951), Buchanan and Tullock (1962), and Olson (1965). A good textbook (now a bit dated) is Mueller (2003). See also Austen-Smith and Banks (2000) and Austen-Smith and Banks (2005). For textbooks in political economy, see Drazen (2000) and Persson and Tabellini (2002).

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